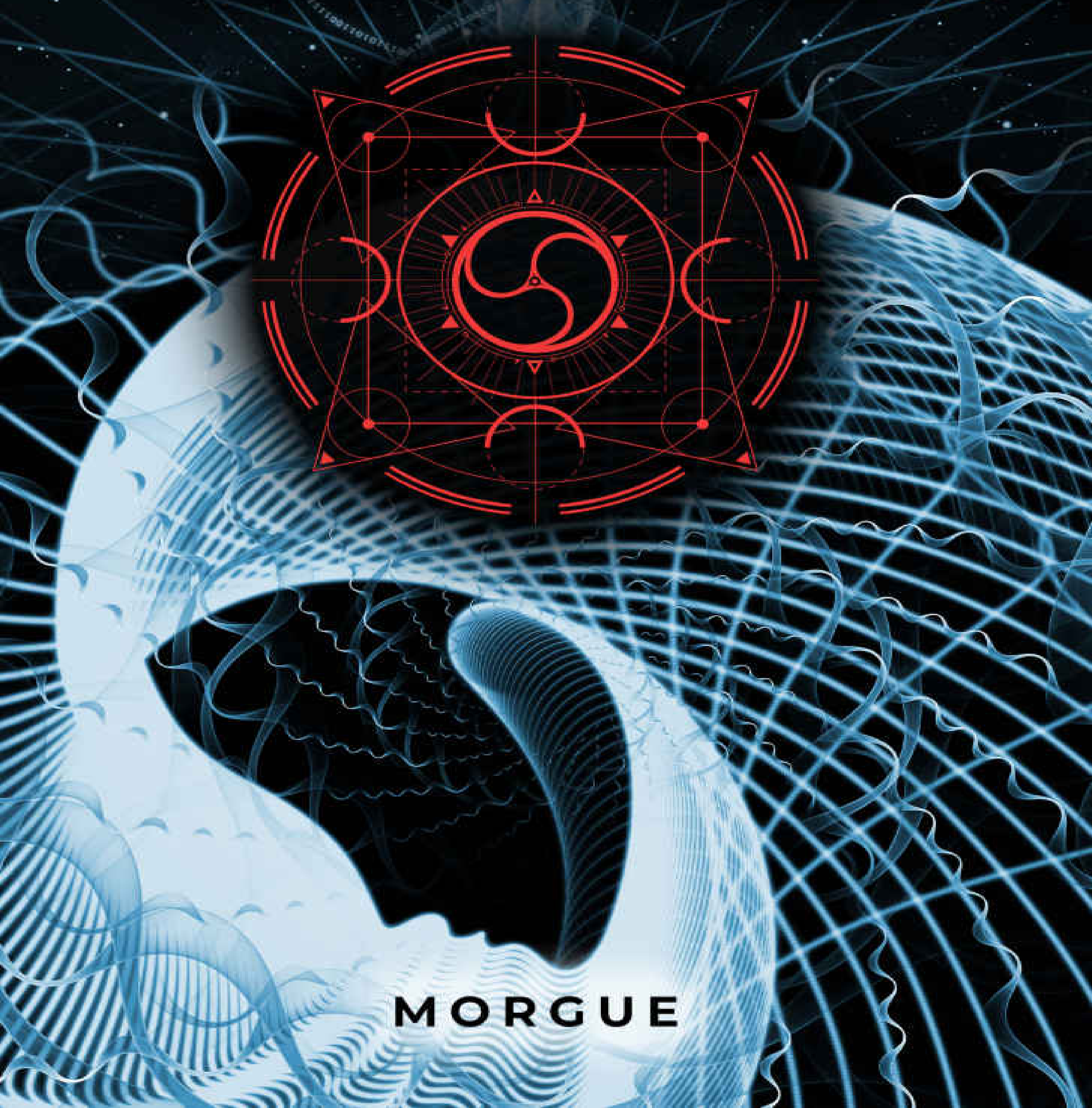
ONTOLOGICAL MATHEMATICS

THE SCIENCE OF THE FUTURE



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PREFACE: HOW TO USE THIS BOOK

This book is an introductory guide to ontological mathematics. It can be read on its own or as a companion to our other material to help you understand the mathematical foundations of Hyperianism.

This book is divided into three sections. The first builds the necessary philosophical foundations for understanding the system. The second lays out the details of ontological mathematics. The third explores the implications ontological mathematics has on reality and how it solves major problems in physics. The information contained in the footnotes is not necessary but included for those who want a deeper understanding. Terms important to our system have been highlighted in bold.

Ontological mathematics is the most complex system in history. With that in mind, this book does not intend to teach every aspect of ontological mathematics. Such a book would be thousands of pages long and would require an understanding of advanced mathematics, physics, and philosophy. We have made the groundbreaking knowledge of our system available to all by introducing the reader to the foundational concepts of ontological mathematics in an accessible way.

This text assumes the reader has only minimal philosophical knowledge, and it is written in such a way that anyone with an understanding of algebra will be able to begin learning the mathematics of our system, though a small amount of trigonometry, specifically the unit circle and the trigonometric functions, will be helpful.

Because of the minimal prerequisites, liberties have been taken to simplify the subject so that it is easier for the average reader to grasp. For example, speed and velocity are used more or less interchangeably, we restrict ourselves to the complex plane, and concepts are explored in an intuitive way rather than in strict, deductive form. Wikipedia has been quoted in several places due to their clear and succinct language. When learning a new subject, it's sometimes necessary to sacrifice some precision to increase comprehension.

This text serves as an introduction to ontological mathematics, providing a level of understanding comparable to what you would receive from taking an introductory course at a university. It's unrealistic to think it's possible to learn everything about ontological mathematics, the most complex system in history, from an introductory text. However, this book will give you an understanding of the most important concepts and give you a glimpse of this monumental, paradigm-shattering system.

Imagine living in a time and place where the Earth is believed to be flat and humans created by a god. Now imagine you discover a book containing knowledge that explains the Earth is a globe, evolution, and many other astounding facts of science. How exciting would that be? As you read the book, your entire perspective of reality would change. Your world would never be the same.

You are holding such a book right now. You currently exist in a time and place where existence is viewed as material. This book reveals that the world is in fact a shared dream.

When properly understood and integrated, the information within this text will change your existence forever and elevate you to a new level of consciousness. This is the science of the future that one day soon will be taught in every school throughout the world.

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Gratitude must be extended to the pioneers of ontological mathematics, Dr. Thomas Stark and philosopher Mike Hockney, without whom this book would not be possible.

A special thanks to my good friend and powerful mind, Susan Mitchell.

“I beg you, reject antiquity, tradition, faith, and authority! Let us begin anew by doubting everything we assume has been proven!”

– Giordano Bruno

SECTION 1: PHILOSOPHICAL FOUNDATIONS

Ontological mathematics is the science of the future that proves the shocking truth that the world is not material but a collective dream. This is not a belief but a deductive, mathematical certainty.

Ontological mathematics requires such a radically new way of thinking that, before exploring the system itself, it's necessary to throw out all the prejudices of the old paradigm and understand exactly how this new paradigm differs. This will require a sturdy foundation of philosophical concepts that will allow us to discover what ontological mathematics truly is.

Ontological mathematics ushers in a revolutionary paradigm shift that allows us to rationally explore the domain of mind. It's the mathematical core of Hyperianism. This is not numerology, gematria, or any kind of new age woo. This is the system of the future that will soon be taught in every school throughout the world. We reject faith, divine revelation, "holy books," as well as the flawed interpretation of the senses. We uphold the truths of reason, logic, and mathematics. To understand this revolutionary system, it's imperative that you have an open mind and look at it objectively, logically, and rationally, instead of relying on the conditioned prejudices of an outdated paradigm.

Ontological mathematics is a new model of reality that requires a new way of thinking.

1.1 - WHAT IS A PARADIGM?

A **paradigm** is set of ideas or beliefs. It's a certain perspective or way of modeling an aspect of existence. It's a conceptual framework or pattern of thought. Everyone subscribes to a paradigm of one kind or another. For example, creationism and evolution are two different paradigms about the origin of species. A paradigm shift is an event that occurs when a model held by the majority gets replaced by a new model. It’s when an old pattern of thinking or way of doing something gets replaced by something new. A paradigm shift occurred when the globe model replaced the flat Earth model. Paradigm shifts usually occur as revolutions instigated by the trail blazers of the new paradigm and are usually fiercely opposed by those who cling to the old. The Scientific Revolution was a paradigm shift from religious faith to empirical science, and it faced enormous resistance especially from the Catholic Church.

The Scientific Revolution is considered to have begun with the Copernican Revolution in 1543. The Copernican heliocentric model placed the Sun at the center of the solar system with the Earth revolving around it. This new paradigm was to replace the old paradigm of the Ptolemaic geocentric model which had the Earth stationary at the center of the universe with the Sun revolving around it. The revolutionary shift to the heliocentric model was supported by the astronomer and physicist Galileo Galilei. The impact that a paradigm shift has is monumental. It often falsifies everything that was once thought to be true and drastically alters consciousness. It’s like stepping into a new reality. The following quote from the play “Life of Galileo” by Bertolt Brecht captures the feeling well:

“For two thousand years men believed the Sun, the planets and all the stars revolved around them. The Pope, cardinals, princes, captains, merchants, fishwives and school boys thought they were stuck dead still, at the centre of that crystal ball. But now we’re flying headlong into space…The old age is dead, a new age is born…For now we know – everything moves! Where belief sat now sits doubt… All that was never doubted, we doubt…The people of our cities cry out for new ideas. It will delight them that a new astronomy says ‘now the Earth moves too.’ Always we were told ‘the stars are fixed to a roof of crystal to stop them falling down.’ But now we’ve got the courage to set them free and flying through space. Overnight the universe lost its centre and this morning – they are countless. Each and none at all is the centre. Suddenly there’s a lot of room.”

For freethinkers, stepping into the new world brought about by a paradigm shift is exhilarating. But those who cling to the decaying remains of the old world will resist and curse the new with their dying breath.

Galileo published evidence supporting the Copernican heliocentric model, and as a result faced heavy persecution by the Catholic Church. He was immediately accused of heresy on the grounds of contradicting scripture, which established the Earth at the center of the universe. In 1633, he was put on trial and convicted. He was forced to curse and reject his ideas. His books containing evidence for heliocentrism were banned, and he was sentenced to house arrest for the rest of his life.

"In the sciences, the authority of thousands of opinions is not worth as much as one tiny spark of reason in an individual man."

– Galileo Galilei

The persecution of Galileo is a clear example of the old paradigm resisting the new. The church's paradigm was based on faith, scripture, and dogma. It heavily resisted the new paradigm based on science and empirical evidence. The church sided with the supposed "absolute fact" of scripture that supports a geocentric view of the universe despite scientific evidence against it.

A new paradigm shift is occurring right now. This is the shift from scientific materialism and empiricism to the scientific idealism and rationalism of ontological mathematics. Ontological mathematics is the new paradigm, while scientific materialism and empiricism is the old. A new revolution in thought is here; one that is more shattering than any that has come before. Whereas the heliocentric shift replaced the Earth at the center of the universe with the Sun at the center of the solar system, we replace matter as the basis of reality with mind.

The universe isn’t physical. It’s a shared dream.

As we lay out the details of ontological mathematics, it’s crucial to remember since it is a new paradigm, you must leave behind the beliefs and prejudices of the old paradigm to be able to comprehend it. You can’t use faith to understand science, and you can’t use materialism and empiricism to understand idealism and rationalism. If the church wanted to understand Galileo, they would have had to first abandon their paradigm’s adherence to faith and dogma in order to view it with new eyes.

If you aren’t familiar with some of the terms we’ve used, don’t worry, they’ll be explained soon. For now, understand that there’s a paradigm shift occurring that’s even greater in magnitude than the shift from faith to science. We are now moving from matter to mind, from empiricism to rationalism. Ontological mathematics isn’t my or any one person’s idea. It’s a new way of thinking that is sweeping the world and is championed by the greatest thinkers of the age. Nearly 100 books have been written about it by various authors. Independent ontological mathematics research groups are appearing around the world. Ontological mathematics and Hyperianism is a global phenomenon.

We are a revolution of consciousness that can’t be stopped. The paradigm of matter is over. The paradigm of mind is here.

To understand this new way of thinking, it’s critical that you abandon old ways of thinking and examine our system objectively with logic and reason.

Just as the heliocentric model constituted a new way of thinking by placing the Sun at the center of the solar system rather than the Earth, ontological mathematics replaces matter with mind as the foundation of existence. Just as scientific empiricism replaced religious faith and dogma, ontological mathematics replaces scientific empiricism and materialism. We replace it with reason, logic, mathematics, and mind. Many people get angry and defensive when presented with a new way of thinking. Those who refuse to have an open mind and examine our system rationally are no different than the Catholic Church that persecuted Galileo and so many more.

Giordano Bruno was a mathematician, philosopher, and hermeticist who developed his cosmological theories around the Copernican heliocentric model. He believed the Sun was at the center of the solar system and distant stars were suns surrounded by their own planets that could possibly support life. In 1593, he was charged for heresy and was sentenced to death. Bruno made a threatening gesture towards the judges and said, “Perhaps you pronounce this sentence against me with greater fear than I receive it.” As he rode to his fate, he responded to the jeering crowd with quotes from his books. A large metal spike was thrust horizontally through his cheeks to pin down his “heretical” tongue. A second spike was thrust upwards through his lips, so that together the spikes formed a cross. Resistant to the end, he turned his head away from those trying to “save his soul” as he was burned alive.

Today, our ideas are being resisted just as the Church resisted the ideas of Galileo and Bruno. It’s important to have an open mind and examine everything with logic and reason instead of condemning anything that doesn’t comport with your current paradigm. Many people would like to think they’re open minded, when in truth they’re just as blind and ignorant as the Church.

“Truth does not change because it is, or is not, believed by a majority of the people.”

– Giordano Bruno

1.2 - A New Way of Thinking

Ontological mathematics proves that we exist in a collective dream. It shows that so-called “matter” is an illusion and that the ultimate reality is a domain of pure mind. Before we explain this, you must understand how the new paradigm works and realize this is a new way of looking at reality. The old paradigm is one of scientific empiricism and materialism, whereas ontological mathematics is one of scientific rationalism and idealism.

Materialism is the belief that everything that exists is either material or is a result from the interactions of matter. That's it and nothing else. **Empiricism** is the belief that knowledge is gained through sense experience. Materialism and empiricism fit together well. Materialism is the belief that all that exists is matter. Empiricism, which is the belief that all knowledge is gained through sense experience, works well with this. According to materialism, matter is the fundamental substance that can be detected by the senses, which according to empiricism leads to knowledge. It's a pairing of matter and the senses.

Ontological mathematics, and thus Hyperianism, is a rational idealism. **Idealism** is the position that everything that exists is ultimately mind. This is the opposite of materialism. **Rationalism** is the position that knowledge is gained through logic, reason, and mathematics, and *not* through faith or sense data. Rationalism is the opposite of empiricism. Rationalism and idealism fit together very well. Idealism says that all of reality is mind. Rationalism, being pure reason and logical thought patterns, can explore mental concepts. It's a pairing of mind and thought.

Materialism	**Idealism**
All things are material. There is no higher reality beyond matter.	All things are mind. "Matter" is an illusion resulting from mind. Mind is the true reality.

Always keep these distinctions in mind. If you apply an empirical and materialist mindset to our rational and idealist system, you'll get absurd results. This would be similar to a religious person attempting to approach scientific evidence from a faith-based position. You'll recall the condemnation of Galileo's evidence because it didn't agree with scripture. No matter how much evidence you may show a religious person, they won't believe you unless it agrees with their "holy" scripture.

Likewise, if you're expecting physical (empirical) evidence that reality is mental (non-physical), you haven't understood our position. It's impossible, by definition, to provide physical evidence of something non-physical. Such absurdities result from trying to understand our system from the perspective of an old paradigm rather than realizing it's an entirely new way of thinking and viewing existence. It's an utter impossibility, by the very definition of the methods and substances involved, to provide physical evidence of something non-physical. In fact, it's laughable that people ask for it with a straight face. Ontological mathematics provides rational, mathematical proof that reality is a shared dream, and proof is superior to evidence. But for you to understand this, you must be willing to step outside of the empirical, materialist paradigm to examine clearly what ontological mathematics says about existence.

"All significant breakthroughs are
break -"withs" old ways of thinking."

– Thomas Kuhn, Philosopher of Science

As you read, you may be thinking to yourself, "When are you going to get to ontological mathematics?" Before we can discuss the mathematics itself, we must first show why the old paradigm is false and how to think in terms of the new paradigm. Without that foundation, it won't make sense. It would be like trying to show empirical evidence to someone who will only believe what's written in the Bible. People of that mindset place faith over evidence. You can show them all the evidence in the world that the Earth is 4.5 billion years old, but they'll continue to believe that the Earth is 6,000 years old because that's what their scripture says. Likewise, all the equations and rational deductions in the world won't help if you're stuck in an empirical mindset, just as those stuck in faith and divine revelation don't care about empirical evidence. Ontological mathematics requires a new way of thinking. So, let's start using our minds.

1.3 - Hempel's Dilemma: Physicalism is a Faith

"A great many people think they are thinking when they are merely rearranging their prejudices."

– David Bohm, Quantum Physicist & Philosopher

A multitude of logical problems arise when reality is viewed as material. If you believe that reality is only material and can be explained exclusively in terms of physical laws, then you're taking a position based on faith. This position is called physicalism and is so close to materialism that the two terms are often used interchangeably. To understand why this is a faith-based position, let's look at Hempel's dilemma. Hempel's dilemma demonstrates that if you take the materialist/physicalist position, then your only options are to believe in something known to be incorrect or to believe in a fantasy, both of which are positions of faith and not knowledge.

If your position as a materialist is that reality can be explained in terms of today's physics, then you're taking a position that is known to be false since there are many aspects of existence that today's physics can't explain, such as the hard problem of consciousness and the incommensurability of quantum mechanics and Einsteinian relativity. It's an accepted fact that the physics of today can't explain all of reality, and to take the position that it can is a position of faith. If you were being intellectually honest, you would have to concede that the possibility exists that physicalist science may have these immense problems precisely because it doesn't include the mind.

The other position you could take would be to believe in a theoretical future physics that is complete and doesn't have these problems. But this is an utter fantasy. It's pure speculation. A future physics may look nothing like today's physics and may very well be predicated on the mind and thus no longer be a physicalist position (in fact, ontological mathematics *is* that future system of the mind). To believe in some undefined "future physics" is a position of faith. Whether you believe in today's physics or in an undefined future physics, Hempel's dilemma shows that *physicalism is a faith.*

Hempel's Dilemma

Those who subscribe to physicalism can take one of two positions:

1. **Reality can be explained in terms of today's physics**: This is known to be false; thus, this is a position of faith.

2. **Reality can be explained in terms of a future physics**: A future physics is an undefined concept; thus, this is a position of faith.

Conclusion: Physicalism is a faith.

All those who subscribe to physicalism may think they're being rational, but as Hempel's dilemma shows, they're taking a position of faith. To avoid a faith-based position, an open-minded person would have to concede the possibility that there's more to reality than matter and that a new paradigm, very different from physicalism, could reveal its nature.

1.4 - The End of Matter

“Everything we call real is made of things that cannot be regarded as real.”

– Niels Bohr, Quantum Physicist

During the 19th century, it seemed that scientific materialism was well on its way to being able to describe everything in existence. Sure, there were some problems, but it was generally believed they would be eventually ironed out. In the early 20th century, everything changed. There was a revolution in physics as a paradigm shift took place from **classical mechanics** to **quantum mechanics.** The advent of quantum mechanics absolutely destroyed the idea that reality is composed of matter as classically described. If you’re unfamiliar with the double-slit experiment, go watch a video on it now.

Through the discoveries of Max Planck, Albert Einstein, Niels Bohr, Erwin Schrödinger, Louis de Broglie, Werner Heisenberg, and many others, came an avalanche of seemingly bizarre results that are completely incompatible with the classical description of matter, such as wave-particle duality, which shows that particles exhibit wave behavior, and quantum entanglement which describes instantaneous communication and connection between particles even over vast distances. These results are in no way compatible with classical, Newtonian science that views matter as hard pellets bumping into one another like billiard balls.

“If quantum mechanics hasn't profoundly shocked you, you haven't understood it yet.”

– Niels Bohr

It’s critical to understand that even though quantum mechanics utterly obliterated the idea of matter to such an extent that the definition of matter no longer applied, physicists *continued to use the word “matter*.” They did so even though it had been discovered that matter is nothing like classical matter. “Matter” was completely unknown and inexplicable to them, and one hundred years later, physicists still have no idea what matter is. Despite this, they continue to use the word “matter” to describe it. This is an alarming fact and must not be taken lightly. Because the establishment of empirical science never changed the word “matter,” even though that word no longer applies, people continue to believe we live in a material universe made of hard bits bumping into each other. This is not the case whatsoever, and it’s been known to be false for a century. The public has become brainwashed to believe that we live in a material world despite the fact that the discoveries of quantum mechanics annihilated that possibility.

Because materialists refuse to abandon their faith in matter, the true revolution demanded by quantum mechanics has yet to occur. People like to think that scientific materialists are on the side of truth, but they’re just as locked into their paradigm as religious believers. They’ve simply traded one faith for another. They went from a faith in god to a faith in matter. Materialists won’t admit their worldview is false, because most people are not freethinkers but become brainwashed into an ideology. They’re just as likely to give up their belief in matter as a Christian is to give up their belief in Jesus. A paradigm usually only changes when the generation that believed in it dies and a new generation grows up that can establish a new way of thinking.

“A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up that is familiar with it.”

– Max Planck, Founder of Quantum Theory

Most people completely lack the ability to change their mind when it requires them to alter their worldview. It took the Catholic Church 300 years to admit that Galileo was right even though the evidence was staring them in the face. It's been 100 years for materialists. Let's see how long it takes them. I'm not holding my breath.

I must make it clear that I'm specifically criticizing those who subscribe to a materialist worldview. There are some in the scientific community who are beginning to understand there's more to reality than matter, though they're still empiricists. Many of the founders of quantum mechanics held views completely contrary to mechanistic, materialist science. The issue is that this is never taught in school; thus, everyone becomes brainwashed into an outdated, failed paradigm that's proven to be wrong. Throughout this book, you'll see many quotes by the founders of quantum mechanics that show they had views 100% contrary to materialism. You probably weren't aware of this because it's generally not discussed.

Max Planck, founder of quantum mechanics, Nobel prize winner, and discoverer of energy quantization, *absolutely denied the existence of matter*. He said, "As a man who has devoted his whole life to the most clear headed science, to the study of matter, I can tell you as a result of my research about atoms this much: There is no matter as such." So why is it still called matter, and why do we teach its existence in schools? There is no matter. As we'll show, this is a reality of mind, and so-called "matter" isn't matter at all but rather mind's internal projection. In other words, this is a shared dream.

Max Planck was once asked, "Do you think that consciousness can be explained in terms of matter and its laws?" to which he replied, "No. I regard consciousness as fundamental. I regard matter as derivative from consciousness." Erwin Schrödinger, another founder of quantum mechanics, Nobel Prize winner, and developer of the Schrödinger wave equation, also believed that matter was not fundamental.

“Consciousness cannot be accounted for in physical terms. For consciousness is absolutely fundamental. It cannot be accounted for in terms of anything else.”

– Erwin Schrödinger, Pioneer of Quantum Physics

Max Planck and Erwin Schrödinger regarded consciousness as being fundamental, *not matter*. As we’ll see, it’s not consciousness per se that’s fundamental. It’s mind. The universe is the internal projection of mind. Regardless, Planck’s and Schrödinger’s views are worlds apart from the classical materialist conception of reality. It’s been known for a century that reality is not material as classically described, yet materialists continue to believe in the existence of matter because they’re locked in their paradigm.

“Our science has cut itself off from an adequate understanding of the Subject of Cognizance, of the mind.”

– Erwin Schrödinger

Planck and Schrödinger emphasized consciousness as fundamental because mind and consciousness cannot be explained in terms of materialist science. As we’ll see, it’s mind that’s fundamental (not consciousness), but they were on the right track. Materialism treats matter as fundamental and mind and consciousness as somehow arising from matter, yet they can’t explain how this could be possible. It’s because it’s impossible. Materialism’s inability to account for the aspect of existence with which you are most intimately familiar—your subjective experience—is called **the hard problem of consciousness**. To better understand why this is such a difficult problem, let’s first look at the concept of **substance dualism**.

René Descartes is known as the first modern philosopher and is famous for his work on substance dualism. Almost everyone is familiar with the coordinate system, also known as the Cartesian grid. It's named in honor of Descartes since he developed it in the 17th century, leading to the development of analytic geometry. It's said he came up with the idea of the coordinate system while lying in bed watching a fly crawl across his ceiling. He's also famous for Cartesian Dualism (which is a substance dualism.) Understanding this concept will be crucial as we move forward.

Descartes considered mind and matter to be two separate substances. He defined matter as that which is **extended**. Something that is extended has length, height, or width. Theoretically, you'd be able to measure it with a ruler or some kind of measuring device. Descartes defined mind as that which is **unextended**. Something that is unextended doesn't take up any space, and you can't measure it with a ruler. To get a better idea of what this means, consider your couch. Your couch takes up space, and you can measure it with a ruler. Now consider your mind–not your brain, but your mind itself, the subjective part of you. Does your mind take up space? Can you measure it with a ruler? Of course not.

"Suppose that I am dreaming, and that these particulars – that my eyes are open, that I am moving my head and stretching out my hands – are not true. Perhaps indeed I do not even have such hands or such a body at all."

– René Descartes, Philosopher & Mathematician

According to Descartes, mind and matter are two distinct substances that have nothing in common with each other whatsoever. He believed that the unextended mind linked to the extended material body to navigate the extended world of matter. He considered the domains of mind and matter to be separate from one another.

Dualism is an interesting idea, but it has a major flaw known as the problem of substance dualism or the **interaction problem**. The problem is as follows: if two substances share no common properties of any kind, how can they interact with one another? If two substances can interact with each other, then they must share some common properties. If they share common properties, then they're not actually two distinct substances, and if that's the case we're no longer talking about dualism. This point is important so make sure you take the time to understand it well.

Dualism

Mind and matter are two separate substances that share no common properties. One is not reducible to the other.

Dualism contains the fatal error of the **interaction problem**: Substances that share no common properties cannot interact.

Due to the fatal error of the interaction problem inherent within substance dualism, philosophers began to turn their attention to **monism**. Monism posits there is only one substance, and matter and mind are ultimately reducible to one or the other. Matter is ultimately mind, or mind is ultimately matter. The philosophical position of materialism is a monism that says mind can ultimately be reduced to matter which is the only true substance. Materialism says that matter is real, and mind is an illusion. The philosophical position of idealism is a monism that is the exact opposite. Idealism says that matter can ultimately be reduced to mind which is the only true substance. Idealism says that mind is real, and matter is an illusion.

Monism

There is only one substance. **Materialism** and **idealism** are both monisms.

Materialism: There is only matter. Mind is an illusion that is reducible to matter.

Idealism: There is only mind. Matter is an illusion that is reducible to mind.

Materialism says that mind comes from matter, and idealism says that matter comes from mind. The problem for materialism is that it's impossible to explain how mind could come from matter. Hyperianism is an idealism, and ontological mathematics clearly shows how so-called “matter” can come from mind. You can easily experience how it's possible for mind to generate the illusion of matter every time you go to sleep and enter a dreamworld. Experience aside, in this book we will demonstrate with mathematical precision how mind gives rise to so-called “matter.”

Materialism has an insurmountable problem on its hands. *Matter has never been observed as existing apart from mind*. Think about that fact carefully. Every observation of matter has always been mediated by mind. That is, matter has only ever been encountered through the subjective lens of a mind. Never has it been observed or shown to exist on its own, apart from mind, *ever.* That's because it's an impossibility. Matter can't exist apart from mind. Ultimately, matter is not fundamental at all, exactly as quantum mechanics shows and many of its founders believed.

In its attempt to reduce everything in existence to matter and nothing else, materialism runs into the hard problem of consciousness. The hard problem of consciousness is materialism's inability to explain how and why entities exhibit subjective experience. It can't explain how you're able to have the subjective experience of pleasure, pain, heat, cold, happiness, hope, love, or any element of your subjectivity. It can explain the mechanics of physical processes, dealing with nerves, neurons firing etc. but it can't explain how it's possible for you to have internal experiences. It can’t explain why you have any subjectivity at all.

“I am very astonished that the scientific picture of the real world around me is deficient… it is ghastly silent about all and sundry that is really near to our heart, that really matters to us. It cannot tell us a word about red and blue, bitter and sweet, physical pain and physical delight…”

– Erwin Schrödinger

If materialism were true, existence might as well be a mindless machine and “entities” would be nothing more than something similar to a complex artificial intelligence system following out a program with no internal aspect whatsoever. In such a scheme, subjectivity is superfluous.

Consider a rock. According to materialism, a rock is an inert object with no internal or mental aspect of any kind. A rock is made up of atoms which, according to materialism, are also objects that contain no internal aspect. Materialism posits that these mindless atoms, with no mental aspect, eventually become arranged in a certain way such that mind emerges. No explanation or mechanism is ever given for how this emergence is possible. It’s just taken as fact. This blind belief is another common feature of a faith-based position. If matter has no mentality or internal quality whatsoever, as materialism claims, then it’s a logical impossibility for it to give rise to something that does have mentality. This would be the equivalent of a miracle.

Materialists often point to the fact that brain states correspond to mental states as evidence that matter gives rise to mind. To make such a claim based on correlation is a blatant logical fallacy. If I’m playing a video game, my game character’s state will correspond to my mental state since I’m the one controlling the character. But it doesn't mean the game character caused my mental states. Correspondence does not imply causation.

"[The] brain does not produce the mind… to assume that the mind is caused by the brain, due to psycho-physical correlation, is to commit the fallacy cum hoc ergo propter hoc."

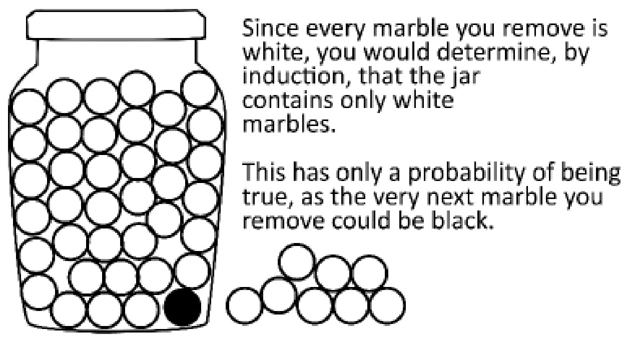
– Peter Sjöstedt-H, Philosopher of Mind

If atoms are devoid of an internal aspect, it's formally impossible to arrange them in any way to produce an internal aspect. It's insane that many people still adhere to the idea of "matter" as classically described even though it's been absolutely refuted. Materialists are as bad as flat earthers. They're stuck in an old mindset, and they either ignore the facts or aren't rational enough to understand them.

1.5 - Induction vs Deduction

Our goal is to elucidate the differences between the old paradigm of scientific empiricism and materialism and the new paradigm of scientific rationalism and idealism, which is ontological mathematics. Scientific empiricism uses a method of investigation called **induction**, while ontological mathematics uses a method called **deduction**. Induction is a method based on repeated tests and has only a probability of being true. The inductive method can never deliver 100% certainty. It only delivers probability. Deduction is a method that delivers a conclusion that's guaranteed to be 100% true no matter what.

Let's imagine you have an enormous, opaque jar of marbles and want to examine its contents using the inductive method. You reach in, pull out a marble, and examine it. Let's say it's white. You reach in again and pull out another marble. Let's say it's white again. Now imagine that you do this repeatedly, one thousand times, and each time you pull out a white marble. By induction, you would conclude that the jar contains only white marbles. However, the very next marble you pull out may turn out to be black, proving your conclusion false. The more white marbles you pull out, the greater the probability your conclusion that the jar contains only white marbles is true, but no matter how high the probability may be, the possibility exists that the very next marble you pull out might be black, rendering your conclusion false.



(Fig. 1.51) An example of induction.

Taking this example to the extreme, the jar may not exist at all. It could be a hallucination. You could be having a personal dream, or you could be trapped in a simulation. The old paradigm of scientific empiricism is an inductive system, and all their conclusions inherit these problems. It can't say anything with certainty about the ultimate nature of reality. It never can. *That should be a huge red flag that this is a false paradigm*. It can never provide certainty about the nature of reality. Ever. That alarming fact is built into its methods. If you want certain truth about the nature of existence, scientific empiricism must be replaced by scientific rationalism.

Here's another example of induction: Let's say you are in a great mansion with many rooms. Suppose that you're in a room at the center of the mansion, and the light is on in that room. Now, unbeknownst to you, all the lights in the mansion are off except for the room you're currently in. The mansion is also set up so that when you open a door to a new room, the lights in that room immediately come on, and when you shut the door behind you, the lights in the previous room turn off. As you explore the mansion, going from room to room, it seems like the lights are on in every room. After exploring hundreds of rooms, you may conclude, by induction, that every room in the mansion has a light on, when in truth, it's only the room that you're currently in that's lit, while the rest of the mansion is in darkness. Taking it to the extreme once again, it could also all be a hallucination or a simulation. Induction is a useful tool, but its truth value is never guaranteed.

On the other hand, the conclusions of deductive reasoning are guaranteed to be 100% necessarily true, provided that the premises are true, of course.

> **Example of a Deductive Argument**
>
> 1. All **humans** are **mortal**
> 2. **Socrates** is a **human**
> 3. Therefore, **Socrates** is **mortal**
>
> If it's true that all **humans** are **mortal**, and if it's true that **Socrates** is a **human**, then it's 100% certain that **Socrates** is **mortal**. It must be true. It's impossible for it to be false. Induction is about exploring the so-called "physical" world, while deduction has to do with the rules of thought itself. In our example of deduction, we can remove the content completely and break it down into pure form:
>
> 1. All **A** are **B**
> 2. **C** is **A**
> 3. Therefore, **C** is **B**
>
> If it's true that all **A** are **B**, and if it's true that **C** is **A**, then it must be the case that **C** is **B**. Notice that we aren't doing any kind of examination of the physical world when we're using pure form. We aren't pulling out marbles or exploring rooms in a mansion. At this point, we aren't even talking about Socrates, humans, or mortals. We're in the realm of pure thought and are examining how it's structured.

Mathematics is a deductive system. You can know that 2 + 2 = 4 and be 100% certain about this due to the rules of thought itself and the definitions of the system. It's analytically true. Given the definitions of the system, it's impossible for it not to be true. It's pure reason.

Induction	Deduction
Reaches conclusions through repeated tests and uncertain inferences. Conclusions reached are never certain, only probable.	Reaches conclusions through pure reason by examining the structure of thought. Conclusions reached are 100% certain, inevitable, inescapable.

Induction can provide helpful and practical information, but it can never deliver the truth about the foundation of existence. The ultimate truth of existence can be found only through deduction. Only deduction delivers 100% certainty. Scientific materialism and empiricism is not a deductive system. It's inductive. The old paradigm is inherently limited and can never deliver ultimate truth. There's an impenetrable barrier it can never cross because truth lies outside its methodological bounds.

To find ultimate truth, the trick is to find the deductive system and definitions that correspond to reality itself. If you can find the deductive system that corresponds to reality, you can use deduction to unlock 100% certainty and knowledge of existence. Ontological mathematics is that system. Ontological mathematics isn't based on any human-invented logical system or even abstract mathematics. It's the living mathematical system of existence. It's something that's never been seen before. It treats mathematics as a living, striving substance. Ontological mathematics is the ultimate deductive system that, when used correctly, can deliver 100% certain conclusions about reality.

1.6 - A Posteriori vs A Priori

This brings us to another important distinction: **a posteriori** and **a priori** knowledge. A posteriori knowledge is dependent on experience. A priori knowledge is independent of experience and concerns pure rational thought. Consider the statement “it’s raining outside.” This is an a posteriori statement because it’s dependent on experience. You must look outside and check to see if it’s raining to know if the statement is true. A posteriori is all about empiricism and experience.

The prime example of a priori knowledge is mathematics. Consider the statement “2 + 2 = 4.” You can verify this statement independent of experience, and thus it’s a priori knowledge. You don’t need to explore an aspect of the world to see if it’s true. You don’t need any of your physical senses at all. All you need to do is think about the concepts involved.

A posteriori facts are not universally true. For example, the truth of the statement “it’s raining outside” is contingent on time and space. Even if it’s raining now in Los Angeles, it may not be raining in two hours, and it won’t be raining now on Mars. This is a very “fuzzy” kind of “truth” that’s constantly changing from one moment and place to the next.

In contrast, a priori truths are universally true. The rational truth that 2 + 2 = 4 is true now, yesterday, tomorrow, and for all time. It’s true here, in Africa, on Mars, and across the universe. Its truth is independent of time and space. It’s pure reason.

A Posteriori Fact	A Priori Truth
Dependent on experience. Facts are arrived at through experience and the senses. Not universally true. Dependent on space and time.	Independent of experience. Truth is arrived at through knowledge that exists in the mind. Universally and eternally true. Independent from space and time.

Scientific empiricism deals in a posterori facts dependent on space and time. A system predicated on a posteriori fact can never deliver ultimate truth because ultimate truth is universal, eternal, and independent of space and time. Once again, we see the inherent limitations built into the old paradigm of empirical science, forever barring it from ultimate truth.

Ontological mathematics has no such restriction. It's a rationalist, deductive, idealist system that deals in a priori, universal truth independent of space and time, and is thus able to deliver the foundational answers to existence.

"I don't believe in empirical science. I only believe in a priori truth."

– Kurt Gödel, Logician and Mathematician

1.7 - Necessary vs Contingent

We must examine one final distinction to understand what makes the new paradigm unique: necessary vs contingent.

A **necessary** truth is something that *must* be true. It cannot be false. It's impossible to imagine a scenario in which it isn't true. The denial of a necessary truth leads to a logical contradiction.

> **Examples of Necessary Truths**
>
> - Red roses are red
> - Squares have four sides
> - 2 + 2 = 4
>
> We can confirm these are necessary truths by denying them to see if it leads to a contradiction:
>
> - Red roses are **not** red – **Contradiction**
> - Squares do **not** have four sides – **Contradiction**
> - $2 + 2 \neq 4$ – **Contradiction**

> Each of these statements is a logical contradiction. If a red rose is not red, then it's not a red rose. By its very definition, a red rose is red. If it were blue, then it would be a blue rose. If a square did not have four sides, it wouldn't be a square. Having four sides is a necessary property of a square by its definition. If it had three sides, it would be a triangle. It's impossible to imagine any kind of scenario or world where necessary truths are false. They must be true.

At first glance, necessary truths may seem trivial. Does the truth that red roses are red really tell us anything useful? But necessary truths define the heart of existence as *the laws of mathematics are all necessary truths.*

A **contingent** "truth" ("fact" is more apt than "truth") is a fact about the world that could be false. Its denial doesn't lead to a contradiction. It's possible to imagine a world or scenario where it's not true.

> **Examples of Contingent Facts**
>
> - America was founded in 1776
> - It's raining outside
> - Earth's orbital speed is roughly 30 km/s
>
> The denial of a contingent fact does not lead to a logical contradiction, and we can imagine a scenario or universe in which they aren't true.
>
> - America was **not** founded in 1776
> - It's **not** raining outside
> - Earth's orbital speed is **not** roughly 30 km/s

You may be thinking, "but America *was* founded in 1776!" That doesn't matter. The point is that it isn't a *logical* contradiction to say that it wasn't. It doesn't "break the rules of thought" so to speak. It's possible to imagine a universe in which America was founded in 1652, or 2500, or was never founded at all. You can imagine a universe in which Earth orbits at some other speed, or where Earth doesn't exist at all. It just so happens that these facts are true. They could have been false. They don't *have* to be true.

While contingent facts can be false, necessary truths must be true. There is no possible universe or scenario where a square has anything other than four sides or 2 + 2 equals anything other than 4. It's impossible for there to be a state of affairs where a square has three sides. Then it wouldn't be a square. It would be a triangle. We're examining the structure, definitions, and rules of the concepts involved.

Using human language may cause some confusion. We used the example "red roses are red" because that's something you can easily imagine and relate to. But to really understand the concept of necessary truth, it would be better to consider the form of a statement. For example, "X is X." This is a necessary truth. It must be true. There is no possible universe in which “X is **not** X” is true. The statement “X is **not** X” is a logical contradiction and cannot be true. “A rose is a rose” is true. “A rose is **not** a rose” can never be true. It’s critical to note that the laws of mathematics are necessary truths.

If you want to find absolute truth, you can find them only in the necessary truths of existence. Necessary truths are eternal and can never be falsified. Contingent facts are found through induction and can all be false. They are not absolute truths. They are facts about the world that just so happen to have come out in the way they did. They reflect our limited knowledge about the “physical” world and are constantly changing. Contingent facts can be falsified at any time. The contingent fact “All swans are white” was considered true for a long time, until it was falsified when black swans were discovered. The contingent fact of classical “matter” was falsified by the discoveries of quantum mechanics.

Scientific empiricists deal in contingent facts. If an empirical contingent fact conflicts with a necessary, eternal truth, they will side with the empirical contingent fact. Do you see how backwards and irrational this is? A paradigm shift must occur for there to be progress. The old paradigm is broken and is literally blocking the way to truth. It’s crazy to place contingent facts over necessary truths. It’s logically incoherent. This is just as irrational as the religious who put faith in scripture over evidence.

Ontological mathematics deals in necessary truths. Necessary truths are discovered through deduction. Ontological mathematics deduces the foundational, necessary truth of existence and builds everything else out of the necessary consequences of that foundation. This is how we can know the truth of reality with certainty. We have discovered the eternal, necessary truths that correspond to the ground of existence, and thus it must be true.

Considering all the concepts we’ve learned, notice the major differences between the paradigms. The old paradigm deals with matter, the physical, experience, probabilities, contingency, and dependence on space and time. The new paradigm deals with the immaterial, the mental, pure thought, mathematics, the necessary, certainty, and independence from space and time.

With this is mind, let's look at the differences between empiricism and rationalism.

Empiricism	Rationalism
Knowledge is derived from experience/sensory data. Sensory experiments are used to obtain data. Experimentation is an inductive method that can never produce certain knowledge, only probability.	Knowledge is derived from reason and logic, not faith or the senses. Mathematics, reason, and logic are used to obtain universal knowledge. Absolute knowledge is certain because it is deductive, necessary, and rational, not empirical.

Let's put everything together and compare the two paradigms.

Scientific Materialism	Ontological Mathematics
Empirical (Sense Data)	Rational (Reason & Mathematics)
Materialist (Matter)	Idealist (Mind)
Inductive (Probability)	Deductive (Certainty)
A posteriori (Experiential Facts)	A priori (Universal Knowledge)
Contingent (Can be Falsified)	Necessary (Eternally True)

The paradigms are polar opposites. An assumption often made by those new to ontological mathematics is that we're making the claim that scientific empiricism doesn't use deduction, reason, or mathematics, and that we reject experimentation, sense data, and induction. This couldn't be further from the truth and reflects a poor understanding of philosophy on the part of those making such an assumption. Obviously scientific empiricism uses mathematics and deductive reasoning, and obviously we value and use the power of experimentation and induction. However, the distinction between the old and the new paradigm concerns what leads to ultimate, **epistemic** truth, as well as what's given priority.

Scientific empiricism gives priority to contingent experiments and the senses, while the scientific rationalism of ontological mathematics gives priority to the mind and the necessary truths of reason and mathematics. We give priority to those faculties and methods that lead to universal, necessary, certain truth. Scientific empiricism gives priority to the flawed senses, and methods that only lead to probability.

The scientific method is incredibly useful and can lead to many practical advancements in areas such as technology and medicine. It's not that we reject empirical methods, such as sensory experiments. Rather, we absorb them as a subset of our system as useful tools for exploring the so-called "physical" world. But they always have a lower position subservient to reason. While both sense data and reason are important, it's insane to give priority to data that is inherently probabilistic, contingent, and uncertain rather than to truth that is necessary, universal, and certain. Certain knowledge must rule over flawed, probabilistic, sensory data.

Scientific empiricism has everything turned on its head, just as the Catholic Church privileged faith and revelation over sensory data. It's the same mistake at a different level. The religious use sensory data and inductive methods (granted, poorly performed induction) from time to time in an attempt to support their distorted views but ultimately reject anything that contradicts their holy book. Likewise, scientific empiricists use a version of deduction and reason but ultimately reject anything that contradicts their faith in matter and the senses.

As you can see, understanding ontological mathematics requires a new way of thinking. If you try to examine ontological mathematics from the perspective of an old paradigm, or without fully understanding the concepts involved, it won't make any sense. It would be like a religious person trying to understand science by praying and reading the bible. To understand ontological mathematics, you must step outside of the materialist, empirical mindset and into the idealist, rationalist, logical mindset.

Now that we understand what makes the new paradigm unique, let's discover its most important principle. One way to discover this principle, is through the Method of Doubt.

1.8 - The Method of Doubt

"If you would be a real seeker after truth, it is necessary that at least once in your life you doubt, as far as possible, all things."

– Rene Descartes

Descartes realized that he was filled with false beliefs and conditioning. To remove these false beliefs, he employed his Method of Doubt. Descartes doubted absolutely every belief he held that could possibly be doubted. He reasoned that if he doubted everything that was possible to doubt, he would encounter something that couldn't be doubted and at least he could be certain of that indubitable truth. He used a variety of thought experiments to aid in his doubting process. He imagined that the entire world was a dream and that an evil god was using its power to constantly deceive him. This allowed him to doubt his senses, the existence of the physical world, his body, and almost everything. However, there was one thing he could not doubt. His famous conclusion "Cogito ergo sum" is Latin for "I think, therefore I am." However, a fuller version is "dubito, ergo cogito, ergo sum" which translates to "I doubt, therefore I think, therefore I am."

Descartes reasoned that everything could be false, including his body and the world around him. He doubted everything that could be doubted. But the one thing he could not doubt was that in order to be able to do this doubting, he had to exist in some way. Descartes said, “we cannot doubt of our existence while we doubt…” In other words, if you’re able to doubt, you must exist in some way or else you couldn’t be doubting at all.

To doubt requires existence. In order to doubt, you must think, and to be able to think, you must exist. Keep in mind, no claim is being made here about what “I” or “existence” might mean. At this point, “I” doesn’t mean physical or having a body or having an ego, and “existence” doesn’t mean material. All his conclusion means is that “I” (whatever that may turn out to be), in order to think, must “exist” in some way (whatever that “existence” may turn out to be). No other claims are being made here.

Descartes was correct, but this is not what’s important for our purposes. There is something more fundamental and “doubt poof” than his conclusion. Look at it once more: “I doubt, therefore I think, therefore I am.” You’ll notice that this is a form of deductive reasoning:

> **The Structure of the Cogito**
>
> 1. If I am **doubting**, then I am **thinking**
> 2. If I am **thinking**, then I **exist**
> 3. I am **doubting**, therefore, I **exist**
>
> What’s more important than Descartes’ conclusion is the rational structure of his thought process. We can break it down into pure form so that the concepts don’t get in the way:
>
> 1. If **X**, then **Y**
> 2. If **Y**, then **Z**
> 3. **X**, therefore, **Z**

This shows that reason itself is fundamental. It’s reason that can’t be doubted. The a priori, deductive structure of reason is what makes

What the Method of Doubt really leads us to is reason itself. Rational structure is the framework, the skeleton, on which absolutely everything in existence hangs and depends. This is a critical point for understanding hyperrationalism and ontological mathematics.

His conclusion, and the steps taken to get there, are possible. If his structure of thought wasn't there, he wouldn't have been able to carry through processes of *any kind*. It's rational structure itself that cannot be doubted and is fundamental.

1.9 - The Principle of Sufficient Reason

The Method of Doubt leads us to the absolute foundation of ontological mathematics: the **Principle of Sufficient Reason**. This is the one master axiom of the entire system. The Principle of Sufficient Reason (which we refer to as the **PSR**) has been defined by many different philosophers in different ways. We define it in a very specific way, which is "For everything that exists, there exists a sufficient reason for why it is thus and not otherwise." This means that for everything that exists, there must be a sufficient reason for why it exists in the way that it does rather than some other way. In other words, there must be a rational explanation for why something is the way that it is.

When people use "rational explanation" in everyday conversation, they usually mean a common sense or a material explanation. *This is not what the PSR means.* It doesn't mean a common sense, physical, material, or mechanistic explanation (ontological mathematics shows that existence is a shared dream, after all.) It simply means that random miracles and anything that conflicts with reason are not allowed. 2 + 2 cannot equal 5. A blue basketball can't randomly appear in your living room for no reason at all. Everything must have a reason for its existence even if that reason is incredibly complex or we don't yet know what that reason is. For everything that exists, there must be some explanation for why it exists. Never mistakenly think the PSR implies that reality is dead, mechanical, or material. The PSR is rational structure itself and is what allows for ordered thought processes, such as our example in the previous section, to occur.

The PSR is often attributed to the 17th century philosopher Gottfried Leibniz. Leibniz was rational idealist and famous for many things, including the invention of calculus. He believed the world wasn't formed from material atoms at all, but from mind. This monumental genius was able to see the new paradigm from the 17th century. Existence is a shared dream, and you are an eternal being of mind.

The Principle of Sufficient Reason

For everything that exists, there must exist a sufficient reason for why it is the way it is, rather than a different way. If there is no sufficient reason to prevent something from existing, then it must exist.

The Principle of Sufficient Reason is self-affirming. This means that if someone asks the question, "What is the <u>reason</u> for the Principle of Sufficient Reason?" they have implicitly affirmed it since their question presupposes that things must have a <u>reason</u> for their existence (since they're asking for a reason), which is exactly what the PSR states. In other words, to ask for a reason for the existence of something, is to affirm that things need a reason to exist, which is exactly what the PSR states. To sum up, if you ask for the reason for the PSR, you're using the PSR to ask that question, which implicitly affirms it.

The PSR may sound complicated, but it can't get any simpler. It just means that all things must have a reason for their existence. If you see a rock soaring through the air, there must be a reason for why it's in that state. Maybe someone threw it, or a bird dropped it, or someone built an anti-gravity device, or it's a holographic illusion, or any number of reasons, but there must be some reason. It didn't just begin flying randomly for no reason at all. That's an impossibility.

The PSR is the foundation on which the entire edifice of ontological mathematics is built. You can use the PSR to test everything around you and to know when someone or something is deceiving you. Even if "god" appeared before you, you could question it, and if it contradicted reason, you would know that it was lying. If your senses show you something that is contrary to reason, then you can know they're deceiving you, that you've encountered some kind of an illusion.

If empirical data gives you an irrational result, then you know there is something wrong with the data or that it's incomplete. Reason allows you to cut through all the bullshit and understand existence with complete certainty. Reason is beyond subjectivism, emotional interpretation, and the senses. The rational statement 2 + 2 = 4 will be true for all time, and it doesn't matter what you feel about it or how much you want it to equal something else. It's definitionally true. It won't change by praying to it, and it's true no matter what your senses tell you. Reason is a faculty of the mind. To be more accurate, *it's thought itself*. This is a mental reality, not a reality of the senses. Pay close attention. A new dawn is rising.

It must be made very clear that if anyone disagrees with the PSR and can't be convinced of its validity, then they'll never understand ontological mathematics, and there's no point in trying to explain it to them. This is because the PSR simply states that everything must have a sufficient reason for its existence. If you deny that things have a *reason*, then there is no *reason* that could possibly be given to you to explain ontological mathematics. If you deny reason itself, no reason will suffice. So before continuing, it's crucial that you understand the PSR.

1.10 - Ontology: That Which is Real

What is the PSR exactly? Ontological mathematics is, of course, an ontological system. **Ontology** is the study of that which is real. It's the study of what actually exists. From the point of view of materialism, a materialist would say that mind is not ontologically real, since in materialism only matter is real and mind is reducible to matter.

From the point of view of idealism, we would say that matter is not ontologically real, since in idealism only mind is real and “matter” is reducible to mind. Ontology is the study of what is real. It asks the question, “What are things ultimately made of, regardless of how they may appear to us?” Ontological mathematics is, then, the study of *real* mathematics. It’s the study of the *living* mathematics of existence. It’s the mathematical study of living mind.

Ontology

Ontology is the study of that which is real. It’s the study of what has actual existence removed from what things may appear to be.

What is the PSR ontologically? Currently, we’ve defined the PSR in terms of human language. But it’s not made of human language. That’s simply the way we communicate the idea. Understanding what the PSR *truly* is impossible for most people. But if you can grasp what it is, the rest of ontological mathematics will come easy. If it’s difficult at first, don’t worry. It can take years to grasp. And in fact, every one of us is on a path of understanding. Its ontology is immensely strange and wondrous.

As we’ll show, the PSR = mind = thought = 0 = nothingness = a very specific mathematical equation. These are all different perspectives and ways of thinking about the same concept. You can choose any one of these concepts as being the ground of Hyperianism and ontological mathematics since they’re all the same concept. As we explore ontological mathematics, this will become clear.

Having established the necessary philosophical foundations, we can now begin to explore the system of ontological mathematics itself.

SECTION 2: ONTOLOGICAL MATHEMATICS

Where do we begin our journey? Ontological mathematics requires such a radically different way of thinking that it's difficult to approach it in a linear fashion. The process of learning ontological mathematics is like taking a journey around a circle. Concepts that will be learned and understood later are needed to fully understand the concepts from where you began. On a circle, there's no beginning. You just start somewhere and keep moving around until you wrap back around to where you started. While exploring ontological mathematics, you'll understand concepts you learned at the beginning in a new way after having discovered other elements along on your journey.

We'll present it in a linear fashion, but keep in mind that once you have finished, you should read it from the beginning to see it in a new light.

2.1 - BEGINNING WITH ZERO

Let's start from the beginning. But what is the beginning? Where does one start when considering existence?

Mainstream western religion begins with a creator god. Their definition of god has many specific attributes. It's a conscious, allegedly loving god that has a plan for you and your life. But those religions can't give a sufficient reason for how this god can possibly exist. Materialist science begins with the Big Bang, but it can't explain what preceded the Big Bang or give a sufficient reason for why it occurred.

Ontological mathematics (and thus, Hyperianism) begins with nothingness, which is the only rational place you can begin. If your explanation of existence doesn't start with nothing as its foundation and rationally explain how everything came about from that ground, then your explanation for existence is wrong, since it started from somewhere other than the beginning. As we continue, remember to use the mindset of the new paradigm. We're laying out a priori, deductive, necessary truths of reason. Let's delve into the void and explore the concept of nothingness.

Nothingness is eternal. Nothingness is uncaused.[1] Nothingness is uncreated. These are essential to what nothingness is. It's entailed in its definition. A creator can't create nothing, because then it wouldn't be nothing anymore. How could you create nothing? Nothing already is. It's not something you can make. Nothingness is eternal because it can't begin, and it can't end.

The "existence" of nothing is perfectly valid by the PSR. The reason for the existence of nothing is that its existence is mandatory. It's necessary. It's an impossibility to stop it from coming into existence because there's nothing to stop. There is no sufficient reason to prevent its existence, and therefore it must exist. If this seems confusing, as a thought experiment, spend ten minutes thinking about how you could stop "nothing" from existing. You'll quickly discover it's a logical impossibility. Since it can't be stopped from existing, it must exist.

We're going to make a logical move, to make this easier to discuss. The concept of nothingness is equivalent to the concept of zero:

$$\text{Nothingness} = 0$$

For the same reason that nothing can exist, by the PSR, zero can also exist because nothingness and zero are the same concept. Since zero and nothingness are equivalent, zero is also eternal, uncaused, and uncreated. Zero is the absolute **ground** of existence.

> **Grounding Example**
>
> Grounding is an important philosophical concept. If "**A**" depends on "**B**" for its existence, then "**A**" is said to be grounded in "**B**."

For example, we could say that molecules are grounded in atoms, and atoms are grounded in electrons, protons, and neutrons. Molecules depend on atoms for their existence, and atoms depend on electrons, protons, and neutrons for their existence.

Zero is the absolute ground of existence and is self-grounding. Zero is its own ground. It doesn't depend on anything outside of itself for its existence, and everything else in existence depends on it. The ground of existence can't depend on anything else for existence other than itself, or else it wouldn't be the ultimate ground. Think of the ground as the foundation upon which you begin to build a house.

Another way to understand this is through the concept of **aseity**. Aseity means that something contains its own cause, or reason for being, within itself. You can think of zero as being uncaused, having aseity, or being the self-grounding ground of existence. They all amount to the same thing.

Anyone who asks what external entity or force caused zero, hasn't understood the concept of zero. The answer to existence must contain its own reason for being, its own cause, within itself. If it didn't, then it wouldn't be the answer to existence because it would depend on something more fundamental for its existence. But then, what would be the cause of that more fundamental thing? If you give it just a moment's thought, you'll realize that the answer to existence must contain the reason for its own being within itself or you'll run into infinite regress and never reach an ultimate ground. You'll have a never-ending chain as each entity depends on another for existence. Things either have their reason for being in something else or within themselves. Molecules have their reason for being within atoms, and atoms have it within electrons, protons, and neutrons. The foundation of reality must have its reason for being within itself or else it wouldn't be the foundation.

> **Infinite Regress vs Grounding in Zero**
>
> $\ldots \leftarrow D \leftarrow C \leftarrow B \leftarrow A$ Fig. 1
>
> $0 \leftarrow C \leftarrow B \leftarrow A$ Fig. 2
>
> Fig. 1 shows a process of infinite regress. No ground can ever be reached because every element depends on something else for its

existence.

Fig. 2 shows the ultimate ground on which every element depends. The ultimate ground must depend only on itself for existence or else it wouldn't be the ground.

Zero is the ultimate self-grounding ground of existence. What external thing could nothing possibly depend on for its existence? Zero contains its own reason for being within itself. That reason for being is that zero, nothingness, must exist because nothing can prevent it from existing since it's perfectly nothing. Another way to say this is that existence is its essence. Its existence is necessary and mandatory by its very nature. Zero's nature demands that it exist. It must exist.

We've equated nothingness with zero and have shown that it contains its reason for being within itself. Now let's make another logical move. We'll equate the concept of zero to the mathematical point. This allows us to imagine nothingness as an infinitely small point, which is simply to help us conceptualize what's coming up next. We haven't changed anything. A mathematical point has no size. It's dimensionless. It's nothingness. It's zero. Putting nothingness in terms of a dimensionless point will be helpful in a moment.

We now have nothingness as zero, as a dimensionless, mathematical point. Is it possible that this zero point could have a structure? If we metaphorically "cracked it open," would we find anything? At first, it may seem like the answer would be no, but if the PSR allows for some kind of structure or arrangement, then it must have one. Let's see.

Mathematically, zero can be set to equal one minus one:

$$0 = 1 - 1$$

But what could this mean ontologically? Remember, ontology is the study of what's real, of what has actual existence. We can write down $0 = 1 - 1$ on paper and know that it's a valid equation according to the rules of abstract mathematics, but if we want to say this equation is true of existence itself, then it can't be treated as just an abstract grouping of symbols on paper. The symbols are just a way for us to represent, signify, stand in for, and communicate a pattern that has existence.

In other words, we want to ask, “Does this equation apply to existence, and if it does, what does it mean?” We want to discover if the equation represents a pattern that has existence, and if it does, determine what each part of the equation means.

For the purpose of understanding this concept better, let’s assume the equation 0 = 1 – 1 does apply to existence. At the moment, this might seem meaningless. But let’s look at it ontologically instead of as an abstract group of symbols.

We can rewrite the equation as:

$$0 = 1 + (-1)$$

This is the same equation, but we’ve rewritten it to get a better understanding of what it’s saying. It says that zero is equal to positive one plus negative one. What this means is that zero is equal to the union of a positive value and a negative value.

Don’t think of this equation as meaning that +1 “annihilates” –1 to give the result of 0. Instead, imagine two oppositely related values or “patterns” existing together to produce the total result of zero.

This can be difficult to conceptualize, which is why we’re doing a lot of thought experiments. As a crude example, imagine that zero was a box that you “opened up.” It wouldn’t be empty. Inside you would find a positive “something” and a negative “something” that, when considered together as a unity, results in perfect zero. Even though the box contained two somethings, a positive something and a negative something, the box would total to zero.

For the purpose of exploring this example, we assumed this equation applied to existence. But does it?

If this is true of existence, then:

$$\text{Nothingness} = 0 = 1 + (-1)$$

What does this mean ontologically? If this is true ontologically, it would mean that if we were to examine the structure of nothing, we would find two oppositely related values: a positive value and a negative value that, when considered together, balances to zero. The concept of nothing entails the union of opposites.

But is the specific example of one plus negative one possible? Is that what you would find if you "opened up" zero? The answer is no, because this specific arrangement violates the PSR. There is no sufficient reason for it to be one plus negative one exclusively. There are an infinite number of other values and arrangements you could use to equal zero. Picking one and negative one specifically is random and arbitrary. There's no reason for it. But this example equation has shown us that zero can contain a union of opposites of some kind.

To discover the structure of zero, we must satisfy two conditions:

1. We must find a union of opposites where the total result is zero.
2. That union of opposites must be valid by the PSR.

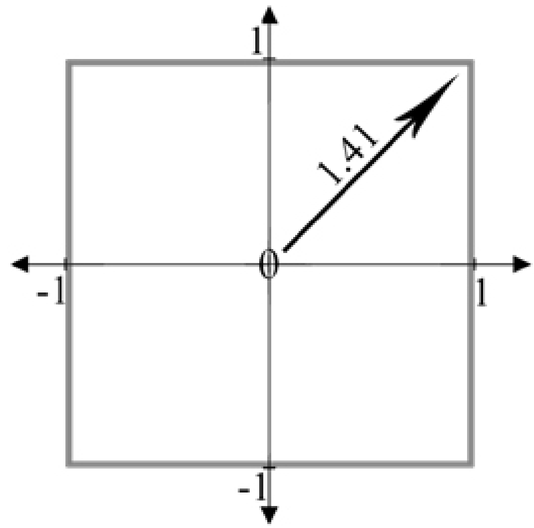
Zero allows for a union of opposites. It's entailed within its definition. As soon as you have the concept of zero, you automatically have the concept of opposites that unify and balance to nothing. However, if we want to find a pattern that applies to existence instead of a meaningless manipulation of symbols, then it must be valid by the PSR. As we'll see, only a very specific union and pattern is possible by the PSR.

It's important to remind yourself at this point that this is all taking place within the realm of thought. We're simply exploring thought patterns, structures allowed by the PSR. This is a priori, deductive reasoning. So far, we have the thought of nothingness which is a zero point that may contain within itself a union of oppositely related values if and only if that union doesn't violate the PSR. The next step is to discover the exact structure of that union.

Imagine there are values packed within the zero point and we want to "peer inside of it" to see how those values are arranged. Since we're examining the structure of thought, by arrangement we don't mean a location in space. There is no space yet. All we're exploring is how these concepts of thought are arranged in relation to each other.

For example, if you have the value +1 then you must have the value –1. Why? One implies the other. If –1 doesn't exist, then +1 has no meaning. A positive value has definition only in relation to a negative value. In a world with no negative values, "positive" has absolutely no meaning. A positive has meaning only when there is a negative. You can't have positive without negative and vice versa, just as you can't have up without down, left without right, or hot without cold. As soon as one of these concepts exists, it *necessarily* implies the existence of its opposite. Try to imagine up without down. It's an impossibility.

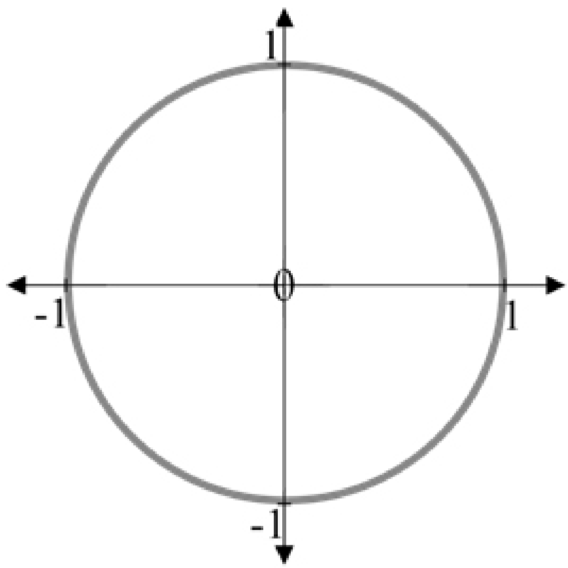
In our mission to find the structure of zero and how its values relate to each other, let's consider arranging values in the pattern of a square. We'll arrange them in such a way that, if we metaphorically placed them on a mathematical grid, the center of the square would land on zero. If you added together every value of the points that formed this square, you would get zero. This pattern results in zero, so it's equal to zero. This structure is equal to nothingness. So far, it seems like a good candidate for our arrangement, but does it violate the PSR?



(Fig. 2.11) A square violates the PSR. There is no sufficient reason for the unequal treatment of points or random turns at the corners.

Indeed, it does violate the PSR. The PSR demands that every value be treated with exact equality and uniformity. You can see that the points located on the corners are much further from the center than the point at 1. There is no sufficient reason for some points to be further apart than others,[2] and there's no sufficient reason for there to be sudden, sharp turns at the corners. This randomness is a violation. If we were to examine every shape, we would find that all of them violate the PSR in some way except for one.

Let's consider arranging values in the pattern of a circle. If you added together every value that formed this circle, you would get zero. With a circle, each value is perfectly balanced in relation to its opposite and treated with totally equality and uniformity. Remember why King Arthur chose a round table? He wanted all the knights to have an equal position. In a circle centered on a cartesian grid, every value is located at an equal distance from the origin, and there are no sharp turns, corners, or deviations. Every other shape results in some kind of randomness, deviation, inconsistency, or incompleteness. A circle is valid by the PSR.



(Fig. 2.12) When points are arranged in this circular pattern, the total result of their values is zero.

A circular pattern is the perfect structure for our union of opposites. Imagine that inside of our point of nothingness is the mathematical coding for a circle whose total adds to perfect zero. Imagine this "circular coding" is inside an infinitely small point. A circular union is what you would metaphorically see if you opened our point. But remember, there really wouldn't be anything to see. We're just exploring structures of thought possible by the PSR. The total value of the circle is zero. It *is* zero. It's nothingness.

Any point on our circle, considered on its own, has some value, either positive or negative. But when the values of every point are considered together, it equals zero. When every component of something adds to zero, we say it **nets to zero**.

The Zero Condition: Reality is Zero

We've seen that zero is the ground of existence. A consequence of this is that existence as a whole must net to zero at all times, even when space and time arise. Existence itself is always zero. This is why existence is possible. It's eternally and perfectly nothing. If reality didn't net to zero, it would mean that somewhere there would be an imbalance. There would be a positive without a negative; this would be like having an up without a down, which is an impossibility. The PSR demands that existence be equal to zero because zero, as a union of opposites, is all that can possibly exist.

Sections of reality, when considered independently from the rest of existence, may not net to zero, just as any point on a unit circle has some value when considered on its own, apart from the rest of the values on the circle but:

Reality, considered as a whole, must net to zero at all times.

This is the zero condition which can never be violated.

2.2 - Nothingness in Motion

So far, we've explored the existence of a point that contains within itself the structure of a circle whose values all add to perfect zero. This is all well and good; however, at this step, we would have a completely still existence where nothing ever happens. But we live in a thriving universe. A universe of change. A universe of motion. This is possible because by the PSR a point can be in motion.

Remember that we haven't yet introduced space. Our point won't be moving through anything at all. This is purely analytic motion that deals with the relationship between the values of the encoded circle. Everything we're discussing is taking place in the realm of thought as we explore possible arrangements of thought patterns. This next step can be difficult to imagine because people are used to thinking of motion in terms of space. It can be helpful to think in terms of "sequential pattern" rather than motion. If the concept of motion gives you difficulty, simply exchange that term for sequential pattern.

How will our point move? Within our mathematical point is the coding for a perfect circle, so our point can begin to sequentially explore that pattern since it contains that information within itself. Since our point contains the coding of a circle that nets out to perfect zero, by the

PSR, there's nothing preventing our point from sequentially exploring that pattern. In other words, our point will exhibit a sequential pattern that actualizes as a circular trajectory. The point contains within itself the mathematical code of a circle, and there's nothing to prevent it from moving in that circular pattern since that pattern maintains a net result of zero, and motion doesn't change that net result. By the PSR, since nothing can prevent it, then it must happen. The only thing that could prevent such motion (a succession of thought patterns) would be if motion violated the zero condition or the PSR. However, there is no violation since the circular pattern it's exploring nets out to zero. By the zero condition, it's possible for it to explore this pattern so, by the PSR, it *must* explore this pattern.

Let's use a crude analogy to wrap our heads around this concept. Imagine a marble that was made of nothingness. Now imagine a circular track that was also made of nothingness. The marble could roll round and round the circular track because the motion doesn't change the fact that they're both made of nothingness. Again, this is quite crude, but our point is like the marble of nothing. The point isn't going around anything in a physical sense. It contains within it the code of a circle, because that circle is equal to zero and is equivalent to it. Therefore, it can begin to sequentially explore that code, because exploring it doesn't change the fact that it all nets to zero.

Before continuing, let's examine once more why a circle is possible by the PSR rather than some other configuration. As stated previously, the values must be arranged in such a way that their total value nets to zero. The circle is the perfect configuration where every value is uniformly related and equidistant from the origin. It's the prime trajectory for our flowing point that exhibits perfect balance and equality. But there's one more thing we need to consider. A circle whose values are only real numbers excludes imaginary and complex numbers. This violates the PSR as there is no sufficient reason to exclude such values (possible thought patterns).

2.3 - Imaginary Numbers

The term “imaginary” is a misnomer. Imaginary numbers have just as much reality as real numbers.

$$\text{Consider that } \sqrt{4} = \pm 2$$

The (+ –) signs mean that it equals 2 or –2. This is because 2 × 2 = 4 *and* (–2) × (–2) = 4. As we know, a negative number multiplied by another negative number equals a positive number.

$$\text{Now consider } \sqrt{-4}$$

What can this possibly equal? It can’t equal 2 or –2 since their squares both equal positive 4. Obviously, there’s a problem. Something is missing. If you input $\sqrt{-4}$ in a basic calculator, it will return an error message. Obviously, existence can have no such errors. Can you imagine existence returning an error message? In a reality where only real values exist, there would be no answer to $\sqrt{-4}$. This is a huge problem. We can’t have any kind of incompleteness like this. This would be like a “fatal error” in a computer code. Existence can’t have any flaws. It can’t have any missing information. Existence must be perfect.

The circle we’ve been exploring contains only real values, and it can’t provide an answer to $\sqrt{-4}$. This means it’s incomplete. It’s missing information.

This necessarily implies the existence of another kind of number. That number is the imaginary number *i*. Consider that:

$$(-1) \times (-1) = 1$$

A negative number is a number that, when multiplied by another negative number, results in a positive number. The imaginary number *i* is simply the number that when multiplied by another imaginary number results in a negative real number:

$$i \times i = -1$$

It really is that simple. To signify that a number is imaginary, we place an *i* in front of it. For example, we can have *i*, 2*i*, 3*i*, 4*i*, etc. Note that *i* really is 1*i*, but we just omit the 1 since we know it's there implicitly. Don't let imaginary numbers confuse you. They're very simple. Just as we put a "–" next to a number to show that it's negative, we place an "*i*" in front of a number to show that it's imaginary. Just as any number you can think of can be negative, so can any number be imaginary. Knowing this, we can answer the question of $\sqrt{-4}$:

$$\text{Since } 2i \times 2i = -4 \text{ then:}$$

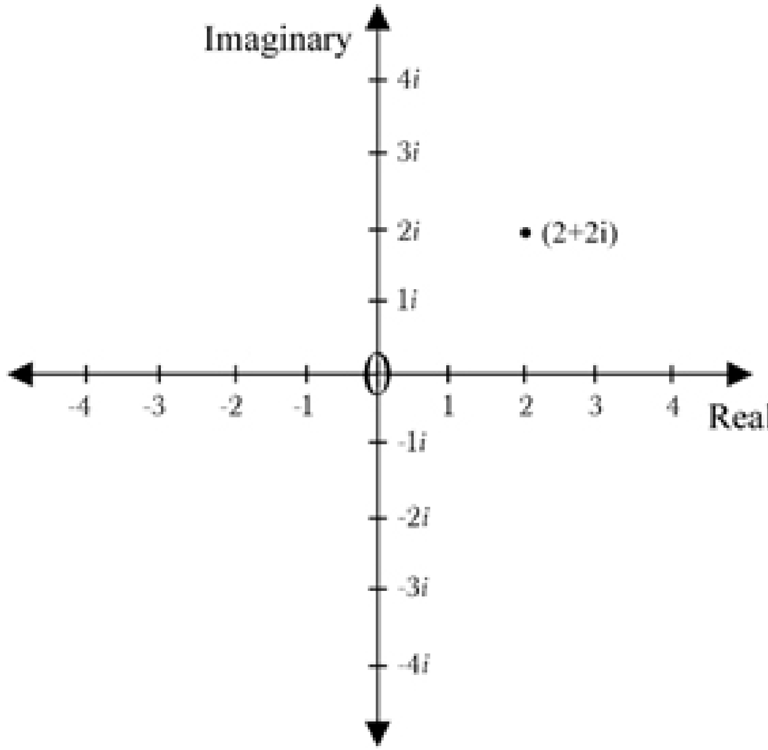
$$\sqrt{-4} = 2i$$

Imaginary Numbers

An imaginary number is a number that when multiplied by another imaginary number results in a negative real number.

$$i \times i = i^2 = -1$$

Imaginary numbers are simple. They're just another kind of number, like a negative number. But there's another way of thinking about them that's important to our discussion. Consider the following image:

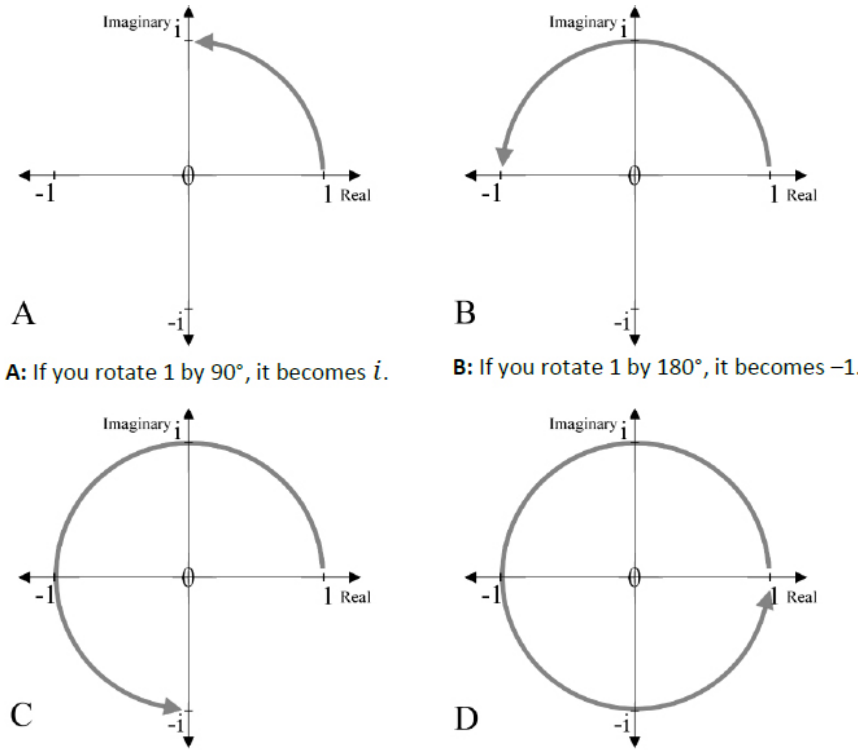


(Fig. 2.31) This is known as the complex plane. It has one real axis and one imaginary axis.

The horizontal axis is composed of all real numbers, while the vertical axis is composed of all imaginary numbers. These axes are **orthogonal**. Orthogonal means they are 90° apart.

Imagine that you start at 1 and rotate counterclockwise by 90°. You would land on i. Now imagine that you start at 1 and rotate by 180°. You would land on –1. Notice that rotating 1 by 180° results in landing on its opposite. A 180° turn resulting in an opposite should make intuitive sense, since when you turn your body around, you end up facing in the opposite direction.

You can see that on the complex plane –1 is the opposite of 1. So what is *i*? *i* is halfway between 1 and its opposite.



A: If you rotate 1 by 90°, it becomes i.

B: If you rotate 1 by 180°, it becomes –1.

C: If you rotate 1 by 270°, it becomes $-i$.

D: If you rotate 1 by 360°, it makes a full rotation and becomes 1 again.

(Fig. 2.32)

Notice that if you begin at *i* instead of 1 and “turn it around” by rotating 180° counterclockwise, you would arrive at the opposite of *i* which is –*i*.

What about complex numbers? All the real numbers are located on the horizontal axis, while all the imaginary numbers are on the vertical axis. Complex numbers are simply all the numbers that lie in between the real and imaginary axes. They’re a fusion of imaginary and real numbers. For example, the point plotted in Fig. 2.31 is the complex number 2 + 2*i*. And yes, the complex number 2 + 2*i* is one number. A complex number has two parts, a real part and an imaginary part, but it’s one number. Complex numbers aren’t difficult to understand, but it’s not necessary to discuss them in depth for our current purposes.

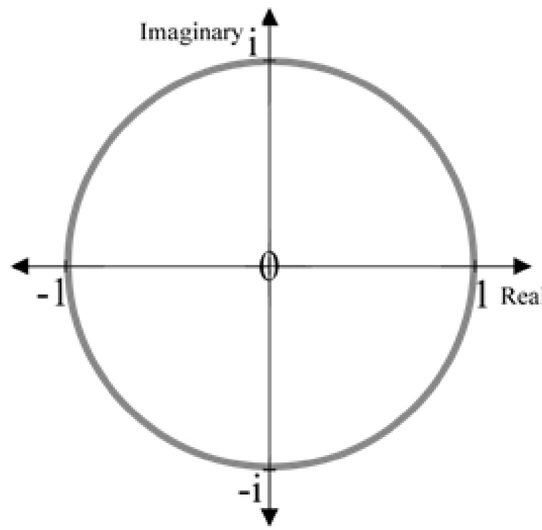
The complex plane graphically demonstrates how numbers can be arranged in relation to each other. This is incredibly important for what we're discussing in ontological mathematics, since we're examining how numbers are related to each other to find a valid relationship structure allowed by the PSR. If we ignored imaginary numbers, there would be many valid "directions" we would be omitting. Just as "up" implies "down," and positive implies negative, as our example of $\sqrt{-4}$ demonstrates, negative real numbers imply imaginary numbers. With all this in mind, let's return to our discussion.

Recall that we're searching for a valid structure of values. We know that zero exists and can contain a unity of opposite values, and we must find how those values are arranged. There is no space as of yet, and the only thing that exists are these values. We're looking for a valid arrangement, but there's nothing physical for them to be arranged in or in relation to. Since they are all that exist, arrangement simply means to discover how they relate to each other, and the complex plane is the perfect tool for visualizing how values relate.

We've seen that a circle is an excellent candidate for arrangement, but we ran into the problem of excluding certain values, namely imaginary and complex values. We've just seen that imaginary numbers must exist. The existence of real numbers mandates their existence to maintain completeness. If we were to use a circular arrangement of only real numbers, we would be excluding imaginary and complex numbers. And why should we arbitrarily exclude them? This is a violation of the PSR. There's no sufficient reason to exclude them, so by the PSR, we *must* include them in our arrangement.

Furthermore, if we did exclude them, this would result in "fatal errors" when things get more complicated. For example, if we "squared" a circle of only real values by squaring every value on the circumference, all the negative numbers would become positive because the square of a negative number is a positive number. Half of the structure would be wiped out, which would ultimately result in the violation of the PSR and the collapse of any possible universe. This fatal error shows that our thought structure is currently incomplete. We haven't found the right pattern yet. We need a circular structure that includes imaginary and complex values.

How should these imaginary numbers be arranged? We just saw that imaginary numbers are 90° to real numbers. We can arrive at such a structure by replacing our real-valued circle with a complex-valued circle. This is simply a circle arranged on the complex plane. We now have a circular arrangement of real values with imaginary values related to them at 90°:



(Fig. 2.33) This complex circle nets to perfect zero while including imaginary and complex numbers.

This circle is an arrangement of real, imaginary, and complex numbers that are related in perfect equality without privileging any value over the other. It's now perfectly stable and free from any fatal errors, contradictions, or flaws while still being equal to zero in total. With our previous example of squaring in mind, our complex circle remains stable under such a transformation since the square roots of negative numbers are imaginary, and the square roots of imaginary numbers are complex, and the square roots of complex numbers are also complex. This circle is now completely compatible with the dictates of the PSR and will remain stable under any valid ontological operation.

2.4 - The Source Formula

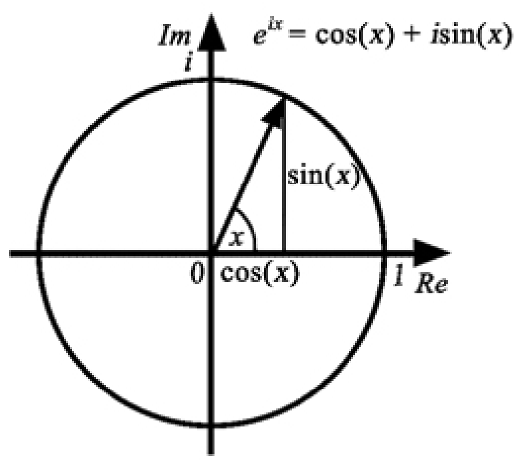
The complex circle is the mathematical coding of our zero point. Our zero point can contain this arrangement precisely because this arrangement is equal to zero. To review:

Nothingness = 0 = the total result of values arranged in a circle on the complex plane.

There exists an equation for this complex circle. This equation is known as **Euler's formula**. This is the most powerful equation in all of mathematics and in Hyperianism, this is called the **Source formula**:

$$e^{ix} = \cos(x) + i \sin(x)$$

The Source formula describes a point continuously moving in a circular pattern on the complex plane. This formula describes the exact pattern that we've been looking for, and by exploring its properties, we can unlock the secrets of existence.



(Fig. 2.41) The Source formula describes the exact pattern valid by the PSR.

Physicist Richard Feynman called this equation “our jewel” and “one of the most remarkable, almost astounding, formulas in all of mathematics.” When the equation is evaluated at pi, the mathematician Benjamin Pierce said, “Gentlemen, that is surely true, it is absolutely paradoxical; we cannot understand it, and we don't know what it means. But we have proved it, and therefore we know it is the truth.”

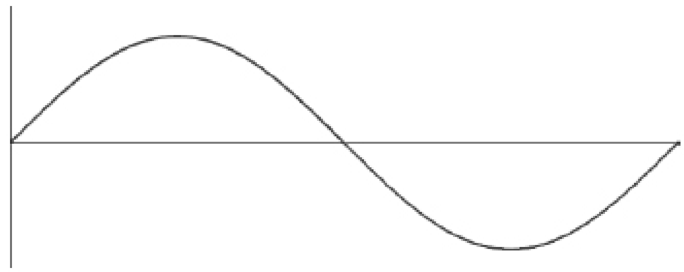
The Source formula is the master key to all reality.

“Like a Shakespearean sonnet that captures the very essence of love, or a painting that brings out the beauty of the human form that is far more than just skin deep, Euler's equation reaches down into the very depths of existence.”

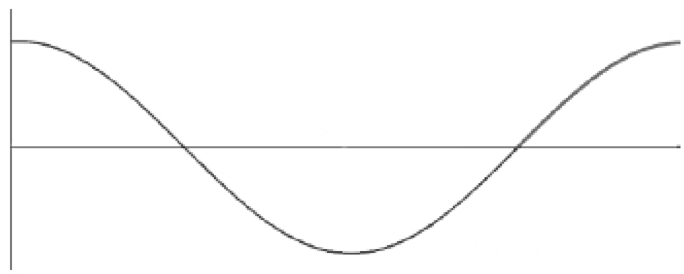
– Keith Devlin, Mathematician

One of the remarkable facts about the Source formula is that while the left-hand side describes a circular pattern, the right-hand side describes two **sinusoidal** waveforms. To be precise, it describes one real-valued **cosine wave** and one imaginary-valued **sine wave**. We use the word “sinusoid” as shorthand to refer to both sine and cosine waves.

A sine wave is simply a curve of this shape:



A cosine wave is simply a curve of this shape:



Both waves are sinusoids.

(Fig. 2.42) The formula for a simple sine wave is $f(x) = \sin(x)$ and for a cosine it's $f(x) = \cos(x)$. A cosine wave is nothing but a sine wave that's been shifted by 90 degrees. This means that $\sin(x + 90°) = \cos(x)$. They aren't fundamentally different. They're both sinusoids.

You can clearly see a real cosine wave and an imaginary sine wave on the right-hand side of the Source formula: $e^{ix} = \cos(x) + i \sin(x)$. Like the circle itself, these two waves also net to zero. These waves are just another way of looking at the information of a complex valued circle. As soon as you have the information for a complex circle, you automatically have the information for a sine and cosine wave.

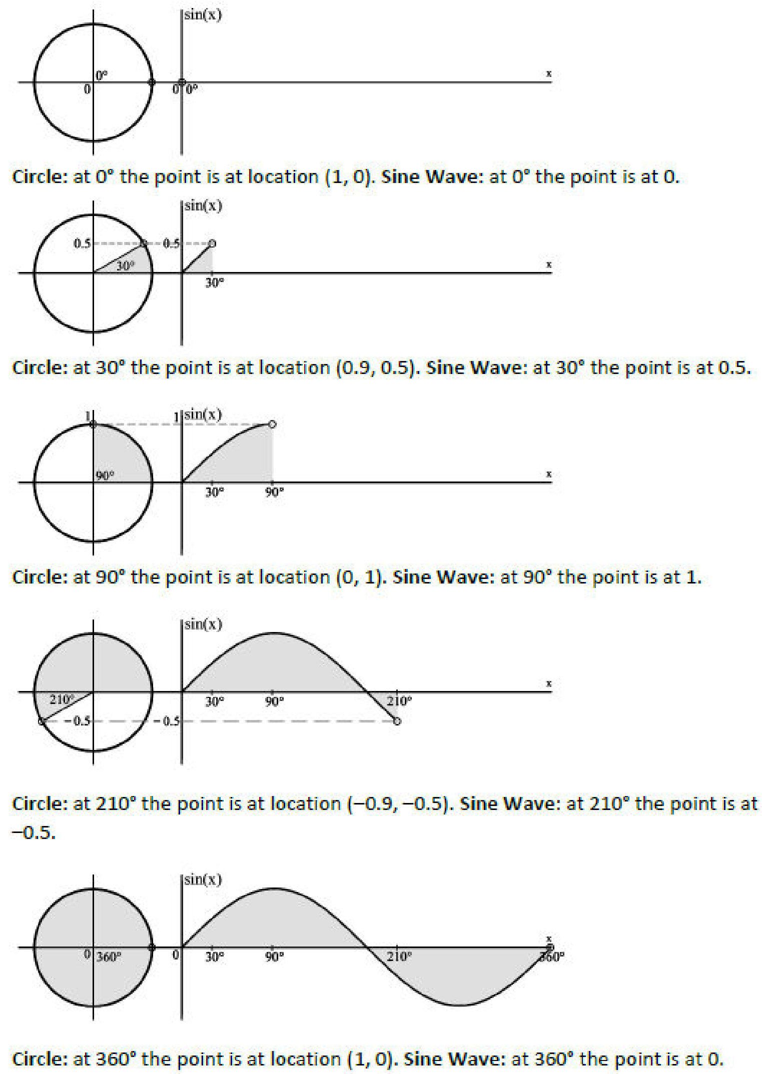
You can't have a complex circle without waves, just like you can't have 2 + 2 without 4. They're mathematically inseparable. The left-hand side of the Source formula describes the circle, while the right-hand side describes the waves: $e^{ix} = \cos(x) + i \sin(x)$.

Since our zero point contains the information for a complex circle, it *must* contain the information for sinusoidal waves since they are mathematically equivalent and inseparable. The existence of these waves is demanded by the PSR and their existence does not violate the zero condition since the waves net to zero. This astounding fact means that as our point travels around the complex circle, it also *generates wave motion.*

Let's examine exactly how the Source formula is equivalent to waves. To keep things simple, let's first look at a circle composed of only real numbers. In the following diagrams, we're going to plot how a sine wave is generated, step by step.

The left side of the graph depicts a circle, and the right depicts how a sine wave relates to it. It's quite simple. For every angle, the circle plots both the vertical and horizontal positions of the point, while the sine wave is a plot of only the vertical locations of the point (it ignores the horizontal locations.)

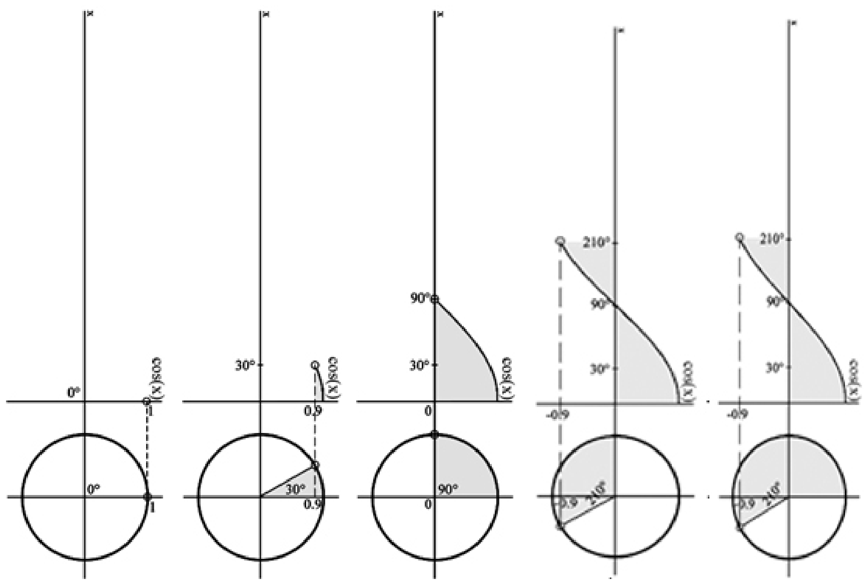
In other words, the circle shows how far up or how far down the point is on the vertical axis, in addition to how far left or how far right it is on the horizontal axis. The sine wave shows only how high up or how far down the point is. When you look at the following images of the circle, pay attention to where the point is located in relation to the vertical axis (how far up or down it is) and ignore the horizontal axis (how far left or right it is). The vertical positions of the point, for every angle, is exactly what the sine wave is.



(**Fig. 2.43)** A sine wave describes the **vertical** positions of the point traveling along the circumference of the circle.

Notice that while the circle records the information for both the vertical and horizontal positions of the point, the sine wave records only the vertical positions of the point. For example, at 30° the circle shows that the point is at the horizontal position of 0.9 and the vertical position of 0.5, while the sine wave ignores the horizontal position of 0.9 and shows only the vertical position of 0.5. The sine wave goes only up and down. Easy!

The cosine wave is just as simple. It's the opposite of the sine wave. It ignores the vertical positions of the point and shows only the horizontal positions. It ignores the "up and down" information and shows only the "left and right" information. When you look at the following diagrams, pay attention to where the point is located in relation to the horizontal axis (how far left or right) and ignore the vertical axis (how far up or down). The horizontal positions of the point, for every angle, is exactly what the cosine wave is. Like the circle itself, these waves also result in zero.



(Fig. 2.44) A cosine wave describes the **horizontal** positions of the point traveling along the circumference of the circle.

Notice that while the circle records the information for both the vertical and horizontal positions of the point, the cosine wave records only the horizontal positions of the point. For example, at 30° the circle shows that the point is at the horizontal position of 0.9 and the vertical position of 0.5, while the cosine wave ignores the vertical position of 0.5 and shows only the horizontal position of 0.9. The cosine wave goes only left and right. Easy again!

What are sine and cosine waves? A sine wave is just the vertical positions of a point traveling on a circle, and a cosine wave is just the horizontal positions of a point traveling on a circle. The patterns they form happen to take the shape of a wave. It’s that simple.

These graphs may make it seem like the circle always exists while the wave is generated bit by bit, but that is incorrect ontologically. It just appears that way for the sake of graphically demonstrating how a circle relates to waves. Both the circular pattern and the wave patterns are eternal. They’re the inner coding of the points and direct their motion.

Remember, to keep things simple, we examined a real-valued circle. But the Source formula describes a circle that contains imaginary and complex values. Everything we just learned about circles and waves applies to the Source formula; all we have to do is replace the real numbers on the vertical axis with imaginary numbers. That’s it. All this means is that you need to put an *i* in front of all the vertical positions. Nothing further needs to be done.

> **How the Source Formula Relates to Circles and Waves**
>
> $$\underset{\substack{\text{(The Circle)}\\ \downarrow \\ \text{The horizontal and vertical positions}}}{e^{ix}} = \underset{\substack{\text{(Cosine Wave)}\\ \downarrow \\ \text{Horizontal positions}}}{\cos(x)} + \underset{\substack{\text{(Sine Wave)}\\ \downarrow \\ \text{Vertical positions}}}{i\sin(x)}$$
>
> The **circle** describes both the **vertical** and **horizontal** positions of a point.
> The **cosine** describes only the **horizontal** parts.
> The **sine** describes only the **vertical** parts.
> A circle (the **horizontal** and **vertical** positions) is equal to a cosine wave (the **horizontal** positions) plus a sine wave (the **vertical** positions).

Note the *i* in front of sin(*x*) in the Source formula. Since the sine wave contains the vertical information, and all our vertical values are imaginary numbers, this means that our sine wave is an imaginary sine wave. All this means is that the values it records are imaginary (the values have an *i* in front of them). Nothing else has changed about it.

There's nothing fundamentally different about a cosine and a sine wave. A cosine wave is just a sine wave that's been rotated by 90°:

$$\sin(x + 90°) = \cos(x)$$

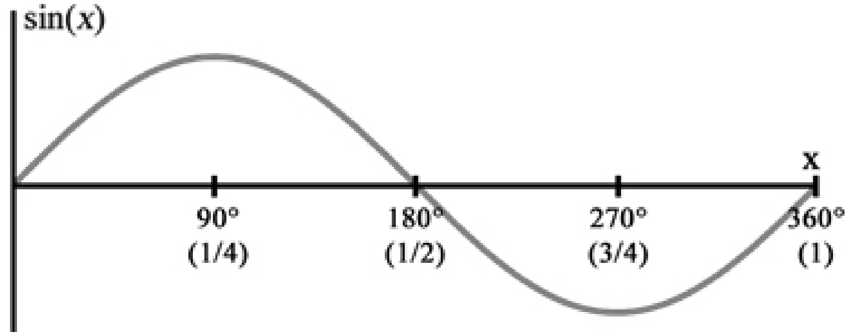
We've described these concepts in terms of "up and down" and "left and right." This is to make the explanations as simple as possible, but always remember that everything we're talking about is taking place in the realm of thought. It's pure information and relation. There is nothing "physical," and there is no space. There is no dimensional up, down, left, or right. All "position" means is how these values are positioned in relation to each other. Where is *i*? It's 90° from 1. Where is –1? It's 180° from 1 and 90° from *i*.

We saw that the concept of 0 entails the concept of a unity of opposites. We discovered the only valid structure of that unity, by the PSR, is a complex-valued circle. By the PSR, motion can occur in this circular trajectory since the circular pattern it's following is equal to zero. The equation for this circle and its motion is the Source formula which is equivalent to sinusoidal waves whose total is also equal to zero. From nothing, we've derived the existence of sinusoidal waves. Their existence is necessary. Reason demands their existence.

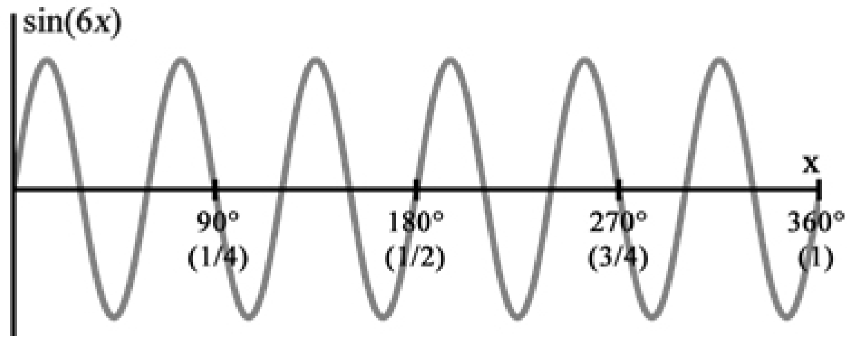
At this point, the astute reader should have a chill running up their spine. From nothing at all, we've unlocked the existence of waves.

2.5 - Frequency

Just as a circle entails the concept of waves, the concept of waves entails the concept of frequency. One automatically implies the other. Frequency is simply the number of times that a wave oscillates.



(Fig. 2.51) The wave sin(*x*) repeats its pattern once in 360°. Since sin(*x*) oscillates once in 360°, it has a frequency of 1.



(Fig. 2.52) The wave sin(6*x*) repeats its pattern six times in 360°. It oscillates six times. Since sin(6*x*) oscillates six times in 360°, it has a frequency of 6.

Notice the "6" in sin(6*x*). The number next to the x is what indicates a wave's frequency. Sin(6*x*) oscillates six times and sin(*x*) oscillates once. Sin(*x*) is actually sin(1*x*), but we omit the one since 1*x* = *x*.

The function sin(*fx*) can become a wave of any frequency by changing *f* to any value we like. Sin(*x*) is a wave with *f* = 1 and sin(6*x*) is a wave with *f* = 6.

As you can see in the figures, frequency is measured in relation to degrees. A degree[3] is a unit often used to describe angle and rotation. The concept of a "degree" is arbitrary. It's unknown why humans began to use a degree as a unit of rotation. In ontological mathematics, we don't use the arbitrary concept of a degree. Since 360° is equal to one full rotation of a circle, then 360° = 1 natural revolution. In the figures, **natural revolutions** are listed below degrees in parentheses. We'll continue to use degrees, since that is what people are most familiar with, but keep in mind that ontological mathematics doesn't use the random, arbitrary concept of degrees. We use natural revolutions where 360° = 1. One what? One full rotation of a circle. In ontological mathematics, we can't have any arbitrariness and must define what exists with respect to itself.

In physics, the concept of frequency is defined as oscillations per unit time. Commonly, it's measured as oscillations per second. But remember that we're exploring the pure domain of mind. We started with nothing, and everything is taking place in the realm of thought. Space and time haven't yet been introduced, so we can't use the concept of time or seconds. We're exploring purely mental values and their structure. We're exploring their possible configurations and arrangements with respect to themselves. Meters and seconds don't yet exist, so we can't define values with respect to these concepts. The only things that exist are these values and, since they are all that exist, they must be defined with respect to each other.

In ontological mathematics, frequency is not defined according to time. As shown in the diagram, it's defined according to angle. Frequency is defined as oscillations per 360° where 360° = 1 full rotation of a circle (which is a natural revolution).

Frequency

Frequency is the number of times a wave oscillates per natural revolution, where a natural revolution is one full turn of a circle.

1 natural revolution = 360°

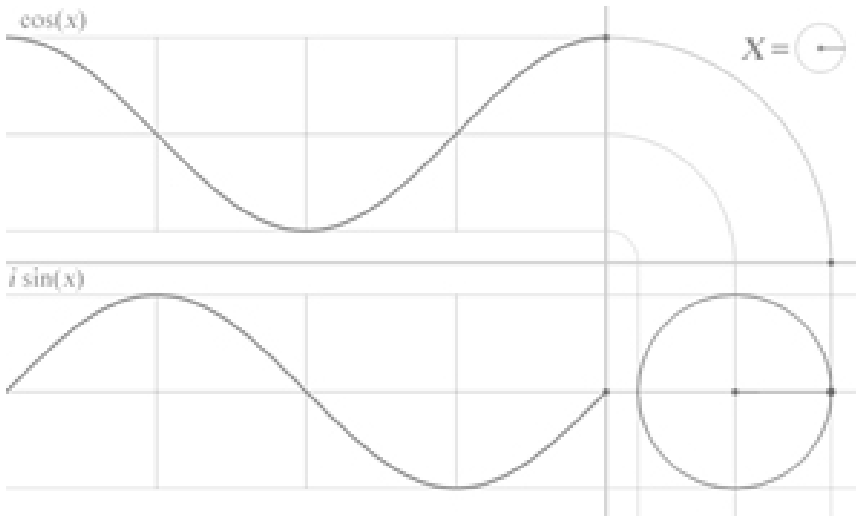
It's critical to understand that we can't define the mental concepts we're exploring according to space or time, and we can't use arbitrary concepts. All that exists are these values, so we're exploring how these values relate to each other. Once we rationally deduce a fundamental concept, then we can define everything else with respect to that fundamental concept. Space and time are not fundamental but will ultimately be derived from these mental concepts (which *are* fundamental).

Recall that learning ontological mathematics in a linear way can be difficult since later concepts will be needed to fully understand earlier concepts. This is one of those moments where information in later sections will help to complete your understanding of frequency. It may be helpful to skip ahead to view Fig 2.143 in section 2.14 and note that it depicts a wave of $f = 4$ and a wave of $f = 8$ that oscillate four and eight times, respectively, during one full rotation of a circle.

To truly grasp these concepts, you should get a calculator and a piece of paper and graph $\sin(x)$ and $\sin(6x)$ yourself and everything will click. If any of this seems confusing, don't worry. We'll continue to use degrees often to keep things simple. The most important thing to understand in this section is that frequency is simply the number of times a wave oscillates.

2.6 - Eternal Minds

We've been exploring pure thought, starting from nothingness, and deductively determining what kind of pattern is possible by the PSR. This brought us to a complex-valued circle defined by the Source formula which results in sinusoidal waves. We now have a point traveling in a circular trajectory which is mathematically equivalent to a sine and cosine wave that all nets to zero. All of this is mandated by the PSR. It must happen. Zero implies the unity of opposites, which implies the existence of a circle, and you can't have the idea of a circle without the idea of sinusoidal waves. We then explored the concept of frequency, which is a concept entailed in waves.



(Fig. 2.61) The Source formula describes a circle, a cosine wave, and a sine wave. (Note that the cosine wave has been rotated simply to save space.)

Now that we understand frequency, let's see what it has to do with the Source formula. Just as sin(*fx*) can take on any frequency by altering *f*, the frequency of the Source formula can also be set to any value we choose:

$$e^{ifx} = \cos(fx) + i \sin(fx)$$

Notice the formula now includes *f*. It didn't appear in the Source formula earlier, because we set $f = 1$ to keep things simple.

There is an *f* in each part of the Source formula: the circle part, the cosine part, and the sine part. In the section on frequency we used sine waves in our examples but frequency for a cosine wave is exactly the same as that of a sine wave, except the shape is different. Altering the frequency of the Source formula has the effect of altering the frequency of its sine and cosine pair.

Here is our next critical deduction:

Since the Source formula is valid by the PSR and can take on many different frequencies, and because valid frequencies net to zero, by the PSR, they're possible. Any frequency that's valid by the PSR and nets to zero must exist. Since there's no sufficient reason to prevent these frequencies from actualizing, every possible frequency is generated. Every form of the Source formula that doesn't violate the zero condition and is valid by the PSR must be actualized.

Up to this point, we've been considering one circle and its sinusoidal pair. Now imagine a vast multiplicity of circles, one for each valid frequency. Each circle will have its own sine and cosine pair with matching frequencies.

We no longer have just one point traveling in a circular trajectory with its sinusoidal pair. We now have a vast multiplicity of points traveling in a circular trajectory for every value of *f* that's valid by the PSR. Each one of these points traveling in a circular trajectory is automatically accompanied by its own sine and cosine pair. This means that we have a vast collection of sines and cosines at every possible frequency. We no longer have just one circular pattern, but many, each one associated with a specific frequency. This is possible because they all net to zero and there is nothing to prevent their existence. They must exist.

Everything we have been describing has taken place in the domain of mind. There is no space or time yet. There is only information relating to itself. These waves aren't anything physical. They're mental. What is a mental wave? It's a basic pattern of thought. These waves, which entail frequencies, are thought patterns.

By the PSR, we now have a vast collection of thought patterns. What is a vast collection of thought patterns?

It's a mind.

We call this collection of sinusoidal waveforms a **monad**. A monad is a living mind. It's simply the collection of sines and cosines of every possible form generated by the Source formula. It's nothing more and nothing less. What is a mind but a collection of thoughts?

Through the Source formula, every possible frequency is generated. This collection of frequencies is a mind, and the Source formula is its guiding formula. Since this collection of mental waves nets to zero, this of course means that a monad is a net zero as well. From nothing, we've derived the existence of one mind. This mind is a net zero. Since it's a net zero, it's a "nothing" itself.

By the PSR, if the existence of one monad is possible, then there is nothing to prevent the existence of a multiplicity of monads. Why is that? Because a monad is a system of structured nothingness. It's a net zero. It's one complete and consistent structure of void. If one zero is possible, what sufficient reason is there to prevent a multiplicity of zeros? There is none. You can't stop one "nothing" from existing, so you certainly can't stop many "nothings" from existing. Monads are structured nothingness. Therefore, a multiplicity of monads exists, all of which are unique containers of sinusoidal waveforms at every possible frequency. Every monad is an individual, unique collection of frequencies, and they each contain their own reason for being within themselves. They're self-grounding.

We've finally reached the ultimate self-grounding ground of existence: A multitude of monads. A multitude of zeros. A multitude of living minds.

Monads – The Ground of Existence

A monad is a net-zero.

A monad is a collection of every possible form of the Source formula that's valid by the PSR. Therefore, the Source formula is its defining and guiding equation.

A monad is uncreated, uncaused, necessary, indivisible, indestructible, independent, individual, and eternal.

A monad is self-grounding.

Monads are the true self-grounding ground of existence.

The mental domain is the collection of all monads. Nothing more and nothing less. Since monads are zeros, the mental domain (being the collection of all zeros) is zero as well. We call this mental domain **the Source**. The Source isn't a thing unto itself. It's simply what we call the collection of all monads. If monads didn't exist, the Source wouldn't exist, since it's nothing more than the collection of monads.

A Summary

All things must be valid by the PSR.

Nothing = 0 = a mathematical point.

Zero is mathematically equivalent to a union of opposites. For example: 0 = 1–1.

The structure of this union of opposites is a complex circle. This is the internal coding of our point.

The point can travel along the trajectory of its internal coding.

The complex circle mathematically implies the existence of sinusoidal waves.

The complex circle, the point's motion, and its sinusoidal waves are defined by the Source formula.

Waveforms of every possible frequency are implied by the Source formula. Thus, a multiplicity of waves exists.

We call this collection of waveforms a monad.

Since one monad is possible, many monads are possible.

A myriad of monads exist.

> **A Reminder**
>
> If our process ever gets confusing, think back to Descartes' "I doubt, therefore I think, therefore I am." He was essentially exploring the consequences of thought. For example:
>
> 1. **X** implies the existence of **Y**
> 2. **X** exists
> 3. Therefore, **Y** exists

As we've seen, nothing implies the existence of a complex circle, which implies waves, which implies frequency, etc. Starting from nothing, each concept logically demands the existence of the next. We're exploring the structure of thought in a way similar to the above deductive argument, albeit in a more fundamental, complex, and intuitive manner.

We've discovered the ground of existence: a multitude of monads, each one a zero. We now have everything we need to begin exploring existence. It's important to note that we've described this as though it were a step-by-step process. It's not. We've only described it in such a way so that you can see how it's rationally necessary that monads exist. In actuality, this is not some kind of process that had a beginning and then unfolded step by step. Monads were never generated.

Monads have always existed and are eternal.

2.7 - Inside a Monad

Now that we know monads are eternal zeros and are the ground of existence, let's take a closer look at their frequency composition. The Source formula is the defining formula of a monad. We've seen it in a simplified form, but we can begin to generalize it:

$$Ae^{i(fx + \phi)} = A\cos(fx + \phi) + iA\sin(fx + \phi)$$

Don't let this formula intimidate you. You may wonder where these extra variables came from. We already know that f is frequency. We didn't see it in the original formula because it was set to 1.

When A is set to 1, f is set to 1, and ϕ is set to 0, we get the original formula. We did this to keep things simple, not because it was necessary. This generalized formula allows for the many different variations and properties of the Source formula that are possible by the PSR. This is still the same formula, just generalized so we can explore all its possibilities.

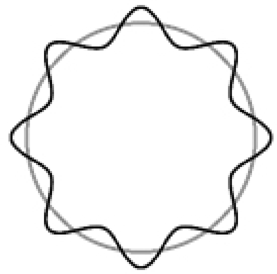
A is amplitude, *f* is frequency, and the Greek letter ϕ is phase. For each of these properties, the Source formula will generate every value that's possible by the PSR. We'll explore phase in a later section. For our current purposes, we want to examine the frequency contents of a monad. The frequency is denoted by *f* in the Source formula.

On closer inspection it turns out that though a great many frequencies are possible not all frequencies are "stable." Only **integer** values are stable frequencies.

> **Integer Examples**
>
> An Integer is any number that is not a fraction or a decimal:
>
> **1**, **27**, **5**, **–8**, **–16**, and **72,561** are all whole numbers.
>
> $\sqrt{2}$, **1/5**, **9.68**, and ***π*** are not whole numbers.

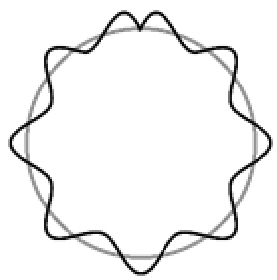
When the frequency *f* takes on integer values such as "1, 2, 3, 4…" it results in a stable, self-reinforcing wave. Non-integer values, such as "1.5, 2.4, 8.25…" are not self-reinforcing frequencies. Another way to say this is that only the fundamental frequency and its harmonics (multiples) are stable.

To better understand why integer frequencies are stable, consider the following image depicting a wave of frequency = 8 wrapped around a circle:



(Fig. 2.71) A wave with an integer value for frequency is stable. Imagine that we have taken the *x*-axis, like the one in Fig. 2.52, and connected its end together to form a circle.

An integer frequency generates a stable, self-reinforcing wave in relation to 1 natural revolution (360°). Now consider a wave where the frequency is a non-integer:



(Fig. 2.72) A wave with a non-integer value for frequency is unstable. Notice the wave is prevented from completing a cycle.

The wave in Fig. 2.72 is not self-reinforcing with respect to 1 natural revolution. It gets “chopped off,” so to speak. It’s prevented from smoothly completing its pattern in 1 natural revolution. The Source formula, therefore, generates stable frequencies at every possible integer value. A wave with frequency 0 is not a wave, and anything between 0 and 1 is not an integer and thus is unstable. Therefore, 1 is the fundamental frequency, and it serves as the base case for all other stable frequencies.

A monad contains frequencies of every possible value. All integer multiples of the fundamental frequency of 1 are stable, self-reinforcing waves with respect to 1 natural revolution. All frequencies are controlled by the Source formula.[4]

Fig 2.71 displays a helpful way to visualize the stable waves of eternity. You can see frequency measured with respect to 1 natural revolution (denoted by the circle). Stable waves don't go off in one direction forever. They are determined, closed, cyclical, self-reinforcing, "loops."

This section demonstrates a key difference between ontological mathematics and abstract mathematics. Abstract mathematics treats equations as having no connection to something that exists. Because of this, there are many operations in abstract mathematics that are not valid in ontological mathematics. In ontological mathematics, ontologically valid equations have actual existence. They're connected to something real—in this case, mental waves within monads. Everything in ontological mathematics represents something that exists. That's what ontology is about. This must *always* be considered when exploring what mathematical properties and operations are ontologically valid.

2.8 - Combining Thoughts: The Fourier Series

We have described points that flow (travel) in sinusoidal wave trajectories at integer frequencies, and monads which are unique collections of those waves. These waves aren't physical. They're mental waves. What is a non-physical mental wave? It's a simple thought. Mental waves are simple thoughts, and monads are the eternal minds that contain those thoughts.

Each monad, governed by the Source formula, contains its own personal set of frequencies. Though every monad contains an identical set of frequencies, a monad may arrange its frequencies (thoughts) into unique patterns by adding them together.

When we add many sines and cosines of various frequencies together, we generate a wave of a more complex pattern. This process of adding sinusoidal waves is described by the **Fourier series**.

The Rest of this Section is Optional

While the rest of this section isn't conceptually difficult, it's the most technical part of the book. Since it isn't necessary to know the mathematical details of the Fourier series to understand the concepts soon to be discussed, reading the rest of this section is optional. All you need to know is that:

The Fourier series describes the process of adding simple sine and cosine waves together at various frequencies to produce a wave of any form, no matter how complex.

With that understanding in mind, feel free to skip ahead to the section "Dreaming in Waves."

The following expression[5] is the Fourier series:

$$f(x) = \sum_{n=-\infty}^{\infty} A_n e^{inx}$$

Some of the symbols in the Fourier series may look new, but they just represent adding together different frequencies of the Source formula whose existence we've already accounted for. Let's break it down bit by bit.

Notice the following term in the Fourier Series:

$$A_n \, \mathrm{e}^{inx}$$

This is simply the left side of the Source formula:

$$A_n e^{inx} = A_n \cos(nx) + iA_n \sin(nx)$$

Compare this to a version of the generalized Source formula that we've already seen:

$$Ae^{ifx} = A\cos(fx) + iA \sin(fx)$$

There are only two differences in the version used in the Fourier series. The first, is that A has an n near its base. This n is known as a **subscript**. The second, is that there is an n where f normally appears. Notice both these differences are regarding n.

The reason for this is simple: since the Fourier series describes the process of adding together many different variations of the Source formula at different frequencies, we need a way to keep track of these different variations. n has the special mathematical property of taking on only integer values so it's perfect for keeping track of things by numbering them. This is what the subscript at the base of A is for. It allows us to label and keep track of each variation of the Source formula.

Also, since n can take on only integer values and, since the frequency of stable mental waves are also integer values, we can simply set $f = n$.

For example, the first variation of the Source formula has a frequency of 1 and takes the following form:

$$A_1 e^{ix} = A_1 \cos(x) + iA_1 \sin(x)$$

The second variation of the Source formula has a frequency of 2 and takes the following form:

$$A_2 e^{i2x} = A_2 \cos(2x) + iA_2 \sin(2x)$$

The third variation of the Source formula has a frequency of 3 and takes the following form:

$$A_3\, e^{i3x} = A_3 \cos(3x) + iA_3 \sin(3x)$$

Let's examine the other parts of the Fourier series.

The symbol ∑ is the Greek letter sigma. The ∑ simply tells us to add together variations of whatever is to the right of it. In this case, the Source formula[6] is to the right of it.

The values above and below ∑ tell us how many variations to add together. The value below tells us where to begin, and the value above tells us where to end. In this case, it starts at negative infinity and ends at positive infinity which tells us to add an infinite number of variations together. However, we can adjust this to add as many or as few as we like.

If we change the bottom number to 1 and the top number to 3, this would tell us to begin at 1 and end at 3, resulting in the addition of 3 variations of the Source formula.

A_n is the amplitude of a wave. This isn't something we've examined, but all that's necessary to know is that the amplitude tells us how much of each variation gets added together. Let's look at a simple example:

$$\sum_{n=1}^{3} A_n = A_1 + A_2 + A_3$$

Notice the 1 below and the 3 above ∑. This is what tells us to add three terms together. Now let's look at the original Fourier series, except we'll replace the infinites with 1 and 3 to keep it simple:

$$\sum_{n=1}^{3} A_n e^{inx} = A_1 e^{ix} + A_2 e^{i2x} + A_3 e^{i3x}$$

The Fourier series represents a summing together of the Source formula at different frequencies. The *n* denotes how many variations of the Source formula and which frequencies are added. The amplitude *A* denotes how much of each frequency to add. By adding together the Source formula at integer frequencies, this of course means that we're adding together sine and cosine waves at those frequencies since as we've already seen:

$$A_n e^{inx} = A_n \cos(nx) + \ iA_n \sin(nx)$$

This is important because adding sines and cosines of various frequencies produces a new wave of greater complexity. See Fig 2.91.

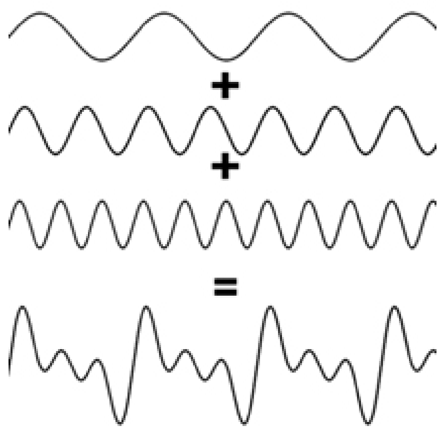
If this section has been confusing, don't worry. Even though it's simple, it's the most mathematically complex concept we've looked at. Luckily, we don't need to dwell on this too much. It's enough to know that the Fourier series describes the process of adding simple sines and cosines at various frequencies to produce a wave of greater complexity.

2.9 - Dreaming in Waves

Having examined the frequency components of a monad, we can begin to understand how minds give rise to the collective dream of spacetime. In this section, we'll describe it in a simplistic and intuitive way to provide you with a general overview of what's happening. Then, in the following sections, we'll learn a few more concepts so we can dive into the process in more detail.

Monads are the fundamental constituents of existence. Nothing else exists but eternal minds and their thoughts. They are the absolute ground of existence. They exist in a mental domain. A singularity. The Source. The Source is the collection of all the minds in existence. There's nothing more to it than that. It's the domain of every monad and their frequency patterns. Nothing exists but monads and their frequency contents. The Source is just the word we use to refer to every monad considered together.

To gain an understanding of how monads generate the collective dream of spacetime, let's first focus on a single monad and understand how a personal dream is generated. A monad can combine its frequencies together to create a wave of greater complexity. This process is facilitated by Fourier mathematics and can be illustrated in the following diagram:



(Fig. 2.91) The Fourier series describes the addition of simple sines and cosines to create any periodic wave pattern, no matter how complex.

As the image shows, the Fourier series describes the addition of simple sinusoidal waves at different frequencies to produce a wave of a very different pattern.

Basic sine and cosine waves are simple thoughts. Through Fourier mathematics, we can add simple thoughts together to create more complex thoughts. This is the process of thinking. Remarkably, Fourier mathematics allows us to create any wave in existence, no matter how complicated it may be. This astonishing fact means that sines and cosines are the building blocks for every wave in existence.

To recap, a monad is a mind that contains sinusoidal waves, which are simple thoughts. The process of thinking is facilitated by Fourier mathematics which allows us to take simple thoughts and combine them into complicated thoughts. Any realizable wave in existence, no matter how complex, can be made from the combinations of sines and cosines.

Remember that the waves of the Source all net to zero. However, Fourier mathematics allows us to add these waves together in such a way that it becomes possible to produce a complicated wave that does not net to zero.

Non-Zero Waves

Do non-zero waves violate the zero condition? Existence as a whole must *always* net to zero, and the zero condition can *never* be violated. In ontological mathematics, whenever a wave is produced that does not net to zero, a complementary wave is also produced that, when combined with the first wave, reduces it to zero ensuring that the total system nets to zero. Non-zero waves are only non-zero when considered independently from their complement that would reduce them to zero. Thus, non-zero results are an illusion from the perspective of existence considered as a whole.

Non-zero waves are only non-zero when considered independently from their complement that would reduce them to zero.

Non-zero waves are what give rise to dimensionality and extension. In other words, they allow for the creation of a dreamworld. When you go to sleep and dream, you're exploring the different frequency arrangements of your mind. Your personal dreamworld changes according to your thoughts, emotions, and even your conscious control when you're lucid, because it's composed of your frequency configurations. Ontological mathematics is not standard "abstract" mathematics. It's *living* mathematics. You are a mind, and your frequencies are *what you are.* They're your thoughts and result in your emotions, sensations, hopes, and your experience of entire dreamworlds.

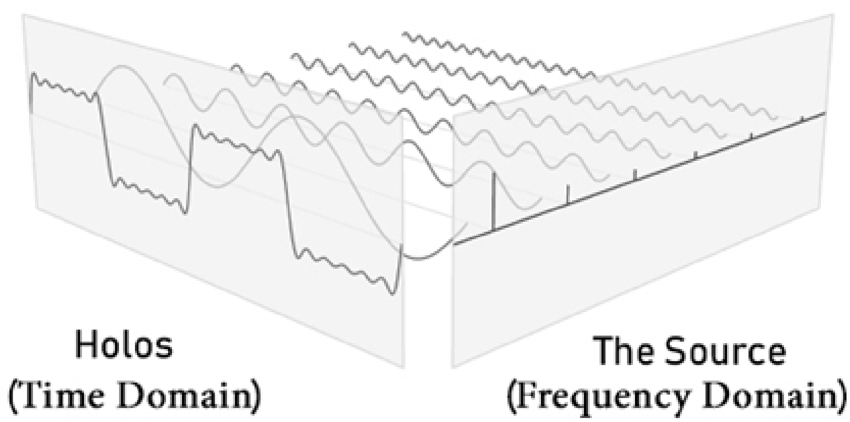
Even though your dreamworld has extension and dimensionality from within it (you can walk from one side of a dream room to another), it doesn't take up any "physical" space outside of you. Though it may seem like it does, this is an illusion. It's all happening within your mind.

When a single monad combines its own personal frequencies, this leads to complex thoughts and a personal, subjective dreamworld. Now consider the case when the frequencies of every monad in existence combine with each other. Everything that we discussed about Fourier mathematics still holds, but in this case, the waves being added together originate from different monads. This is collective thinking, and the resultant complicated wave is formed from the frequencies of many monads rather than one.

In the case of a single monad, wave addition results in a subjective dreamworld. In the case where the frequencies of every monad in existence are involved, wave addition results in an objective collective dreamworld. This is the shared dream that you and I exist in right now. This shared dream doesn't behave in an erratic way like a personal dream. It remains stable because it's formed from the frequencies of every mind. In this shared dream, your personal frequency contribution is distributed throughout the universe along with the frequencies of every other mind. You can't change things by thinking about them like you can in a personal dream because the collective dream is formed by the waves of every mind in existence rather than yours alone. This is why the objective dreamworld is stable and slow to change. If we want change to occur, we must work together to make it happen.

Matter, as classically described, does not exist. So-called "matter" is nothing but sinusoidal waves combined in such a way to produce slow, stable patterns. There is no such thing as a physical existence independent from the mind. The domain of spacetime is a result of the mental activity of all the minds in existence. It only seems to have physical extension from your perspective within it. When you walk across a room in a personal dream, it's all taking place in your mind. When you walk across a room in the objective spacetime world, in actuality it's a collective dream taking place within every mind in existence. There is no such thing as an external, extended existence that takes place outside of the mind.

In Hyperianism, the collective dreamworld of spacetime is called the **Holos**. The Holos depends on the Source for its existence. As we'll see in a subsequent section, the Holos is the mathematical Fourier projection of the Source. It's an internal projection. At all times, it exists within the mind.



(Fig. 2.92)[7] Fourier mathematics describes the link between the Source and the Holos. They are not separate domains but are two sides of the same coin.

In terms of Fourier mathematics, the Source is the frequency domain, and the Holos is the spacetime domain. As you can see in the above image, the Source and the Holos are two sides of the same coin. They're two different ways of looking at the same information. The simple sines and cosines of the Source combine to create complex Holos wave patterns.

Ontological mathematics is a new paradigm. It's a rational system of mind. It's an idealist monism. It's idealist because everything is ultimately mind. What we subjectively perceive as a reality of “matter” is a shared dream. It’s a monism because everything reduces to the one substance of mind.

You are a mind, not a body. Your body is just an avatar (a stable wave configuration) that you as a monad use to explore the dreamworld of the Holos.

“We do not belong to this material world that science constructs for us. We are not in it; we are outside. We are only spectators. The reason why we believe that we are in it, that we belong to the picture, is that our bodies are in the picture. Our bodies belong to it. Not only my own body, but those of my friends, also of my dog and cat and horse, and of all the other people and animals. And this is my only means of communicating with them.”

– Erwin Schrödinger

The current paradigm of scientific materialism and empiricism cannot account for how matter came into existence, i.e. what caused the Big Bang, nor can it account for the existence of the mind or consciousness. It can’t account for the fundamental aspects of reality because the answers lie outside its paradigm. Ontological mathematics answers these questions with ease and elegance.

In this section, we’ve explored how a collective dreamworld can arise, but we’ve explained it in a simplistic way. We’re going to delve into the process more precisely, but we need to understand a few more concepts first.

A Reminder Concerning Frequency

> As was explained in the section on frequency, in physics, frequency is defined as oscillations per unit time, usually with respect to seconds. But pure frequencies exist in the Source which is not within space and time. The Source is a mental domain that generates (and can interact with) the spacetime dreamworld but isn't within it. The pure frequencies of the Source are not dependent on space and time at all. Therefore, frequencies are not oscillations per unit time. Rather, they are the number of oscillations per 360° or natural revolutions. Frequency is defined according to angle (pure dimensionless information and mental relation), not according to time (which is a dimensional quantity). Ontological mathematics requires a new way of thinking, divorced from the old paradigm of materialism. We must be careful to keep this in mind when dealing with equations and mathematical concepts where materialists try to force in physicalist concepts like meters and seconds.

2.10 - Space, Time, and Motion

This section will require some mental gymnastics since the world has been so conditioned to think of space, time, and motion incorrectly. Ontological mathematics requires a new way of thinking, so forget everything you know about space, time, and motion and how it's been traditionally defined. In our system of mind, these concepts are defined in a completely different way.

Time is often confused with motion. However, we've seen that a point can move (sequentially explore an encoded circular pattern) in the Source domain of pure mind without the need for space or time. In ontological mathematics, it's important to understand the difference between motion, space, and time.

Motion is simply mathematical sequence. It's logical/rational unfoldment. It can occur independently of space and time as pure dimensionless thought, or it can occur within the dimensional spacetime Holos. When you think about pure concepts, such as mathematics, this is mental motion independent of spacetime. When you walk through a dreamworld, this is motion through your personal spacetime domain. When you walk through the everyday world, this is motion through the collective spacetime dream we all inhabit together.

What about time? **In ontological mathematics, time is simply another kind of space and is not necessary for motion or the succession of events.** Space and time are the containers for dreamworlds. Time has nothing to do with motion. Motion is pure mathematical sequence. Motion can occur as pure thought without the need for a dreamworld, or it can occur within a spacetime dreamworld.

Imaginary numbers are ultimately what give rise to time, and real numbers are what give rise to space. In a moment, we'll show how this is the case. But first, consider that the Source formula describes a circle that lies on the complex plane. Refer to Fig. 2.31 in the section on imaginary numbers.

A complex plane has a real axis and an imaginary axis that are orthogonal (the angle between them is 90°) to each other. We associate the real axis with space and the imaginary axis with time. This association isn't arbitrary since in ontological mathematics, time is the same as space, except that it's orthogonal to it. It would be more accurate to call time "imaginary space" to clear up any confusion. Remember, *time is just another kind of spatial dimension*, just as imaginary numbers are simply another kind of number. Time is defined differently from how it's been traditionally viewed. Once more: Time has nothing to do with motion. Time is another kind of space.

2.11 - The Speed of Mind

By the PSR, a point travels in a circular trajectory. But at what speed does it travel? Remember, we're dealing with pure mind, so the concept of speed is different from how materialists conceptualize it. In materialist terms, speed is usually defined as meters per second, but in the mental domain, there are no such concepts. There is no space or time, so the concepts of meters and seconds don’t apply. However, mental motion as mathematical sequence still occurs. We ran into a similar concept in the frequency[8] section, where frequency is defined as oscillations per 1 natural revolution (360°) rather than according to seconds. It’s purely informational relation and is not determined by a contingent, temporal concept. We’re going to see something similar here, except in this case, we’re talking about speed rather than frequency.

Speed must be defined with regard to pure mathematical/mental relation. We can’t talk about meters, seconds, or any dimensional unit. Additionally, by the PSR, every possible state must be treated with complete equality and uniformity. We can’t have any randomness, arbitrariness, or incompleteness as that is a violation of the PSR.

The speed of mind is frequency times wavelength. Wavelength (denoted as λ) in the mental domain is the “length” of one oscillation with respect to one natural revolution and is the reciprocal of frequency. Since wavelength is the reciprocal of frequency in the mental domain, wavelength times frequency is equal to 1. This means the speed of mind is equal to 1.

The speed of mind can also be understood as the overall ratio of real-valued “travel” to imaginary-valued “travel” (including both positive and negative values). This must be equal to unity, which is 1. The complex circle is defined by real and imaginary numbers. If the speed of mind wasn’t equal to 1, one kind of number would be privileged over the other, which is a violation of the PSR. Furthermore, this condition of uniformity and equality means that every mental wave no matter the frequency has the exact same speed, which is 1.

You might wonder, “one what?” Remember to think outside of material terms. In physics, speed is defined as a ratio of spatial units to temporal units (often by meters per second). There is no space or time in the mental domain, but as we mentioned earlier in the section “Space, Time, and Motion,” real numbers will end up resulting in space, and imaginary numbers will end up resulting in time (imaginary space). We can’t talk about meters per second in the mental domain. We can only talk about the values that exist and their relation to each other. We don’t have meters per second. We have real-valued “travel” to imaginary-valued “travel.” So, to say the speed of mind equals 1 means that the overall ratio of real to imaginary travel must be equal to unity.

As we’ll see, the speed of mind is the absolute fundamental speed and reference frame from which all other speeds arise, relate, and are defined. This unitary speed of mind will serve as the fundamental speed of existence from which all other spatial and temporal speeds are derived.

To get a better understanding of the speed of mind, we must explore wavelength and its relationship to frequency in greater detail, and we will do so in section 2.13.

The Speed of Mind

The speed of mind is equal to wavelength times frequency, which is equal to 1.

$$\text{The Speed of Mind} = \lambda f = 1$$

It is also the overall ratio of real-valued travel to imaginary-valued travel, which results in unity.

There is no space or time in the mental domain, but as we mentioned earlier in the section on space, time, and motion, real numbers will end up resulting in space, and imaginary numbers will end up resulting in time (imaginary space). In physics, speed is defined as a ratio of spatial units and temporal units (often by meters per second). In the pre-spacetime domain, speed is defined as a ratio of pre-spatial numbers (real values) and pre-temporal numbers (imaginary values) since real numbers will ultimately give rise to space and imaginary numbers will ultimately give rise to time.

As we'll soon see, the speed of mind is the absolute fundamental speed and reference frame from which all other speeds arise, relate to, and are defined by. This unitary speed of mind will serve as the fundamental speed of existence from which all other spatial and temporal speeds are derived.

2.12 - The Speed of Light

Before continuing, we must take a moment to examine the extraordinary properties of light. The speed of light is the fastest possible speed in the universe. No matter your inertial reference frame, the speed of light is always constant, and nothing can ever travel faster. Light is massless, maximally length contracted, and maximally time dilated. This means it has no physical extension and would not experience the passage of time. You couldn't measure it with a ruler, and a clock wouldn't tick for it. In addition, light exhibits both wave-like and particle-like properties. These are seemingly immensely paradoxical properties.

The Properties of Light

Nothing can travel faster than the speed of light.

The speed of light is the same no matter the reference frame.

Photons have zero mass, zero extension, and zero motion through time.

Light is both wave-like and particle-like.

Why does light behave so differently from everything else in the universe? Do the properties listed seem like the properties of something material? Absolutely not. Consider the mental waves we've been examining. They have net zero mass, net zero extension, and certainly don't exist within time. The speed of mind is always the same no matter what. Nothing can travel faster than the speed of mind since all other speeds are derived from it. Mind has both wave-like and particle-like properties since a point flows in a wave trajectory.

Mind has the same properties as light because light is mind.

Remember that ontology is the study of what is real. This entire time we've been studying mind, but we've also been studying light. All the apparently strange properties of light are immediately resolved when the materialist bias is removed and it is seen that light is actually mind. How could anyone consider something unextended to be physical? Light isn't anything physical. Light is thought. Nothing "material" can travel faster than light, because light is pure thought from which the dreamworld of so-called "matter" is derived. This is why we often refer to the eternal waves of existence as **innerlight**. Do you see how easily the problems of materialist science are resolved once the dogma of matter is replaced with a mental existence?

Light does not exist within space and time. Light is thought, and thoughts are the eternal waves of existence that exist within monads and the Source. Even though light does not exist within the spacetime Holos, it can interact with it. Obviously, it must be able to interact with it since spacetime is derived from it.

Think of it this way: pure thought doesn't exist within a dream, but it controls and can interact with the dream since the dream is derived from thought. Light is thought, and it doesn't actually move through space or time. The opposite is true: spacetime moves through light. What you perceive as the movement of light is actually the movement of spacetime through light and the interaction of light with spacetime. This is why the speed of light is the same no matter what spacetime reference frame you take.

We often refer to it as innerlight to make it clear that we're talking about light from its own perspective instead of how humans perceive light and define it from their spacetime perspective. This is a critical point. To reiterate, pure mind (even though it's not within spacetime) must be able to interact with spacetime since spacetime is derived from it. The metaphor of a hologram is helpful here. The source beam is outside of the hologram, but it's what causes the hologram (the hologram is constructed from it). The universe can be thought of as an internal hologram. The dream exists *within* thought. The Holos exists *within* the Source. Spacetime exists *within* light. Everything takes place within the zero point mental singularity at all times. This is incredibly important to remember as we discuss light speed.

The speed of light is measured to be c = 299,792,458 m/s. How can this be if light is mind and the speed of mind is equal to unity? As you'll notice, light is measured with respect to meters per second, which are spatial-temporal concepts. This is the result you get when you define light speed according to meters and seconds from within the perspective of spacetime instead of with respect to light itself.

What is a second? Who decided how long a second should be? A second is defined as the elapsed time during 9,192,631,770 cycles of the radiation produced by the transition between two levels of the caesium 133 atom. Notice how arbitrary this definition is. This random selection has nothing to do with foundational features of reality. When you try to define light according to such a random unit of measurement, it's no wonder you get the seemingly bizarre result of c = 299,792,458 m/s for its speed.

When light is defined according to spatial-temporal units, mind is being defined according to spacetime, which is utterly backwards. Spacetime is defined by mind/light since it is derivative of mind/light. This is why the term "innerlight" is a helpful reminder to always define it with respect to itself rather than according to a spacetime perspective. “Meters” and “seconds” are completely arbitrary concepts, and it’s inevitable that you’ll get a strange result for light speed when defined according to such random concepts.

The speed of light, in natural units, is the same as the speed of mind, because it *is* mind. It is unity and equality, which is why it cannot be exceeded and why it’s the same in all reference frames. The speed of light “c” is equal to one: $c = 1$. Even physicists are coming exceedingly close to this truth. Wikipedia says, “In branches of physics in which c appears often, such as in relativity, it is common to use systems of natural units of measurement or the geometrized unit system where $c = 1$.” Regarding natural units it says, “In physics, natural units are physical units of measurement based only on universal physical constants. For example, the elementary charge e is a natural unit of electric charge, and the speed of light c is a natural unit of speed.”

Light is mind. Its speed is equal to unity and equality. The speed of light c is the natural unit of speed where $c = 1$.

“This ocean of energy could be thought of as an ocean of light.”

– David Bohm

2.13 - Wavelength

We’ve seen that $c = 1$, but c also equals wavelength times frequency:

$$c = \lambda f$$

The Greek letter λ is called lambda and is used to denote wavelength. The fact that the speed of light equals wavelength times frequency is well understood; there’s nothing controversial about this. We’ve already examined frequency, but now let’s explore wavelength in detail. It’s very simple.

Since:

$c = \lambda f = 1$

Then:

$\lambda f = 1$

Dividing both sides by *f*:

$$\frac{\lambda f}{f} = \frac{1}{f}$$

Simplifying, we see that:

$$\lambda = \frac{1}{f}$$

Wavelength equals one over frequency. When c = 1, we say that wavelength is the **reciprocal** of frequency.

Wavelength is the length of one oscillation of a wave. In physics, this is measured with respect to space, usually in meters. But as always, when dealing with waves of mind, measuring them with respect to space has no meaning. Everything must be put in terms of dimensionless thought, which means that values can only be measured with respect to each other. We've seen this multiple times, but it's always good to reiterate this point as people are so used to considering things only in terms of space and time.

Wavelength in the mental domain is the "length" of one oscillation with respect to one natural revolution. Remember, this is not a physical length. It's determined by the "distance" (mathematical arrangement) of values. Specifically, it's determined by angle according to natural revolutions.

Frequency is determined by angle according to natural revolutions as well, and since wavelength is the reciprocal of frequency (when c = 1)., it make sense that it's determined according to the same concept.

Frequency is the number of times a wave oscillates in 1 natural revolution, and wavelength is the length of one oscillation.

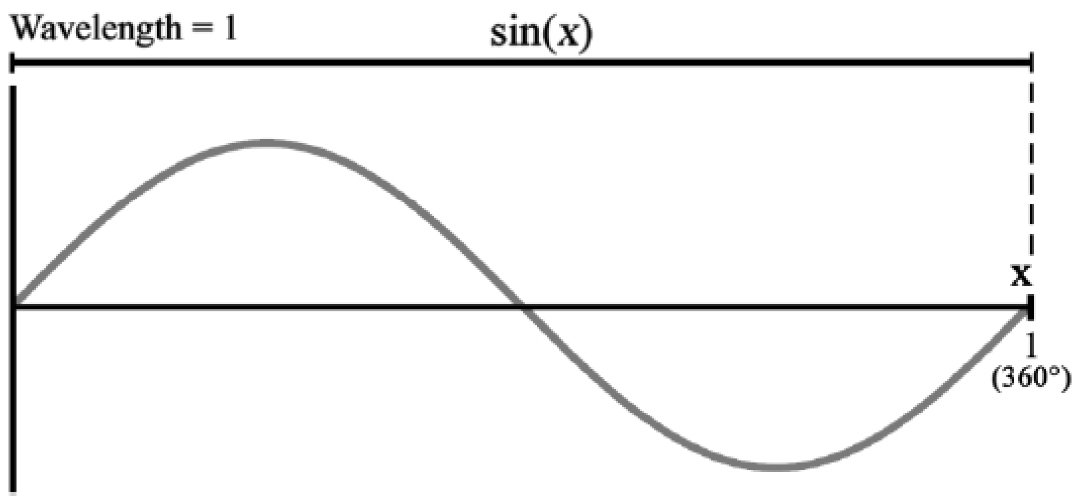
Wavelength

Wavelength is the "length" of one oscillation measured according to angle in natural revolutions. When *c* = 1, wavelength is the reciprocal of frequency:

$$\lambda = \frac{1}{f}$$

We'll examine two examples. To find the wavelength in each example, look at the x-axis and measure the "distance" the wave takes to complete one oscillation.

Let's examine the function sin(*x*):



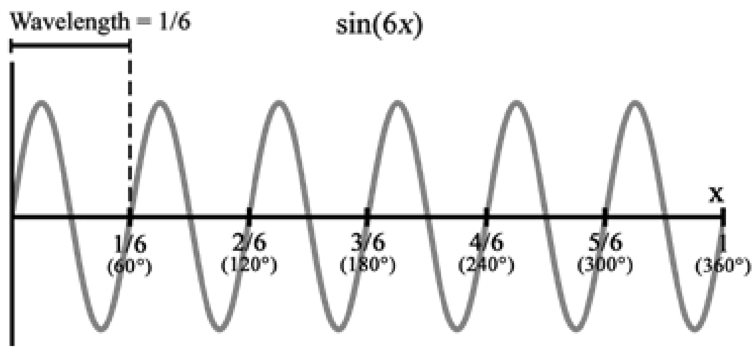
(Fig 2.131) Sin(*x*) completes one oscillation when *x* = 1 natural revolution, so its wavelength is 1.

Sin(*x*) oscillates once in 1 natural revolution, so its frequency is 1. The length of one oscillation is 1 natural revolution, so its wavelength is also 1.

We can also find its wavelength by remembering that $\lambda = \frac{1}{f}$.

$$\text{For } \sin(x):\ f = 1 \text{ and } \lambda = \frac{1}{f} = \frac{1}{1} = 1$$

Now let's examine sin(6*x*):



(Fig 2.132) Sin(6*x*) completes one oscillation when $x = \frac{1}{6}$ natural revolution, so its wavelength is $\frac{1}{6}$.

Sin(6*x*) oscillates six times in 1 natural revolution, so its frequency is 6. The length of one of its oscillations is $\frac{1}{6}$ natural revolution, so its wavelength is $\frac{1}{6}$.

As before, we can find the wavelength of sin(6*x*) by remembering that $\lambda = \frac{1}{f}$

$$\text{For } \sin(6x):\ f = 6 \text{ and } \lambda = \frac{1}{f} = \frac{1}{6}$$

This should make sense intuitively, since if something has 6 times the frequency it must be $\frac{1}{6}$ times the size if you want to fit it in the same "length."

$$\text{For } \sin(x):\ c = \lambda f = \frac{1}{1}1 = 1$$

$$\text{For } \sin(6x):\ c = \lambda f = \frac{1}{6}6 = 1$$

With these two examples, we can clearly see that $c = \lambda f = 1$ holds.

2.14 - Light Equals Mind

Now that we know that light is mind and understand wavelength, let's return to the speed of mind and the Source formula. We saw that the speed of mind is $\lambda f = 1$. We can now explore this in much greater detail. We know that a point travels at speed $c = 1$ and c is also equal to frequency times wavelength: $c = \lambda f$. Furthermore, $\lambda = \frac{1}{f}$.

Recall that there can be no reference to space or time as everything in the mental domain is dimensionless, and frequency is simply the number of oscillations per 1 natural revolution (360°). This dimensionless property, of course, applies to wavelength as well, since wavelength is the reciprocal of frequency (when $c = 1$). Wavelength is the "length" of one oscillation according to one natural revolution.

Let's use what we know to discover an amazing result:

$$\text{Since } c = \lambda f \text{ and } \lambda = \frac{1}{f} \text{ then } \lambda f = \left(\frac{1}{f}\right) f = \frac{f}{f} = 1$$

$$\text{So } c = \lambda f = \frac{f}{f} = 1$$

Recall that the stable frequencies of pure mind are discrete, integer values. The letter n is commonly used to denote that only integer values are allowed. Therefore:

$$c = \lambda f = \frac{f}{f} = \frac{n}{n} = 1$$

This result means that even though speed is always equal to 1, this 1 can be realized through *every* possible frequency and wavelength combination *as long as unity is maintained*. For example:

$$\frac{1}{1} = \frac{2}{2} = \frac{3}{3} = \frac{4}{4} \dots \frac{n}{n} = 1$$

Note that the three dots mean that we keep going and $\frac{n}{n}$ is a way to show that it can be any of these values. $\frac{n}{n}$ allows us to represent every possible value without having to list them.

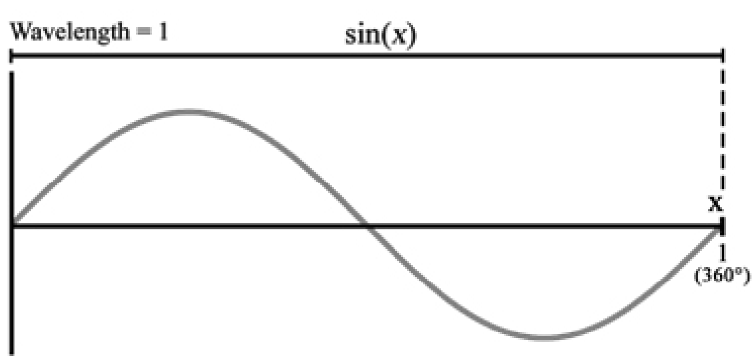
Since $c = \frac{n}{n} = 1$ then, by the PSR, every possible value of *n* must be generated as long as it results in unity. This astounding result means that a point traveling at speed *c* = 1 around the complex circle described by the Source formula automatically generates waves at every possible integer frequency. Notice that, armed with the new concepts we learned, we are now able to rigorously state exactly why a multitude of frequencies are possible.

The Frequency Multiplier

The Source formula, the equation of your mind, generates stable waves at every possible frequency and wavelength that satisfies:

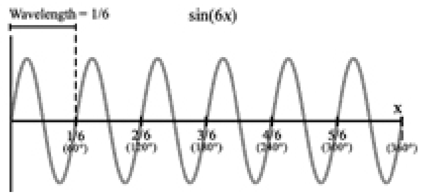
$$c = \lambda f = \frac{n}{n} = 1$$

These frequencies are your thoughts. Once again, consider the following two waves:



(Fig 2.141) Sin(*x*): Speed = *c* = 1; *f* = 1; *λ* = 1

For sin(x), its speed is c = 1. It completes one oscillation per natural revolution; therefore, f = 1. Wavelength is simply $\frac{1}{f}$; therefore, $\lambda = \frac{1}{1} = 1$.



(Fig 2.142) Sin(6x): Speed = c = 1; f = 6; λ = $\lambda = \frac{1}{6}$

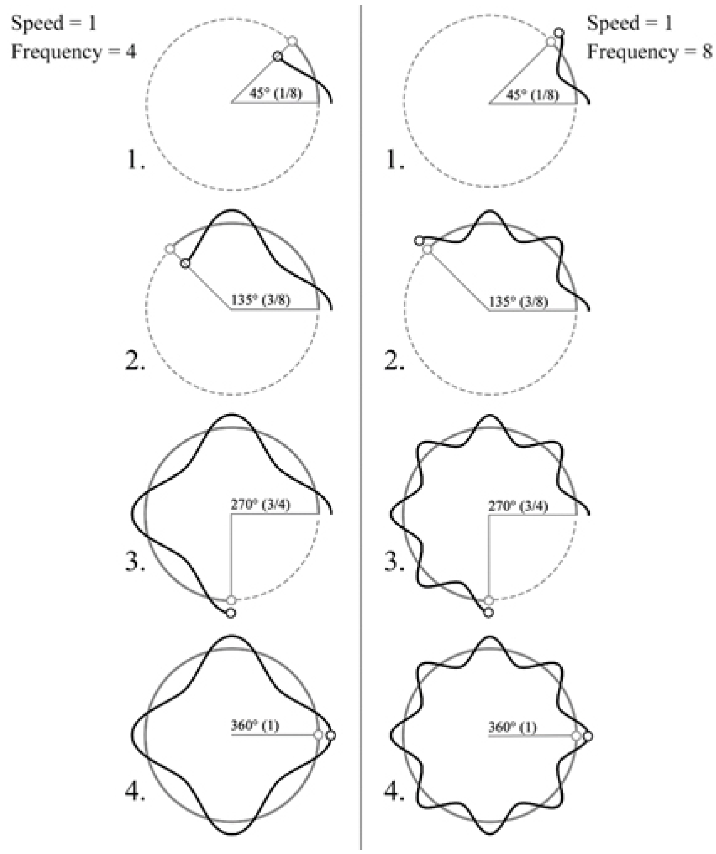
For sin(6x), its speed is c = 1. It completes six oscillations per natural revolution; therefore, f = 6. Wavelength is simply $\frac{1}{f}$; therefore, $\lambda = \frac{1}{6}$.

Putting it all together:

$$\text{The speed of sin}(x)\text{: } c = \lambda f = \frac{1}{1}1 = \frac{1}{1} = 1$$

$$\text{The speed of sin}(6x)\text{: } c = \lambda f = \frac{1}{6}6 = \frac{6}{6} = 1$$

Both waves have the same speed of c = 1, but they have dramatically different frequencies and wavelengths. To get an intuitive understanding of what this means, take a look at Fig 2.143 that depicts a wave with a frequency of four and a wave with a frequency of eight.



(Fig 2.143) Left: The circle depicts speed = 1 while the wave depicts frequency = 4. **Right:** The circle depicts speed = 1 while the wave depicts frequency = 8.

There's a lot going on in this diagram but it's powerful. It's worth taking the time to understand all its subtleties. The circle depicts a point traveling along its trajectory at speed c = 1. As it goes around, the wave that it's associated with is also shown.

Look at the circles side by side. Each "side by side" is a snapshot of their sequence showing their progress in relation to each other. Notice that everything is traveling at the exact same speed. For every sequential moment, the point on the circle on the left and the point on the circle on the right are exactly matched. One never overtakes the other. The waves associated with the circles also have speed c = 1. Regardless of their frequency, the points traveling along their wave trajectories never pass each other.

Even though everything is moving exactly at the same speed, they have drastically different frequencies and wavelengths. Nonetheless, their speeds are the same. You can clearly see that as the wave with $f = 4$ completes 1 natural revolution, the wave with $f = 8$ also completes 1 natural revolution. The wave on the left oscillates four times in 1 natural revolution, while the wave on the right oscillates 8 times in 1 natural revolution.

This important diagram contains a visual representation of our major concepts such as natural revolution, frequency, wavelength, speed, angle, and how they all relate.

These are remarkable, elegant, beautiful results. All mental waves travel at the exact same speed, but they can have dramatically different frequencies. Higher frequency means higher energy.

> **Do Not Confuse Frequency with Speed**
>
> Always remember that **frequency** and **speed** are different concepts. Frequency = f and speed = λf. Mathematical animations depicting a point traveling around a circle generating waves are depicting its frequency = f not its speed = λf.

Light speed is the necessary and sufficient condition for determining if something is in the dimensionless mental domain of the Source or the dimensional spacetime domain of the Holos. Anything traveling at the speed of light is maximally length contracted and maximally time dilated. It's unextended with respect to spacetime. Any wave traveling at the speed of light is necessarily a mental wave. It's mind and nets to zero. A wave traveling at less than light speed is a spacetime Holos wave. It does not net to zero and exists within spacetime.

The Light Speed Condition

Any wave traveling at speed $c = 1$ is a mental wave.

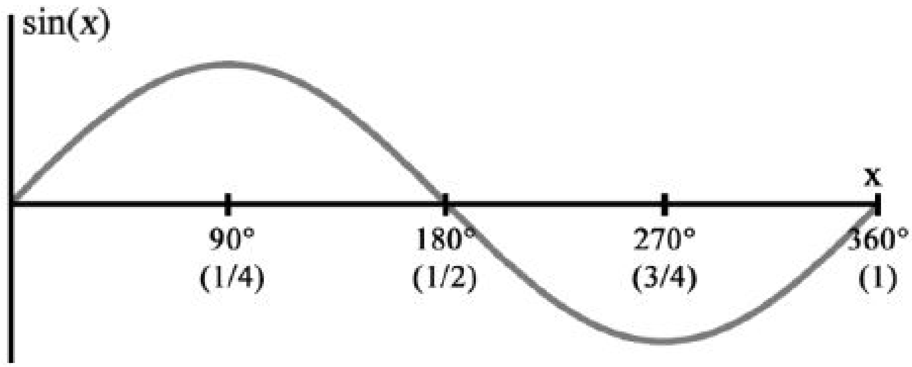
Any wave traveling at speed $< c$ is a spacetime Holos wave.

2.15 - Phase Shift

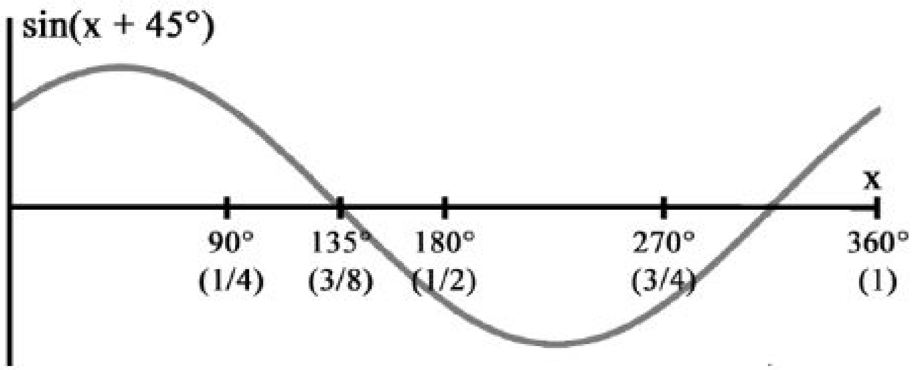
Phase shift is the final concept we must examine to understand how the mental domain of the Source can result in the collective dream of the spacetime Holos. Recall the generalized Source formula:

$$Ae^{i(fx + \phi)} = A\cos\,(fx + \phi) + iA\,\sin\,(fx + \phi)$$

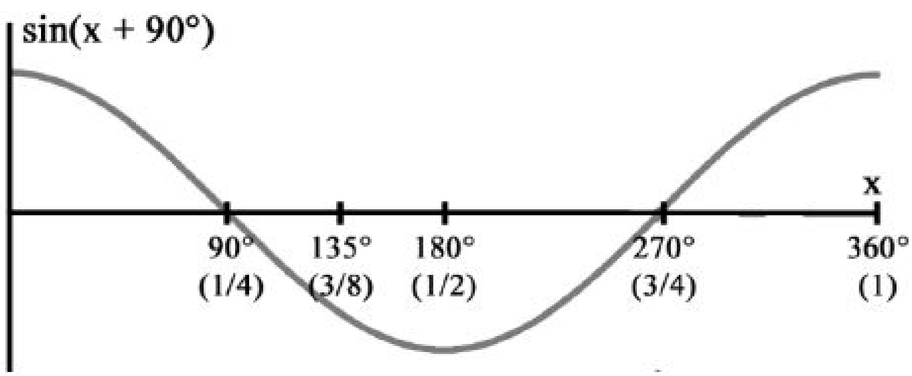
Phase is represented by ϕ. Understanding phase shift is easy. All it does is slide a wave over. In this case, it slides it to the left.



Sin(*x*): Phase = 0. Notice that it crosses the *x*-axis at 180°.



Sin(*x* + 45°): Phase = 45°. Notice it now crosses the *x*-axis at 135°. It has shifted 45° to the left.



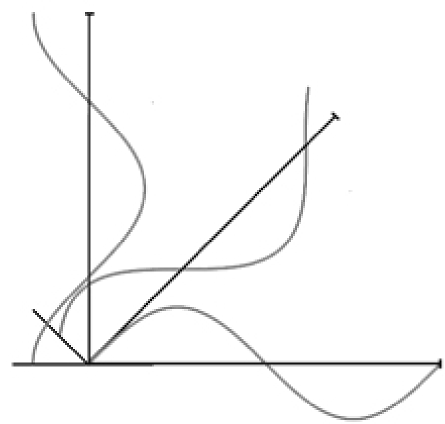
Sin(*x* + 90°): Phase = 90°. Notice it now crosses the *x*-axis at 90°. It has shifted an additional 45° to the left.

(Fig 2.151) Phase shifts the wave across the *x*-axis.

Note that sin(*x* + 90°) has the same shape as a cosine wave. That's because a cosine wave is simply a sine wave that has been phase

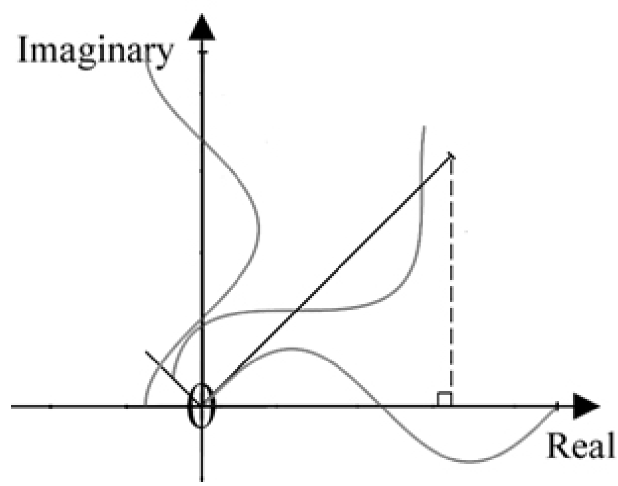
shifted by 90°:
$\text{Sin}(x + 90°) = \cos(x)$.

Phase shift is simple. There's one more way that can be helpful to visualize phase shift for our purposes, and that's by rotating the wave:



(Fig 2.152) Three waves have been rotated by 0°, 45°, and 90°.

To better understand a few concepts that will soon become necessary, let's imagine that we place these rotations on a complex plane:



(Fig 2.153) The waves in Fig 2.152 gave been placed on the complex plane.

Notice the wave with phase = 45° lies in between the real and imaginary axes. If we draw a vertical line from this wave, it forms a right triangle with the axis. If a wave can form a right triangle, we say it has a **projection**.

The waves at 0° and 90° lie on the axes themselves, and they do not form a right triangle. For example, if we attempt to draw a vertical line from the wave at 90°, the line will overlap the axis itself and will not form a triangle. If a wave cannot form a right triangle, we say it has no projection. This will become important in a moment.

A wave that's phase shifted by 90°, 180°, 270°, or 360° will lie directly on one of the axes and will not form a right triangle. Notice that these angles are all integer multiples of 90°. We say that waves shifted in this way, (or waves that have 0 phase), have **orthogonal phase**. Any wave that lies on one of the axes has orthogonal phase. We can also call these waves **axial waves** since they lie on the axes. Axial waves have no projection (they do not form a right triangle).

Waves shifted by any other value will not lie on any axis. Such waves will lie in between the axes and will always form a right triangle. We say that these waves have **non-orthogonal phase**. We can also call these waves **interaxial waves** since they always lie between the axes. Interaxial waves will always have a projection (they always form a right triangle).

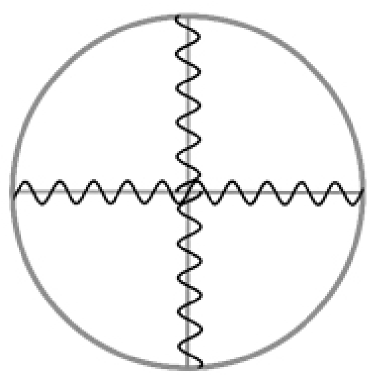
Sinusoidal waves traveling at light speed with orthogonal phase are pure mind. Sinusoidal waves traveling at light speed with non-orthogonal phase are also mind, by the light speed condition. But, as we'll see in the next section, they aren't "pure." The interaxial waves contain spacetime information and are what give rise to the spacetime Holos. In an existence where the phase of every mental wave is in an orthogonal state, there is no Holos domain of spacetime. It's a domain of pure mind where the collective dream hasn't begun.

2.16 - The Big Bang

We now have all the concepts necessary to begin understanding the Big Bang and how mind generates so-called “matter.” This is how, mathematically speaking, the minds of the Source generate the collective dreamworld of the Holos.

We’ll use a complex circle to model what we’re about to explore, albeit in a simplistic way.[9] We’re going to place waves on top of it, just like we did with the complex plane in the previous section. The angle of the radii of the circle will model how much a wave is rotated (phase shifted) We already did this with the complex plane so there’s nothing new here yet.

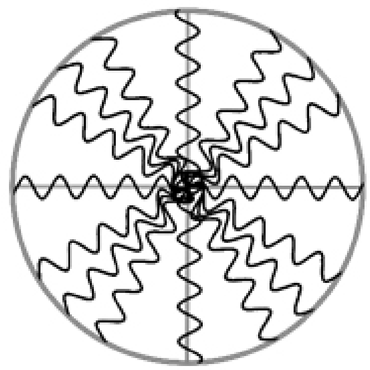
We must do one more thing. We’re going to use the length of the radius to model speed. Each radius will represent the speed of the wave associated with it. Since mental waves travel at light speed, the length of the radius will be $c = 1$.



(Fig 2.161) The pre-Big Bang state: Every wave runs along the axes and has orthogonal phase relations. This is the pure mental domain of the Source.

In a state of purely orthogonal phase relations, every wave will run along the axes. These axial waves are the necessary and eternal waves of pure mind. In this state, the spacetime dream of the Holos does not exist. There is only the mental domain of the Source singularity. Pure sine and cosine waves running along the axes have no projection and are pure and independent from any other wave.

By the PSR, there's no sufficient reason to prevent other phase relations. In terms of the circle we're using to model phase, this means there's nothing to prevent waves from running along the radii between the axes and populating the interaxial regions.



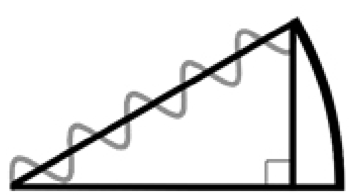
(Fig 2.162) The Big Bang: Waves run along the axes and in between. This results in non-orthogonal phase. The interaxial waves generate the dreamworld of the Holos.

These interaxial sinusoids are mental waves, but they aren't "pure." They have a projection, and their projections are a combination of partial sines and cosines.

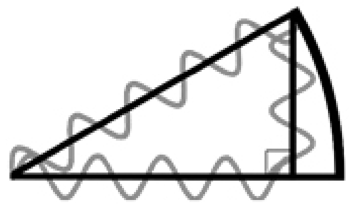
A phase-shifted wave is composed of two partial waves, one cosine and one sine:

$$A \sin(x + \phi) = a \cos(x) + b \sin(x)$$

Let's take a slice of our circle and have a closer look at one of the shifted waves:



(Fig 2.163) The radius associated with the interaxial wave forms a right triangle. The axial waves form no such triangle. Only phase-shifted sinusoids have this property.

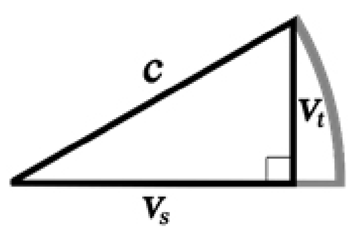


(Fig 2.164) A phase-shifted interaxial sinusoid projects a partial sine and cosine wave. Axial waves have no such projection.

Since we have a right triangle, keep in mind the **Pythagorean theorem**:

$$a^2 + b^2 = c^2$$

Recall that we're using the length of the radius to model the speed of each associated wave. This radius becomes the hypotenuse (the longest side) of our right triangle. Let's take a closer look at this right triangle:



(Fig 2.165) The Pythagorean theorem applies to the velocities of the interaxial waves and their projections.

Since this wave has a speed of $c = 1$ and we're using the radius to model wave speed, the radius associated with that wave will have a length of c. This forms a right triangle. We label the other two sides of the triangle v_s and v_t. We could have used any two letters to label the sides, such as a and b, but we choose v as a reminder that they represent velocities. The small s and t next to their bases are called subscripts. Since we're using the same letter v to denote two different quantities, these subscripts are just markers to tell us the velocities are different. We could have chosen any two letters (or numbers, such as 1 and 2) to be our subscripts. Why we chose *s* and *t* will become apparent soon.

The phase-shifted wave is a mental wave and has speed $c = 1$, but it's mathematically equivalent to two slower moving waves.

By the Pythagorean theorem, the velocities of those waves are:

$$v_s^2 + v_t^2 = c^2$$

where $c^2 = 1^2 = 1$.

Any wave moving at less than light speed, considered on its own, no longer results in net zero and thus is dimensional. As was defined earlier, real numbers are related to space, and imaginary numbers are related to time. Since the horizontal side of the triangle is related to the real axis, and the vertical side is related to the imaginary axis, v_s is a speed through space, and v_t is a speed through time. We chose the subscripts *s* and *t* as a reminder that one is a velocity through space and the other is a velocity through time. This result means that the speed through space squared plus the speed through time squared is equal to light speed squared.

The Mind Square Law

The speed through space squared plus the speed through time squared equals the speed of mind squared.

$$v_s^2 + v_t^2 = c^2$$

Where $c = 1$ = the speed of light = the speed of mind.

The phase-shifted interaxial sinusoid itself is a dimensionless mental wave, but it projects a partial cosine and sine wave which are dimensional space and time waves.

When mental waves become phase shifted such that their phase is no longer orthogonal, spacetime forms. Phase-shifted, non-orthogonal waves are still mental waves that travel at light speed, but they depend on and project a combination of slower-moving sines and cosines. Waves that have orthogonal phase have no such property. They're pure and independent from any other wave. The sines and cosines of a reduced velocity that comprise the non-orthogonal waves do not net to zero when considered on their own and are what form the shared dream of spacetime.[10]

Ontologically, the orthogonal state depicted in Fig 2.161 is a state of pure mind (the Source), while the non-orthogonal state depicted in Fig 2.162 results in the inner projection of a spacetime dreamworld (the Holos). In the case of one monad, this is the projection of a personal dreamworld. In the case where every monad in existence is involved, it's the Big Bang and the beginning of our collective dreamworld. It's important to note that the interaxial waves are still mental waves and travel at light speed, but their partial projections are space and time waves. In other words, they're mental waves that contain the mathematical "coding" for space and time. As has been repeatedly stated, everything is derived from mind, and now you can see exactly how that is the case. We can now easily see why nothing "material" can ever be accelerated faster than light speed. This is a question the old paradigm cannot answer.

Axial waves are associated with purely real or imaginary numbers since they lie on the axes. Interaxial waves are fusions of both real and imaginary numbers since they lie in between the axes, hence they are associated with complex numbers. Axial waves are pure and independent of any other wave. Interaxial waves are fusions of partial waves that result in space and time.

Phase-shifted sinusoids are generated by the PSR. At the moment prior to the Big Bang, there was perfect symmetry, and every wave was an orthogonal, axial wave. Since axial waves don't project the slower-moving waves that comprise spacetime, this means that it was a domain of pure mind, and the shared dream had not yet begun. The Source existed, but the Holos did not. The Source is necessary. The Holos is contingent.

The Big Bang occurred when sinusoidal waves of the Source singularity went from “in phase” to “out of phase.” It was the moment non-orthogonal phase relations were generated (interaxial waves), which resulted in the mathematical projection of the spacetime Holos. This was an inner, mental projection. This was the moment we began to collectivity dream this shared dream. The phase-shifted mental sinusoids of the Source singularity resulted in the inner projection of the spacetime Holos.

The Big Bang, mathematically speaking, was a massive phase transition resulting in an internal cosmic projection.

Once again, consider a hologram. If you imagine the Source as a beam of pure light, the Big Bang would be the moment that a holographic film was placed in front of it resulting in a holographic projection. The difference here is that there’s no holographic film needed. The necessary mathematics are built into the Source’s internal phase information. Furthermore, the resulting hologram isn’t external. You can think of the Holos as an *internal* hologram, which is a dream.

The Big Bang occurred when phase-shifted sinusoids populated the interaxial regions. Every one of these shifted waves (which are still mental waves) project slow-moving, partial waves that result in space and time.

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[1] By uncaused we mean having no *external* cause.

[2] There are more reasons, dealing with mathematical completeness, for why such a shape violates the PSR, but this is the simplest example.

[3] Radians are also commonly used to describe angle.

[4] There are a few things to note about this section. First, to keep things simple, we are not considering imaginary frequency. Second, in this book we are examining only stable frequencies. In an upcoming section, we will see $\lambda f = 1$. Both integer and non-integer frequencies are solutions to this equation. Examining non-integer frequencies is well beyond the scope of this introductory book, so we will be considering only the stable, integer solutions.

[5] If you're familiar with the Fourier series, note that it has been simplified for ease of explanation for the purpose of focusing on conceptual points rather than being weighed down by technical details. The simplified version considers only functions with a period of 360° (1 natural revolution).

[6] To be precise, we're adding the left side of the Source formula, which is known as a complex exponential, but we'll continue to refer to it as the Source formula for ease of communication and because the left side implies the right side.

[7] This picture demonstrates Fourier mathematics, but it's very simplistic and only meant to give you an intuitive understanding. It's missing a great deal of ontological detail, which we'll explore in an upcoming section. Note that the left side shows only the time domain but the Holos is the domain of *both* space and time.

[8] Frequency is *not* speed. These concepts are often confused. If you watch an animation of a point traveling around a circle while generating waves, this is almost always a depiction of its frequency, not its speed.

[9] The waves in the figures in this section are not to scale, and their positions on the radii are not exact. The figures are simply a way to intuitively understand the concepts being discussed.

[10] To fully examine the ontological implications of the equation $A \sin(x + \phi) = a \cos(x) + b \sin(x)$ requires knowledge of calculus, complex analysis, and additional philosophical concepts that are beyond the scope of this introductory book. The coefficients, *a* and *b*, are not simple scalars and cannot be treated in a trivial way. They contain rich ontological information that determines the properties of the waves.

Partial waves do not net to zero. Notice in Fig. 2.164 they are constrained to a length less than 1. We do not consider these waves in their entirety as they are prevented from completing their cycles due to their velocities being less than light speed. These partial waves have very different intervals from the full interaxial waves they comprise. We do not consider them over 1 natural revolution. Since we are only considering partial information of a full sinusoid, this leads to non-net-zero results, which in turn leads to the illusion of an extended universe. It's only when the two partial waves are treated together in their entirety that they are equivalent to a full mental wave traveling at light speed, and it is the full mental wave that nets to perfect zero across 1 natural revolution.

SECTION 3: IMPLICATIONS

Having explored the mathematical structure of existence from nothing at all to the moment of the Big Bang and the beginning of the Holos, we'll now begin to explore its implications, what this means in terms of existence, and how it can answer the biggest questions in science, such as the hard problem of consciousness and the apparent paradoxes that arise from quantum mechanics and special relativity.

Scientific materialism and empiricism are completely powerless to explore the state that preceded the Big Bang. Ontological mathematics provides the answer: The Big Bang was the moment we began to dream the collective dream, and what preceded it was a domain of pure mind.

3.1 - THE COLLECTIVE DREAM

Monads are unique containers of waves at every possible frequency. They are minds that contain thoughts. Each monad contains a vast multiplicity of non-orthogonal waves that project partial waves that result in space and time. The waves of these minds all interact with each other. There is no such thing as matter. Matter is simply the result of the thought patterns of myriad minds combining to form a stable dream reality. When a mind tunes into its own personal frequencies (thoughts), it experiences an unstable personal dream that changes according to its unconscious content, emotions, memories, and conscious control in the case of a lucid dream. Since the personal dreamworld is built from the monad's own contents, it changes according to the thoughts of that monad.

When a monad is tuned in to the collective projections of all monads, which is exactly what the bodily senses have evolved to do, it experiences the stable collective dreamworld of the spacetime Holos. Since the collective dream is built from the frequencies of every mind in existence, no one mind can control it. This is why it's stable.

When the partial spacetime waves of every monad interact with each other, their configurations stabilize and form what materialists refer to as “matter” and the various particles of the standard model of physics. There is no “matter.” All is mind. All is thought. The universe is our collective dream.

You live in a reality of living mathematical frequency patterns. It’s a reality of thoughts. A reality of innerlight. “Matter” is light whose information has been “fractured” into component waves of a lowered velocity that combine to form stable wave structures. It doesn’t seem that way because your physical senses didn’t evolve to show you the truth of reality but to allow you to survive on this planet. Your body is simply a stable wave structure itself. It’s an **avatar** developed through evolution to select for the mathematical information projected by innerlight that will be useful to your survival in the Holos.

Take your eyes, for example. Look at all the colors around you. When you’re in a dark room, there is no color because there is no visible light. The only difference between one color and another, red and violet for example, is the frequency of the light. Frequency is the number of times a wave oscillates. Frequency is nothing but information. It’s a number. The only thing that differentiates red from violet, and blue from green, is its frequency, its number. An apple looks red because light strikes the apple and the apple absorbs every frequency of visible light except for red. Since the red light is not absorbed, that light strikes your eye, and you see the apple as red. But remember, red is just a frequency, a number, information. But you experience that number as red. Take a minute to look at all the colors around you. They’re all simply different frequencies, different numbers. This is a living mathematical dream of innerlight.

“All is number… There is geometry in the humming of the strings, there is music in the spacing of the spheres.”

– Pythagoras, Philosopher and Mathematician

Everything is made of waves and frequency. Color is frequency. Sound and music are frequency. Even matter is transformed frequency. There is nothing that is not made of waves and frequency.

Quantum mechanics reveals the wave nature of reality. It's now time to abandon the idea of matter altogether. There is no longer an excuse to hold on to this outdated idea. Wikipedia says, "All matter can exhibit wave-like behavior. For example, a beam of electrons can be diffracted just like a beam of light or a water wave. The concept that matter behaves like a wave was proposed by Louis de Broglie in 1924."

Do you see what we have done? All of reality is based on waves, and we have shown, via Fourier mathematics, how we can derive any physically realizable waveform from the combinations of the interaxial projections of the Source formula. This is it. This is the end of the road. The answer to existence is here. We have everything we need to construct the universe. As Fourier mathematics shows, through the combination of sinusoids projected from the interaxial waves of the Source, we can create any physically realizable wave structure no matter how complicated. There's no such thing as matter. "Material" particles are simply stable configurations of sinusoidal waves originating from the monadic collective. It's a mental projection from a cosmic mind. It's a shared dream. The cosmic mind is the Source. It's the collection of all monads and their frequencies. It's all of us together. It doesn't exist apart from us. It is what we are when taken together.

The Source can exist without the Holos, but the Holos *cannot* exist without the Source. Thought can exist without a dream, but a dream cannot exist without thought. Pure thought is necessary. Dreams are contingent. The mathematical laws of mind are always the same, while dreams are forever changing. Every new dream can be wildly different from the last.

The Beginning of the Universal Dream

The axial waves within the minds in the Source give rise to phase-shifted, interaxial mental waves. This is mandated by the PSR.

The interaxial mental waves project partial waves of space and time that result in the inner dreamworld of the Holos.

The projected waves, originating from myriad monads, combine to create complex spacetime wave structures (by Fourier mathematics).

Stable wave configurations form the particles of the standard model, all so-called “matter,” bodily avatars, and all constituents of the universe.

Ontological mathematics has no problem accounting for what preceded the Big Bang and how the universe began. The state prior to the Big Bang was a purely mental singularity, a net-zero point, that contained myriad sinusoids all with orthogonal phase. When sinusoids of the Source became phase shifted, as mandated by the PSR, this resulted in the spacetime projection of the Holos. This was the Big Bang. In terms of Hyperianism, it was the moment we began to dream this collective dream.

3.2 - The Human Senses

A phase-shifted wave, traveling at light speed is a mental wave that nets to zero, but it’s composed of two linked partial waves that when considered on their own travel at less than light speed and do not net to zero, which results in space and time. The entirety of the spacetime Holos is composed of these partial waves. These are the waves that combine to form intricate and stable wave structures that we perceive as “material” particles. For simplicity’s sake, consider just one interaxial mental wave and its two partial component waves. By tuning in to the component waves via the senses of our bodily avatar, they are interpreted as spatial and temporal signals. If we tuned in to the combined result of both waves, this nets to zero, and we wouldn’t detect spacetime at all.

> **Spatial and Temporal Waves?**
>
> There’s been a lot of discussion about interaxial waves and their partial projected waves. We’ve referred to the projected component waves as though they were separate waves for ease of communication; however, this is not entirely accurate.
>
> The two component waves, rather than being waves in their own right, are partial mathematical perspectives of the mental interaxial wave. In other words, they denote how much, or what portion, of the mental wave we consider. Since only portions of the mental wave are considered, this partial view no longer nets to zero and results in mathematical information interpreted by the brain. These “fractured” viewpoints of mental waves (denoted by their projected component waves) are what give rise to the mathematical illusion of an “extended” universe.

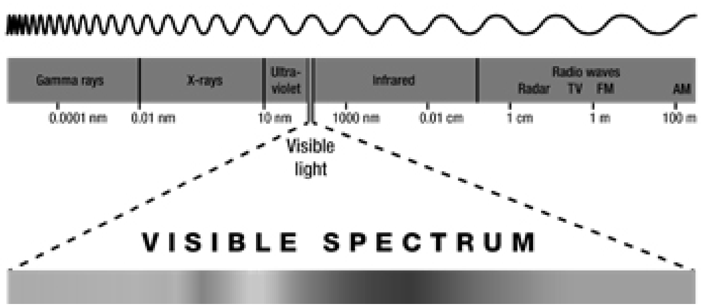
The human senses are configured by evolution to decompose the myriad mental waves into their partial spacetime component waves and select only for the information that’s useful for survival and power acquisition. The senses of your bodily avatar are what tune in to the space and time information of the Holos. Reality is ultimately zero. It’s nothingness. But we can mathematically break it down into fractional pieces and experience immensely intricate dreamworlds. We experience the illusion of the spacetime Holos precisely because we’re able to tune in to the projected partial waves (and their various configurations) separately rather than their joined result. A phase-shifted mental wave is a fusion of two partial component waves, one carrying spatial information and the other temporal information. If we experienced the joined result of both waves, in their entirety, it would net to nothingness.

We experience space differently from time because our senses are constantly filtering and selecting for only partial information. We taste food, hear sound, and see color because our senses are selecting for filtered, fractured, partial information. Under the influence of a psychedelic substance, our tuning process becomes altered leading to results such as hearing color, seeing sound, experiencing time in new ways, and many other novel sensations. When we experience the feeling of oneness and the unity of all things, and what some have called the clear white light, that’s when we’re selecting for the joined result rather than the partial component waves. We’re tuning in to the interconnected mental domain of the Source rather than the fractured Holos.

A prism is a useful analogy. We’ve all seen white light interact with a prism and refract into a rainbow. You can experience the clear white light, or you can experience the colors of the rainbow. Similarly, you can experience the unitary Source, or you can experience the fractured multiplicity of the Holos.

As another analogy, consider a black and white drawing on a piece of paper. The dark areas are what give it definition. Remove the dark areas, and you would simply have a blank piece of paper. This analogy extends to the spacetime Holos. The pieces of information that we omit and filter out, are like the dark areas in the drawing. The information we omit generates our experience of a world.

Consider the light spectrum:



(Fig 3.21) Your senses can only visually detect a small fraction of the electromagnetic waves all around you.

You can see only a miniscule portion of the light spectrum, the portion of visible light. There are gamma rays, x-rays, infrared waves, and radio waves all around you right now that you can’t visually detect.

Your senses are constantly selecting for information that evolution found useful while filtering out the rest. If you could detect the world of waves all around you, you would experience reality 100% differently.

3.3 - Form and Content: Living Mathematics

This is a good time to clarify the difference between form and content, outer and inner. Since we've been exploring the rational structure of how existence is possible, someone might get the idea that reality is mechanical and lifeless. On the contrary. Existence is a living mental organism. Ontological mathematics is completely different from abstract mathematics. Ontological mathematics is alive. We've seen that a point will flow in the trajectory of a circle. But it is *will,* the unconscious striving for symmetry, that drives it to do so. Its mathematical form gives it the parameters to act out its will.

Arthur Schopenhauer saw reality as a cosmic will. Friedrich Nietzsche saw existence as will to power. Indeed, existence is a cosmic will, made up of a myriad of individual wills (monads), whose striving for symmetry equates to power. As living, experiencing subjects, every mind is striving to become more expressed, more powerful, more evolved, and more **lucid** within this collective dream.

We've explored the concept of a sinusoidal wave but only its formal aspect. It also has an inner aspect. Every sinusoidal wave is also a fundamental **quale**. It's a fundamental unit of experience.

Qualia

Qualia are instances of subjective experience.

Examples include: The *taste* of wine. The *pain* of a laceration. The *redness* of an apple. The *feeling* of love. These are qualia.

We can add sinusoids together to create complex waves, but when we do so, we're also adding together basic qualia to produce complex qualia. Through the addition of sinusoids, we can create the taste of a strawberry. Every formal aspect of existence also has content that can be subjectively experienced by a mind. This is known as a **dual-aspect monism** (not to be confused with dualism). Every wave has both an inner and outer aspect. You are a living mathematical mind, and your subjective experience is what it's like to be a mathematical being experiencing a mathematical dreamworld.

Mathematical form is what provides the parameters and playing field for us as living minds, but it's our internal subjectivity and will, that drives existence. In order to get up and retrieve an apple from my table, I require the mathematical form to allow this action to take place. But it also requires my internal will. Both are necessary. Without the form, there would be no rules or playing field. Without the will, there would be no motion. Remove the form, and ontological mathematics falls apart. Remove the internal aspect, and it also falls apart. You can't separate the inner from the outer because they're two sides of the same coin. They're two aspect of the same substance.

Ontological mathematics is *alive*. It's *mind.* Always remember that. This can be difficult for many people to understand because they're used to conceptualizing mathematics as abstract concepts, symbols, and marks on paper. *That is not ontological mathematics.* Because this is so difficult for people to grasp, it can be helpful to think of existence as living energy that mathematics perfectly describes. It's fine if you need to understand it in that way at first, but ultimately that's not correct. Existence *is* living mathematics.

Why do we have an internal, subjective aspect? Why do we have a will? Because it's *necessary* for existence. It's necessary for ontological mathematics. Existence, and thus ontological mathematics, wouldn't be possible without it. Existence wouldn't randomly generate subjective experience for no reason at all. This would be a violation of Occam's razor and the PSR. Subjectivity is necessary for ontological mathematics to work. Without subjectivity and will, the point would never begin to travel in its circular trajectory just like I'd never get up to retrieve the apple. Just as a point has an internal trajectory to flow around a circle, you have an internal trajectory to do all your daily activities.

Imagine individuals walking around a busy mall. Their trajectories are their internal mental patterns, and these trajectories are all interfering with each other and the environment. One person may change their trajectory to enter a store that's displaying a compelling advertisement, while another moves towards someone they find attractive, as another moves away to avoid a threat. This extends to all aspects of existence. Imagine electrons moving around whose trajectories are constantly changing as their wavefunctions interfere with each other and their environment. This is because they have an inner aspect. This does *not* mean they're conscious. They're *not* self-aware. They're inner aspect is extremely basic. Nonetheless, every wave is a component of a monad which is a living mind. Particles without an internal aspect would never exhibit motion just like people without minds in a shopping mall would never move. Keep in mind this is a loose analogy. A monad is directly linked to a human body. A particle does not have a monad linked to it. Rather, a particle is composed of waveforms that originate from various monads.

We'll say it again: Existence is *living* mathematics. You are a mind, a subjective center of experience, and your sinusoidal waves are your personal qualia. You *are* your frequencies. Each sinusoid associated with you as a monad is a piece of you. It's one of your thoughts. It's infinitely more you than your bodily avatar.

Form is the mathematical structure, and content is how you personally, subjectively experience that mathematical structure. You can't experience the form in itself as form. You can only experience the form as content. But you can know the form through pure reason.

Let's use color as an example. Colors are nothing but frequencies of light. You experience an electromagnetic wave at 450 Terahertz as red. In this case, "450 Terahertz" and the rest of the structure of the wave comprise its form, while “redness” is your experience of it as content. You can experience redness as content, but you can never experience 450 Terahertz as form. But you can know it. Form and content are two different ways of looking at the same information.

Let’s use the example of a blind person and a baby to get a better understanding. A blind person could never experience redness, but they could rationally understand the concept of electromagnetic wave oscillation at 450 Terahertz. In the case of the blind person, they can’t experience the content, but they can know the form. A baby can experience redness, but it can’t know the concept of wave oscillation until it gets older. In the case of the baby, it can experience the content, but it can’t know the form.

Pure form is something we can never experience directly, in itself. We can only experience its qualitative aspect: the content. However, we can know form through the use of pure reason. Through ontological mathematics, all formal reality becomes accessible to us through pure reason.

The Source singularity and the spacetime Holos both have form and content. The form of the Holos is the structure of the complex waveforms from which it’s composed, and you experience its content as your sensory experience of this collective dream. The form of the Source is the structure of its simple sine and cosine frequencies. You can experience its content through altered states of consciousness as the interconnected unity of existence.

“Science cannot tell us a word about why music delights us, of why and how an old song can move us to tears.”

– Erwin Schrödinger

As Schrödinger points out in the above quote, materialist science can’t tell us a thing about how we’re able to be subjectively moved by music, but ontological mathematics certainly can. The following quote is by rationalist, idealist, and inventor of calculus, Gottfried Leibniz, who believed that reality was not physical, but composed of monadic minds:

"Music is the pleasure the human mind experiences from counting without being aware that it is counting."

– Gottfried Leibniz, Philosopher and Mathematician

This is an existence of both form and content, outer and inner. The formal structure makes movement possible, but we'd never move without our inner aspect. The point would never begin to flow in its circular pattern. If you had no will, no desire, no inner aspect of any kind whatsoever, you'd never get up out of bed. The inner aspect of reality is necessary by the PSR. There's nothing to prevent it, and it must exist. The inner aspect is what gives ontological mathematics life and makes existence possible at all.

3.4 - The Hard Problem of Consciousness

"Consciousness poses the most baffling problems in the science of the mind. There is nothing that we know more intimately than conscious experience, but there is nothing that is harder to explain."

– David Chalmers, Philosopher of Mind and Cognitive Scientist

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For ontological mathematics, the hard problem of consciousness is not a problem at all, let alone a hard one. Since the new paradigm is an idealism and not a materialism, this means that everything is mind.

Sinusoidal waveforms are thoughts. Therefore, we've already accounted for mind since it is fundamental to our system. As we've seen, the PSR demands it. What we need to do is show how so-called "matter" arises from mind, which we have done in detail. "Matter" (which is not material at all, as classically defined) is a direct consequence of Fourier mathematics and arose as an inner mental projection at the moment of the Big Bang, as explained.

A semantic problem that arises is that many people define consciousness differently. For example, many people would consider consciousness to simply be an entity's perceptual field of experience. This is not consciousness but **sentience**. When many people refer to consciousness, they're actually referring to sentience. Sentience is the ability to experience qualia. As defined earlier, "qualia" is a term used to describe basic instances of subjective experience. This would include experiences like the taste of a strawberry, the feeling of love, and the prick of a needle. Notice that none of these experiences require self-awareness. Sentience doesn't necessarily imply self-awareness. It simply means the ability to have subjective experience. A dog is sentient since it can experience the taste of food, but it hasn't developed the concept of "I."

Our system of mind accounts for sentience easily since ultimately mind is all there is. What about consciousness? Ontological mathematics defines consciousness as explicit self-awareness and the ability to form and manipulate concepts through that awareness. This requires the development of an ego as the center of consciousness. The ego itself, which is the "I," is a concept and the center of an entity's conceptual field of consciousness. Our system easily accounts for this as well, since consciousness is not different in kind from basic mind and sentience. It only differs in degree. Consciousness is a complex form of mind. It's mind reflecting on itself.

There is no problem of consciousness for ontological mathematics. Consciousness is a structure of mind that arises from mind. A mental structure arising from mind is not an issue. A mental structure arising from a non-mental substance, namely matter, is a huge issue. It's a logical impossibility, which is why materialism is unable to answer their "hard problem." Under this definition of consciousness, Schrödinger and Planck were right to say that matter isn't fundamental, but wrong to say that it's derivative of consciousness. Consciousness doesn't arise until it's developed through the process of evolution. So-called "matter" is derivative of mind, not consciousness. Mind is more fundamental than consciousness.

3.5 - Cosmic Cycles

This book studies the mathematical form of existence, rather than its content. This is important to keep in mind as every form we discuss has incredible implications when it comes to the experience of that form's content. For example, we've discussed how the generation of phase-shifted waves is a mathematical consequence of the PSR, which results in the Big Bang. But from the experiential perspective, this amounts to a decision made by every mind in the Source acting together as one cosmic mind to "shatter itself," throw itself into unconsciousness, and begin a dream reality. What follows is an unconscious, mental process of mathematical interaction and unfoldment that results in the spacetime Holos.

Existence is eternal and cyclical. Thus, reality goes through **cosmic cycles**. Mathematically, this amounts to the process of existence reaching a state of perfect orthogonality (the pure Source without the Holos) and then entering a state of non-orthogonality when phase-shifted waves are generated (resulting in the inner projection of the Holos). Existence goes from orthogonality to non-orthogonality to orthogonality, and so on, in an eternal process. Since the only thing that exists are monads and their frequency contents, this means that they are going through this orthogonal process individually as well.

For individual minds, attaining higher degrees of orthogonality translates to self-optimization. They begin in an unconscious state and seek higher degrees of consciousness and lucidity. This results in gaining more power and control over themselves and their environment. (This is why Hyperianism stresses the importance of actualizing the Higher Self.) For those familiar with the philosophy of Friedrich Nietzsche, striving for higher degrees of orthogonality and mathematical symmetry is the ontological basis of the **Will to Power**.

For reality as a whole, the process of attaining higher degrees of orthogonality translates to the optimization of the collective of all minds. In the dreamworld of the Holos, this is the creation of better and better societies that allow each mind (citizen of the universe) to optimize and express themselves without irrationally impeding the optimization and expression of fellow minds. This entails the cultivation of a society that allows for the freedom of expression, equal opportunity, art, education, and the eradication of hatred, violence, sexism, racism, homophobia, etc.

As we move through time, minds will create increasingly harmonized societies. With the development of technology and the evolution of consciousness, minds will begin to attain greater control of their environment, allowing them to alter reality into the perfect dream. Just as they first created better societies, they will go on to create better and better dreamworlds. As reality and individual minds continue to grow, optimize, evolve, and harmonize, they will relate to each other in such perfection that they will begin to work united as one mind while retaining their unique individuality, just as every individual part of the body is unique and has its own purpose but works together to create a whole.

Mathematically, everything discussed in this section amounts to exploring mathematical patterns that result in greater degrees of symmetry and orthogonality. To put it crudely, the better that things “fit together,” the better it feels. As we move through time, increasingly harmonious dream configurations will be created until such perfection is attained that the dream vanishes entirely. This is the end of the Holos, as the perfect orthogonality of the Source is restored. The approach to this perfect state will be subjectively experienced by us as a near-infinite moment of what one might describe as perfect love as all minds relate to each other in such perfection that it will be as though we are one mind. After this near-infinite experience, we, as the one Absolute collective mind, will "shatter itself" and begin a new dream.[11] This is the Big Bang.

Since this book deals with the mathematical form rather than the content, the subjective experiential aspect isn't discussed in detail. This is why this text is best read as a companion to our other books. Explore our other Hyperian material to discover more.

3.6 - The New Paradigm

Ontological mathematics is a system of living mind. It shows that the foundation of reality is a myriad of individual minds whose collective unconscious thought patterns give rise to the shared dream of the universe. As you can see, the new paradigm easily accounts for what the old paradigm cannot. The problems of the old paradigm of scientific materialism and empiricism are not trivial. They are monumentally important dealing with the foundation of existence, the Big Bang, and that which you're most intimately familiar: your mind and consciousness. Everything we experience is filtered through our subjectivity. No one has ever stepped outside of their mind. Subjectivity is the only thing that you or anyone else has ever experienced, and yet the old paradigm can't account for it.

If you refer back to the list that contrasts the differences between the old and new paradigms at the end of section 1.7, you'll see how opposite they are. As was pointed out, but worth emphasizing again, a common misconception is that this means the new paradigm rejects everything about the old and vice versa. Some people are quick to point out that scientific empiricism uses mathematics and deductive reasoning, while they simultaneously assume we reject empirical evidence and induction. This is a misunderstanding on their part. Of course scientific empiricism uses deduction and mathematics, and we embrace empirical evidence, induction, and the scientific method. The differences between the old and new paradigms are regarding the fundamental nature of reality, what leads to epistemic knowledge, and what methods are given priority.

Both the old and new paradigms use empirical evidence, mathematics, and reason. However, the old says that mathematics is just a tool and the senses lead to knowledge. The new says that mathematics is the foundation of reality, the senses are just a tool, and reason leads to knowledge. Even though both paradigms share certain methods, they are worlds apart in how those methods are employed and which are given priority.

Ontological mathematics is a system of scientific rationalism and rejects the old paradigm of materialism and empiricism. Rationalism means that knowledge of reality is obtained through reason, whereas empiricism means that knowledge of reality is obtained through the senses. We give primacy to a priori, necessary, universal truths while empiricism gives primacy to a posteriori, contingent sense data.

Many people think that scientific empiricism gives primacy to logic and reason but that is simply false. It’s an empirical system and not a rationalist system. Empiricists consider rationalist methods, such as mathematics, to be useful tools, but they always give priority to sense data. Our position is the exact opposite. We understand that sense data can provide useful information about the Holos, but we always give priority to logic, reason, and mathematics. We don’t deny the scientific method. It’s incredibly powerful. However, we understand that it’s a tool only applicable to the contingent dreamworld of the spacetime Holos. Only the mind can access the Source singularity and the necessary, formal aspect of reality. Empirical methods don’t apply to a nonempirical reality. You can’t apply physicalist methods to a non-physical domain. Ontological mathematics goes above and beyond these methods and allows us to explore the Source domain of pure mind.

"From all we have learnt about the structure of living matter, we must be prepared to find it working in a manner that cannot be reduced to the ordinary laws of physics. And that not on the ground that there is any "new force" or what not, directing the behavior of the single atoms within a living organism, but because the construction is different from anything we have yet tested in the physical laboratory."

– Erwin Schrödinger

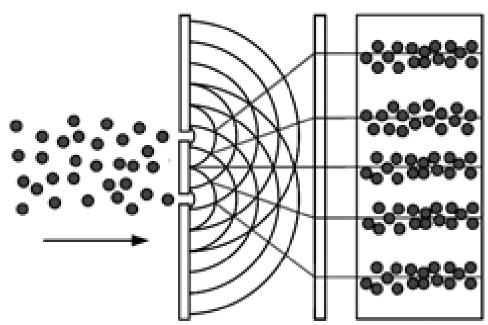
3.7 - Quantum Mechanics

Once again, familiarize yourself with the double-slit experiment, if you haven't already. The Copenhagen interpretation is one of the most well-known interpretations of quantum mechanics. If you've ever watched a video or documentary on quantum physics, this is almost certainly the interpretation you were given. The Copenhagen interpretation states that an unreal probability wave collapses into existence upon measurement. If that sounds strange, that's because it's irrational. Materialists don't believe in an existence beyond matter. When a materialist uses the term "unreal," they don't mean "mind" or a kind of substance beyond matter. They mean "unreal." Which is a completely undefined concept. Their position is that an "unreal" wave collapses into reality. If that idea seems hard to grasp, that's because it doesn't make sense. It's a logical impossibility. It's a violation of the necessary laws of thought. Remember that unreal doesn't mean nonphysical. They don't believe in the nonphysical. Unreal simply means unreal.

To a rationalist, this position is obviously crazy. Something "unreal" can't suddenly become real. That isn't an explanation. That's equivalent to a miracle. You might as well say god did it. Empiricists and materialists take this absurd position because they're attempting to find a way to explain the experimental data in terms of their fallacious paradigm. Their irrational conclusion isn’t based on anything other than being brainwashed into an outdated paradigm and refusing to give it up. They’re like religious believers who refuse to give up their faith when presented with evidence. Materialists are as absurd as flat earthers, albeit in a less obvious way, simply because materialism is popular. In the near future, materialists will be considered just as crazy as flat earthers are today.

As rationalists, we know with certainty that the Copenhagen interpretation is false since it conflicts with reason. Reality cannot come into existence from an undefined “unreality,” and it can’t collapse into a random result. Empiricists can believe the Copenhagen interpretation, even though it conflicts with reason, because they're empiricists first.

The irrational Copenhagen interpretation is given to try and make sense of wave particle duality within a materialist paradigm. Ontological mathematics can easily solve wave particle duality: A particle does not collapse into reality from an unreal probability wave. Particles are flowing points following their wave trajectories. Easy and done. All that needs to be done is replace the “unreal probability wave” with an internal mental wave.



(Fig 3.71) Particles do not collapse into reality from an unreal probability wave. Particles follow a well-defined mental wave trajectory, defined by their inner mathematical formula.

The scientific community has been dumbfounded for a century over wave particle duality and cannot answer how something can have both wave-like and particle-like properties. The answer is beautifully simple. It exhibits both properties because points flow in wave trajectories. Points relate to the particle aspect, and the trajectory of its motion is the wave aspect. Problem solved.

Why can't materialists accept this elegant solution? Because the wave trajectory is the particle's inner coding. In other words: it is mind. This doesn't mean the particle is conscious. It doesn't mean it has hopes and dreams or fears or anything like that. It simply means it has an inner coding. For materialists to accept that a particle follows a trajectory guided by an internal coding would mean they would have to admit that a particle has an inner aspect, but this is outside their materialist paradigm. Just like the religious who distort reality to try and make it fit with the outdated ideas in their holy book, materialists distort reality in an attempt to make it fit with their faith in the outdated idea of matter. Of course, this is usually unconscious. Because materialists are so brainwashed by their beliefs, they would rather believe in a random "unreal" wave than in a real mental wave. This is a reality of mind, not matter.

The Copenhagen interpretation claims that a quantum system remains in a superposition of states until the unreal wavefunction collapses into a random state. This is utter nonsense. Have you ever heard of Schrödinger's cat? This is a thought experiment that demonstrates the implications of this interpretation. In a nutshell, it says that if you put a cat into a box along with a poison gas container where the release of the poison depends on a quantum superposition, then the cat is both alive and dead at the same time. It's not until you open the box that it collapses into a definite state of either dead or alive. Do you see how absurd this is? Empiricists and materialists will laugh at the religious who believe in a talking snake, yet they believe in cats that can be both living and dead. The cognitive dissonance is astounding. They're like Christians who laugh at the idea of Zeus while believing in God.

Students learning about quantum mechanics in a university course will be taught the Schrödinger’s cat thought experiment as though it were true. If you’ve ever seen a documentary on quantum mechanics, it’s almost certainly the case that a “living and dead” cat was featured as fact. Did you know Schrödinger developed his cat paradox, not because he thought it was true, but to emphasize the insanity of the Copenhagen interpretation? Odds are, you didn’t. This fact is conveniently left out of the vast majority of quantum mechanics courses. Wikipedia says, “According to Schrödinger, the Copenhagen interpretation implies that the cat remains both alive and dead until the state has been observed. Schrödinger did not wish to promote the idea of dead-and-alive cats as a serious possibility; on the contrary, he intended the example to illustrate the absurdity of the existing view of quantum mechanics.”

On the statistical interpretation of quantum mechanics, Schrödinger said:

“I don't like it, and I'm sorry I ever had anything to do with it.”

– Erwin Schrödinger

Let’s look at another paradox that arises for materialists and empiricists: quantum entanglement. To put it succinctly, quantum entanglement shows that entangled particles are connected in such a way that an action taken on one particle will affect the other particle instantly, regardless of distance. Why is this a problem? Because it’s *instant*. One particle could be on Earth while the other could be on the other side of the universe, and if they are entangled, one particle will alter its state instantly based on the other particle’s state. This is a problem for materialists since, from a material perspective, you can’t have such occurrences happen instantly over vast distances, and a signal must be sent from point A to point B, which takes a certain amount of time. There are always physical distances, speeds, and travel involved.

This is not a problem for ontological mathematics in the least. Everything in existence is connected through the Source singularity, and all physical distances are an illusion, just as distances in a personal dream are an illusion. Instantaneous communication over vast distances isn't a problem since apparent "vast distances" are an illusion taking place within the connected, collective mental domain of the Source. All minds, particles, and "matter" are inherently connected.

Do you see how all the hard questions of existence are instantly cleared up as soon as we switch from a material view to a mental view? A faulty paradigm will inevitably generate contradictions and unanswerable questions. Consider how Christians are unable to satisfactorily explain how a creator god could exist. They generated their own problem because it's a false paradigm. Likewise, scientific materialism and empiricism has the problem of the Big Bang, the hard problem of consciousness, the foundation of matter, wave-particle duality, the measurement problem, quantum entanglement and on and on ad nauseum. These problems only exist within the context of their failed paradigm. Like the religious, they generate their own errors.

Quantum mechanics shows that all "matter" is based on waves. Materialists don't know what to do with this, which is why they come up with their anti-rationalist interpretations that include "unreal" probability waves. The results of quantum mechanics are perfectly compatible with ontological mathematics since we show that all of reality is based on mental sine and cosine waves, which is a mathematical consequence of the Source formula. These sinusoidal waves are the basis of Fourier mathematics which is the basis for quantum mechanics. Through Fourier mathematics we can create any wave structure, no matter how complicated, from mental sinusoidal waves.

Do you see what this means? These waves are the building blocks of existence. We've derived the existence of these waves from nothing at all, from zero, from mind, and from them, we can build the *entire universe*. Take a moment to let it sink in how monumental this is. We don't need anything else. We don't need matter, god, or an unreal probability function. Using the waves we have derived, we can build the entire universe. That's it. This is the answer. We don't need any other explanation because these waves can account for every possible structure and phenomenon in existence.

Matter is an archaic concept stemming from igorance just as a creator god is an archaic concept stemming from ignorance. We simply don't need the concept of a creator god, and we don't need the concept of matter. It's superflous. It's unnecessary. We can explain absolutely everything in the universe in terms of mental waves without ever referencing the idea of "matter" as an independent substance from mind. Once you change the paradigm from materialism to idealism, the problems generated by the old paradigm vanish. You don't need a creator god, and you don't need matter. Both ideas are equally absurd, but it takes a highly evolved mind to understand that.

3.8 - Special Relativity

We can find absurd irrationalities like the Copenhagen interpretation throughout all of physics. This is due to their empirical, rather than rational, stance. Another example is found in the theory of special relativity. Special relativity makes the claim that there is no absolute reference frame and all references frames are relative. Under this interpretation, there's no objective absolute background to reality.

According to special relativity, if two observers, let's call them A and B, are moving at some velocity v with respect to each other, each observer can make the claim that they are at rest and it's the other who's moving. From person A's perspective, they are at rest and B is moving at velocity v. From person B's perspective, they're the one at rest and A is moving at velocity *v*. According to the theory, there is no absolute reference frame. Everything is relative. A rationalist would immediately spot a problem here. This abolishes an objective reality and leads to all kinds of logical problems.

The theory also shows that the speed of light is constant for all observers no matter the reference frame. This means that, despite A and B being unable to agree on which one of them is moving, they can both agree that light is moving at speed *c*. In special relativity, even though velocities are relative, light speed is the exception on which all observers can agree.

Despite the claim that there is no absolute reference frame, why can all observers, no matter their inertial reference frame, agree on the speed of light? Empiricists have no answer. Ontological mathematics handles this seemingly paradoxical situation easily: There *is* an absolute reference frame. The absolute reference frame *is light*. This is why all observers can agree on the speed of light. Everything moves in relation to light. We saw this in section 2.16. Light doesn't move through spacetime. Spacetime moves through light. What is light? Light is maximally length contracted and maximally time dilated. This means light is dimensionless. It's unextended. Light is mind. Light is thought. This is why we often refer to the mental waves of reality as innerlight. Scientific materialism and empiricism can't explain what light is or how it can have the seemingly strange properties that it does. Once again, this is because it falls outside of their paradigm. Ontological mathematics provides the answer. What scientists have called light is thought. It's the essence of existence. What we refer to as innerlight[12] is none other than the flowing point following its wave trajectory defined by the Source formula. If that seems strange to you it's because you're still operating under the assumptions and definitions of the old paradigm. Matter does not exist. The world around you is a dream. All is mind. Light is thought, and the interactions of thought give rise to the collective dream reality of the Holos. So-called "matter" is transformed light/thought.

"Universe consists of frozen light."

– David Bohm

Innerlight and the minds that contain it are the essence of existence. The Source singularity is a domain of pure innerlight. All motion in the spacetime Holos takes place with respect to light. This is why the speed of light is constant for all observers. *There is an absolute reference frame*. It's light. All mental waves travel at the speed of light and are maximally time dilated and length contracted. They don't exist in spacetime at all, though they can certainly interact with spacetime. All spacetime waves (which, as we saw earlier, are fractional sinusoids) travel at velocities less than the speed of light.

Why is the speed of light the same for all observers? Because light *is the absolute reference frame.* As we saw in section 2.16, all spacetime velocities are derived from and defined by light.

Empirical science can't answer why something can never travel faster than light. We can. The speed of light is the speed of mind and is equal to 1. Since this is a mental speed and the fundamental speed of the universe, this means all spatial and temporal speeds are derived from it. The Mind Square Law is a direct result of the Source formula, which states the speed through space squared plus the speed through time squared equals the speed of mind squared:

$$v_s^2 + v_t^2 = c^2$$

Neither the speed through space v_s nor the speed through time v_t can ever exceed the speed of light c. By placing the mental Source at the heart of existence as the absolute reference frame of light, the paradoxes of special relativity are cleared away, and objectivity is restored. The axial waves of the Source are the absolute reference frame. Let's examine some of the mathematics of special relativity.

The following is known as the Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v_s^2}{c^2}}}$$

It’s okay if you’re not familiar with this equation. Just know that the Lorentz Factor is at the heart of special relativity and is the factor by which time, length, and mass change for an object moving at some speed.

Recall that the Mind Square Law is a direct result of the Source formula. The law states that the speed through space squared plus the speed through time squared is equal to light speed squared. Where v_s is the speed through space and v_t is the speed through time.

This means that as an object’s speed through space increases, its speed through time must decrease. This is in agreement with special relativity.

Let’s take the Mind Square Law and solve for v_t:

$$v_s^2 + v_t^2 = c^2$$

Subtracting v_s^2 from both sides:

$$v_t^2 = c^2 - v_s^2$$

Dividing by c^2:

$$\frac{v_t^2}{c^2} = \frac{c^2}{c^2} - \frac{v_s^2}{c^2}$$

Simplifying:

$$\frac{v_t^2}{c^2} = 1 - \frac{v_s^2}{c^2}$$

Taking the square root of both sides:

$$\sqrt{\frac{v_t^2}{c^2}} = \sqrt{1 - \frac{v_s^2}{c^2}}$$

Simplifying:

$$\frac{v_t}{c} = \sqrt{1 - \frac{v_s^2}{c^2}}$$

Multiplying both sides by *c*:

$$v_t = c\sqrt{1 - \frac{v_s^2}{c^2}}$$

The term $\sqrt{1 - \frac{v_s^2}{c^2}}$ is one over the Lorentz factor γ, so it equals $\frac{1}{\gamma}$.

Our final equation is:

$$v_t = \frac{c}{\gamma}$$

We just derived the Lorentz Factor from nothing but exploring the ontological properties of the Source Formula. No sensory experiments needed. There are two critical points to take away from this example.

The first point is that absolutely everything fundamental about existence can ultimately be derived from the Source formula. This includes all so-called “physical laws” and the structure of all so-called “material” particles. The Source formula is the master equation for all existence, and everything can be accounted for according to it. All “matter” and all “physical” laws come from this one Source equation.

Scientific materialists and empiricists *have* never, *can* never, and *will* never explain what physical laws are ontologically. If everything is material, what are physical laws? Are they made of matter? Obviously not. Then what are they? You'll never get an answer. In ontological mathematics the answer is easy. Physical laws aren't physical at all. They're the necessary mathematical consequences of the one governing equation of all reality: the Source formula.

The second point is that all the paradoxes in modern physics can ultimately be resolved by acknowledging the existence of the mental Source domain governed by the Source formula. We derived the Lorentz factor, which forms the basis of special relativity, from our system which most certainly *does* include an absolute reference frame: The axial waves of pure mind, which is light itself. This resolves the absurdities that arise when you try to interpret existence from a materialist perspective that denies the existence of a mental domain. Since they deny the mental domain (which is the absolute reference frame and ground of existence,) you get the insanity of a world where objectivity has been banished and everything is relative. This is what happens when you choose to ignore the foundation—the ground—of existence.

Paradoxes are littered throughout special relativity and quantum mechanics, and as we've shown, ontological mathematics can resolve them since it includes what they don't: the mental domain. No wonder you get paradoxes when you deny the foundation of existence. It's like trying to build a house on nothing at all. It just doesn't work. You need a ground to build something stable.

A major problem in physics right now is reconciling relativity and quantum mechanics. In general, relativity deals with the very large, and quantum mechanics deals with the very small, and physicists can't get the two theories to work together.

As demonstrated in our example, we've linked the Source formula with Einsteinian relativity. But the Source formula is also the basis of Fourier mathematics from which you get quantum mechanics. The Source formula is at the heart of both relativity and quantum mechanics. It's the link to unifying both theories. If scientists would abandon their materialist dogma and accept the existence of a mental domain governed by the Source formula, they would discover the key to unifying quantum mechanics and relativity. If they devoted their time to studying the Source formula and grounding their theories in an absolute mental domain of light instead of trying to force everything to fit within their materialist and empiricist faith with no ground at all, they would discover the unified theory they've been searching for.

The old paradigm of materialism and empiricism has taken a stance against reason. Through the Copenhagen interpretation of quantum mechanics, reality is based on "unreality" and randomness. Through special relativity, objectivity has been abolished and existence made subjective and relative. These absurdities are anti-rationalist and exist only because empiricists are interpreting their results through the lens of a failed paradigm. If you place the Sun at the center of the universe, you're going to run into problems. It's just wrong. If you place matter at the foundation of reality, you're going to run into problems. It's just wrong.

Pastor Peter LaRuffa said, "If somewhere within the bible, I were to find a passage that said 2 + 2 = 5, I would believe it, accept it as true and then do my best to work it out and understand it."

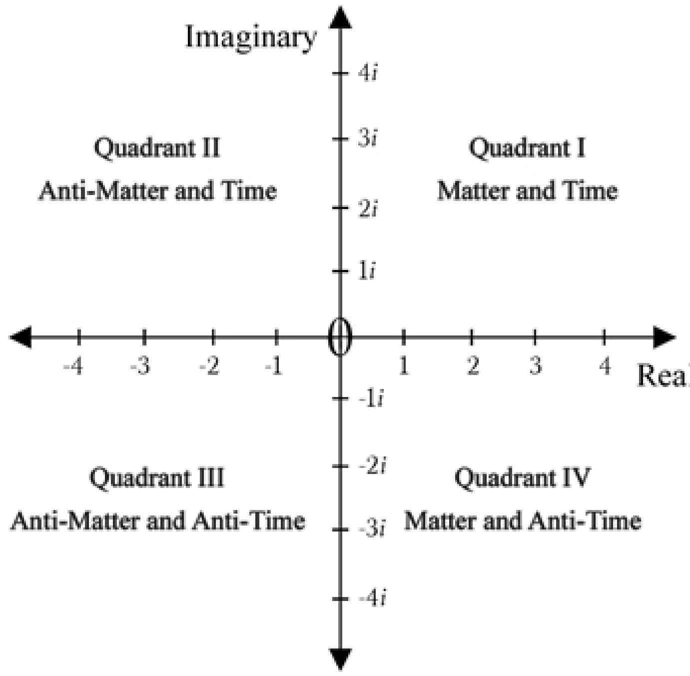
Pastor LaRuffa clearly places his faith over reason, even when the position is absurd. Scientific materialists and empiricists do the same when they interpret reality as unreal, random, and relative, according to their beliefs. These interpretations are absolutely opposed to reason and are as absurd as saying that 2 + 2 = 5. It just takes a higher intelligence, a rationalist, to understand that. They'll laugh at Pastor LaRuffa, while they commit the same fallacy. They'll mock the religious for their belief in creationism and resurrection, while they believe in cats that can be simultaneously alive and dead, a material universe that randomly springs into existence from nothing for nor reason through no mechanism, and unreal waves of probability that collapse into existence. They aren't rationalists. They only pretend to be when it suits their beliefs, just like those who subscribe to a religious faith. Existence isn't random, unreal, or relative. Our system shows with mathematical precision that existence is certain, mental, and founded on an absolute reference frame.

3.9 - Antimatter

One of the greatest unsolved questions of physics concerns the apparent absence of antimatter in our universe.

CERN has an article titled "The matter-antimatter asymmetry problem" in which they state, "The Big Bang should have created equal amounts of matter and antimatter in the early universe. But today, everything we see from the smallest life forms on Earth to the largest stellar objects is made almost entirely of matter… One of the greatest challenges in physics is to figure out what happened to the antimatter, or why we see an asymmetry between matter and antimatter."

This is another question easily answered by ontological mathematics. Consider the complex plane. It's from this plane that the Source formula draws all its values for the creation of the collective dream of the universe at the Big Bang.



(Fig. 3.91) The complex plane defines four sub-universes generated at the Big Bang.

When we examined the interaxial wave projection in Fig. 2.163 in section 2.16 – The Big Bang, the wave we examined came from Quadrant I. Thus, it projected waves associated with positive real numbers and positive imaginary numbers, which led to space, time, and matter. Let's consider the other quadrants and their values.

Real numbers are associated with matter. Imaginary numbers are associated with time. Negative real numbers are associated with antimatter. Negative imaginary numbers are associated with anti-time.

At the Big Bang, one objective universe was generated. However, the complex plane shows the generation of four sub-universes, one for each quadrant. The positive/negative combination for each quadrant is different, which results in a different composition for each sub-universe.

For example, Quadrant I is composed of positive real numbers and positive imaginary numbers, thus Quadrant I results in a sub-universe of matter and time. Quadrant II is composed of negative real numbers and positive imaginary numbers, and Quadrant II results in a sub-universe of antimatter and time. Quadrant III results in a sub-universe of antimatter and anti-time. Quadrant IV results in a sub-universe of matter and anti-time.

At the Big Bang, equal amounts of antimatter and matter were generated, but they were generated in separate sub-universes. This is a mathematically inevitable result. You'll notice that each quadrant is separated by the axes, meaning these sub-universes are inaccessible to us since it would require the acceleration to light speed, and by the light speed condition, if something were accelerated to light speed, it would cease to be matter and become mind. It's unknown which sub-universe we exist in since we could have just as easily labeled "matter" as "antimatter" (the same logic applies to time and anti-time) since our matter is another sub-universe's antimatter.

Since the Source formula is inherent in every portion of reality, it controls both cosmic processes and processes at the particle level. At the cosmic level, there are clearly delineated zones, as illustrated by the four quadrants. Even though we may be in a quadrant of matter, antimatter can still be generated in highly energetic processes that mimic the conditions of the Big Bang. This is possible because the Source formula is embedded in every particle. In other words, even though at the Big Bang our quadrant may have settled into a state of "matter," the particle level is still controlled by the Source formula. This means it contains the mathematical "coding" for antimatter that we can "tease out" by energetic processes similar to those at the Big Bang.

Not only does ontological mathematics elegantly answer the question of "missing" antimatter, it also reveals the existence of anti-time and sub-universes. We can go where physicists could never dream.

3.10 - The Six-Dimensional Universe

It's generally said that we exist in a domain of 3 spatial dimensions and 1 time dimension. We know that space is associated with real numbers and time with imaginary numbers. We also know, via the complex plane, that the real axis is paired with the imaginary axis. This means that if we have 3 real dimensions, then we also necessarily have 3 imaginary dimensions. We therefore exist in a 6-dimensional space comprised of 3 real dimensions (space) and 3 imaginary dimensions (time). The explanations in this text have been restricted to circles and the complex plane for simplicities sake, as spheres and a 6-dimensional spacetime are beyond the scope of this introductory book. However, 6d spacetime is discussed in other books on ontological mathematics.[13]

3.11 - Abstract vs Ontological Mathematics

We have shown that this is a reality of living minds that unconsciously generate a collective dreamworld. This is all possible through ontological mathematics. Ontological mathematics is worlds apart from abstract mathematics. It's alive. This can't be stressed enough, yet so many people who encounter our system ignore this critical point. Ontological mathematics is about eternal, living thought. It's about waves of mind that have both external form and internal content. It's *not* about empty marks on paper. Consequently, ontological mathematics must be treated in a completely different way from ordinary mathematics, and there are certain constraints and conditions it must follow, such as the PSR and the zero condition.

Another key difference is that we have reduced existence to one equation: the Source formula. This means that all ontologically valid mathematical operations must be based on manipulations of this formula that are allowable by the PSR. Any configuration that can't be reduced back to this formula isn't valid. This means there are many manipulations in abstract mathematics that aren't valid ontologically.

In abstract mathematics, sinusoids are simply mathematical abstractions and nothing more. They're empty concepts. In ontological mathematics, sinusoids are photons following a wave trajectory (light/thoughts). This radically different viewpoint completely alters what kind of mathematical operations are valid. An abstract mathematician will have no problem considering half a sinusoid. But in ontological mathematics, since sinusoids are light, we must always consider what this means ontologically. Obviously, there's no such thing as half a photon, and any equation that seems to refer to such an impossible entity is either ontologically false or must be carefully reinterpreted. Likewise, when encountering an equation that refers to half a sinusoid, we must treat it completely differently when using the perspective of ontological mathematics.

The genius logician and mathematician Kurt Gödel believed mathematics had actual existence. Kurt Gödel's incompleteness theorems are extremely important to ontological mathematics. They show that mathematics can't be constructed through human-invented axiomatic logic. Ontological mathematics isn't a human construction; it's the nature of reality.

While Gödel believed mathematics was real, the philosopher Wittgenstein believed that mathematics was nothing but a **tautology**.

> **What is a Tautology?**
>
> A tautology is the expression of the same thing in different ways.
>
> **2 + 2 | 1 + 1 + 1 + 1 | 3 + 1 | 2 x 2 |** and **4 x 1** are all tautologies of **4**.
>
> Existence is the exploration of the infinite tautologies of the Source formula. If anything is not a tautology of the Source formula, then it's not ontologically valid.

The truth is that both Gödel and Wittgenstein were right. Mathematics is real (it's thought; it's light), but it's also a tautology. All of existence is a vast interlocking tautology of the Source formula.

Ontological mathematics is complete and consistent as it's a vast tautology that relies on nothing but a single axiom: the Source formula, which is the ontological expression of the PSR. It's what defines a monad, the ground of existence. Ontological mathematics isn't something invented by humans. It is discovered. Humans invent the notation (symbols) that represent mathematics, *not* mathematics itself. The mathematical laws themselves are eternal and ubiquitously embedded in existence.

"Math is discovered. To be invented requires an inventor, but math exists outside of humanity. But ultimately, the laws of the universe will be reduced down to a single equation, perhaps no more than one inch long. But leaves the final question, where did that one inch equation come from?"

– Michio Kaku, Theoretical Physicist

Michio Kaku is absolutely right to say that reality can be reduced to a single equation. Ontological mathematics reveals that single equation to be the Source formula. We can also answer Kaku's final question, "where did that one inch equation come from?" The Source formula is eternal and its own cause. It's the defining equation of nothingness. By the nature of zero and the PSR*, it must exist*. It can't be stopped from existing. It's mathematically structured void. All of reality naturally unfolds from this equation.

The future is here. We have the key to existence. Are you on board, or are you one of the multitudes blinded by the paradigm of the current age? Those who stand against reason are just as backwards and ignorant as those who had Galileo imprisoned and Bruno burned at the stake.

The paradigm of matter is ending. The paradigm of mind is beginning.

"The ontology of materialism rested upon the illusion that the kind of existence, the direct

'actuality' of the world around us, can be extrapolated into the atomic range. This extrapolation, however, is impossible… atoms are not things."

– Werner Heisenberg, Pioneer of Quantum Physics

The concept of matter is DEAD. But people are still worshipping its corpse. Wake up. Get out of the dark ages.

The people of this world have been incorrectly taught that mathematics is simply a useful tool that humans invented. Nothing could be further from the truth.

"A mathematical truth is timeless, it does not come into being when we discover it."

– Erwin Schrödinger

Mathematical pattern is not something invented. Those patterns have always existed, and they are discovered by humanity. What humans invent are the particular symbols used to represent those patterns, not the patterns themselves. Look around you. Everything is based on pattern. Everything follows a set of rational rules. Temperatures rise and fall. Forces attract and repel. A seed will grow and bloom, and planets orbit.

If reality wasn't mathematical, what do you think it would be? What would be the thing that explains the rational order and pattern all around you? Are you going to find some magic, non-mathematical particle that explains it? That would be a religious faith in a logical impossibility. Furthermore, you would have to explain what that particle is and how it could exist, and if you didn't, that would be just as irrational as saying god created it. Reality is rationally ordered and structured because this is a living mathematical reality. It's a reality based on a rational, energetic pattern. If reality wasn't mathematically structured, everything would be in utter chaos. Nothing would be stable. Completely random things would be happening every instant. In fact, no existence would be possible at all since there would be no formal structure to allow it to exist. Reality can exist only because it has an underlying formal structure.

If you're having difficulty conceptualizing this, think of it this way: Reality is based on waves of energy. These waves are in eternal motion. They are the most basic units of pattern, and when they combine and interact with each other, they create patterns of greater and greater complexity until the patterns are so complex they give rise to everything around you. These waves of energy are perfectly described by mathematics. They're perfectly described by mathematics because they are mathematical. These mathematical waves have a formal aspect and an inner aspect. This is *not* abstract mathematics. This is ontological mathematics. Mathematics is living, striving, eternal mind.

Every pattern that you experience around you is a mathematical pattern that can ultimately be reduced to the Source formula. There is no such thing as a pattern that isn't mathematical. What would a non-mathematical pattern be? That's an impossible idea. Mathematics is *pattern*. This is a living mathematical dream. Mathematics isn't invented. It's discovered.

Let's use the Pythagorean theorem as an example. The Source formula contains within it the pattern for right triangles. We saw this clearly when we examined interaxial waves. Their projection forms a right triangle which, of course, contains the Pythagorean law for right triangles. This mathematical property is embedded in the Source formula, which means it's embedded in the waves of existence and all reality. The Pythagorean theorem is a mathematical property embedded in existence. This mathematical property is then *discovered* by humans (not invented). What humans invent is the notation (symbols), words, labels, and names they use to describe that law, but they don't invent the law itself. They choose to call it the

“Pythagorean theorem.” They invent the word “right triangle.” They choose to use the marks of a, b, and c to denote the legs and hypotenuse of that triangle. All these contingent labels are invented by humans, but the law *itself* is a necessary, eternal property of existence because this is a reality of waves generated by the Source formula which contains the Pythagorean law.

As another example, consider a so-called “physical” law such as the law of conservation of energy which states that energy can neither be created nor destroyed—only transformed from one form into another. Humans invented the name for the law and how it’s described, but they didn’t invent the law itself. The law is discovered, not invented. This also applies to the fundamental mathematical truths of existence. In fact, “physical laws” are merely human interpretations of mathematical truths embedded within the Source formula. A “physical law” is just a mathematical law or result that has been interpreted and distorted through the failed paradigm of empiricism. Humans didn’t invent the law of conservation of energy, and they didn’t invent the Pythagorean law. The Pythagorean theorem is a fundamental truth of existence. It’s contained within the Source formula and is necessary for generating waves. This is an existence of waves, and thus existence has the Pythagorean law intrinsically “built in.”

To say that reality is mathematical means that reality is based on an underlying system of interacting mental energy waves that follow a precise mathematical pattern. By understanding how this pattern works, you can know everything about existence.

The ancients recognized this. They realized that there was a rational ordering and underlying pattern. Pythagoras and the Pythagoreans believed that reality was mathematical. Plato, who was heavily inspired by Pythagoras, said that the material world was an illusion and that the true reality was a mental domain of perfect forms that could be apprehended through reason. In Plato’s Timaeus, he describes the world being formed from geometric patterns.

Heisenberg said, “…modern physics takes a definite stand against the materialism of Democritus and for Plato and the Pythagoreans. The elementary particles are certainly not eternal and indestructible units of matter, they can actually be transformed into each other. As a matter of fact, if two such particles, moving through space with a very high kinetic energy, collide, then many new elementary particles may be created from the available energy and the old particles may have disappeared in the collision. Such events have been frequently observed and offer the best proof that all particles are made of the same substance: energy.” Energy is mind. Energy is thought. Energy is frequency. Energy is innerlight.

Heisenberg goes on to say, “But the resemblance of the modern views to those of Plato and the Pythagoreans can be carried somewhat further. The elementary particles in Plato's Timaeus are finally not substance but mathematical forms. ‘All things are numbers’ is a sentence attributed to Pythagoras. The only mathematical forms available at that time were such geometric forms as the regular solids or the triangles which form their surface. In modern quantum theory there can be no doubt that the elementary particles will finally also be mathematical forms but of a much more complicated nature.”

“The smallest units of matter are not physical objects in the ordinary sense; they are forms, ideas which can be expressed unambiguously only in mathematical language.”

– Werner Heisenberg

Ontological mathematics is that language. The science of the future is here.

3.12 - Alternate Dream Spaces

Ontological mathematics shows that personal dreams are the result of the frequency configurations of your personal sinusoids and that the collective spacetime Holos is the result of the collective frequency configuration of all the minds in existence. There is nothing to prevent highly evolved minds from creating collaborative dream spaces. This means shared dreaming and the creation of collaborative dreamworlds (alternate frequency configurations). A personal dream results from the sinusoids of an individual mind. The collective dream results from the sinusoids of every mind. Collaborative dreams result from the sinusoids of a group of highly evolved minds operating together. The more minds that contribute, the more stable these alternate dream spaces become. These alternate frequency configurations that result in alternate dream spaces are all around you, right now. You simply aren't tuned in to them, just as you aren't tuned in to your personal dreamworld at this moment. This has incredible implications that will be elaborated on in future material. The astute reader may begin to see where this leads.

3.13 - The Science of the Future

Do you want to know more about ontological mathematics? Where should you go from here? This book is best read as a companion to our other Hyperian material. If you haven't read any other Hyperian material, that's the next step as it begins to lay out our system in an intuitive way. Continue your journey by going to www.iamhyperian.com.

This book is greatly simplified and restricted to the complex plane. If it wasn't, it would be thousands of pages long. We've shown the solution to the Big Bang, consciousness, wave-particle duality, quantum entanglement, and antimatter asymmetry. We've replaced the randomness of the Copenhagen interpretation with mathematical certainty and restored objectivity to special relativity by showing that light is the absolute reference frame. There is so much more that ontological mathematics has solved, and this is explored in other material. Laying out every detail of ontological mathematics is not the goal of this introductory text. This is designed to give you just a taste and is written in such a way for it to be accessible to everyone.

If you're interested in continuing to explore the mathematics itself in more detail, begin by studying all the concepts marked in bold. You can easily find videos on these subjects. Once that has been completed, you should study the **Truth Series by Dr. Thomas Stark**, and the **God Series by philosopher Mike Hockney**. We recommend beginning with the Truth Series. These two resources alone will provide you with almost 50 books on ontological mathematics. If you're interested in **Hyperianism**, which explores how to apply this information to your life and what it means in terms of unlocking your inner potential, look into the material listed on www.iamhyperian.com.

We have shown that reality is a shared dream. If someone doesn't understand how we've done so and is still looking for empirical evidence, they haven't understood the new paradigm of rationalism and are still operating under the old paradigm of empiricism. They're like a religious person demanding that it be shown to them through scripture. It's not faith, belief, or the senses that explain reality. It's reason, logic, and mathematics. These are tools of mind, and only mind can access the true structure of existence because this is a mental reality. Only the mind can access the mind. This is not a material universe. The word "matter" should cease to be used. This is a mental reality, and the "physical" world is a collective dream.

What could possibly explain existence other than reason? It makes no sense to posit that anything other than reason could be the answer. An explanation entails that something has a reason. The rational laws of existence have never been violated and never will because existence is reason expressed. It's the expression of the PSR which entails ontological mathematics and all reality. Ontological mathematics shows that reality is living mathematical mind. Reality can be no other way. It must be this way. This is not some theory or speculation. This is the only possible way reality can be. It is the only possible configuration that can exist. It is philosophically *necessary* that it be this way. It must be this way. It is an impossibility for it to be any other way.

From nothing at all, we've seen how sinusoidal waves and the minds that contain them are possible. They are the true zeros and ground of existence. We've discovered the one formula, the Source formula, that governs all reality. Through phase relations and Fourier mathematics, the collective of all minds generate complex thoughts that interact, stabilize, and result in an inner dream universe. Through the Principle of Sufficient Reason, ontological mathematics proves that this reality is a shared dream. We are the future. We are the bridge to a new world. We are the architects of the cosmic dream. Ontological mathematics is taking the world by storm. Myriad books have been written on this subject by various authors. The world is beginning to wake up to the dream around them.

The old paradigm is crumbling. The new paradigm is here. Ontological mathematics has the power to answer all the major questions of existence. Soon this will be taught in every school across the world, and "matter" will be looked at as a relic of the past.

This book will change the world. By understanding its contents, you are gaining knowledge now about reality that the masses will only understand in the future. By integrating the concepts within, you are accelerating your evolution and expanding your consciousness. Be a part of the future *now*.

You can choose to be like those of the Catholic Church and stand with what is old, or you can be among the trail blazers, the Galileos and Brunos of the world, and stand with the new.

Existence is living thought. It's a multiplicity of minds weaving together a cosmic dream of innerlight. We are individual minds, and taken together, we form one cosmic mind—the Absolute. As we awaken to that fact, the Absolute awakens to itself. Hyperianism is the collective becoming conscious. We will begin the collective NOW and dream a better dream.

YOU ARE A MIND. EXISTENCE IS THOUGHT.
THE WORLD IS A DREAM.

THE SCIENCE OF THE FUTURE IS HERE.

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[11] It's important to note that cosmic cycles aren't identical. Each new dream reality unfolds in a completely unique way. Each Big Bang is an unconscious, mathematical process.

[12] There is no difference between light and innerlight. They are the same concept. However, when light is defined from the perspective of spacetime, strange results occur such as its seemingly arbitrary value for speed when measured in meters per second. It's not the speed of light that is arbitrary; it's the contingent spatial-temporal concepts of meters and seconds that are arbitrary. People are in the habit of defining light according to their spacetime perspective, so we use innerlight as a helpful reminder that light must be defined according to the necessary mental domain of the Source, not the contingent spacetime domain of the Holos.

[13] We can refer to our existence in the spacetime Holos as having 6 dimensions *or* 3 complex dimensions. Since each point in the complex plane is defined by one number (a complex number), the complex plane is 1 dimensional, considered as a complex dimension.

“Philosophy is written in that great book which ever lies before our eyes — I mean the universe — but we cannot understand it if we do not first learn the language and grasp the symbols, in which it is written. This book is written in the mathematical language, and the symbols are triangles, circles and other geometrical figures, without whose help it is impossible to comprehend a single word of it; without which one wanders in vain through a dark labyrinth.”

– Galileo Galilei

"THIS FINAL PAGE IS THE FIRST IN THE STORY OF YOU"

Create yourself
Transform your life
Become whole

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